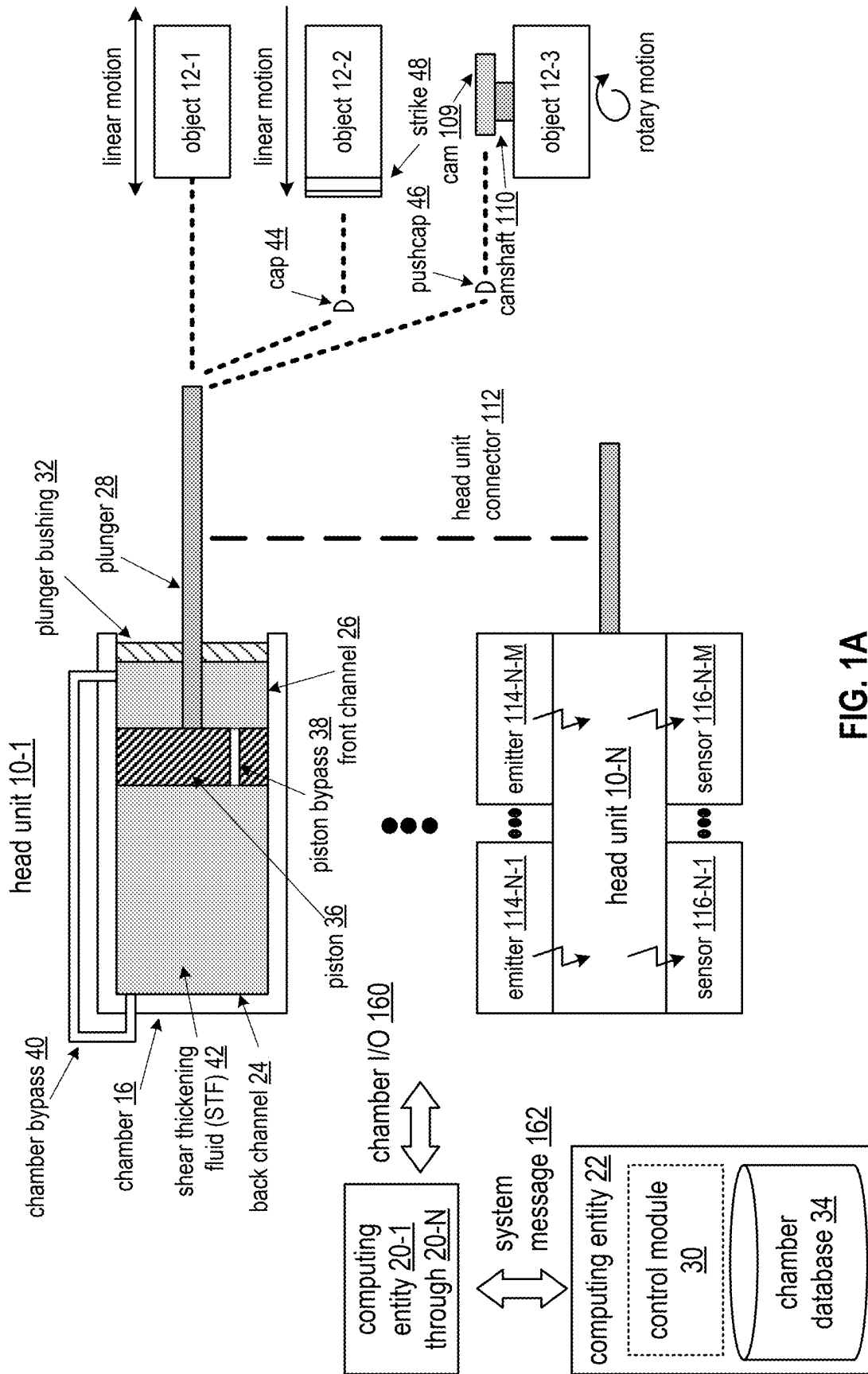




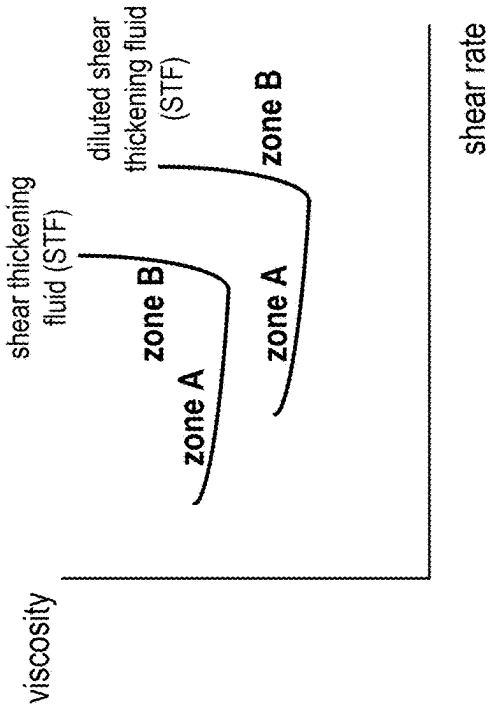
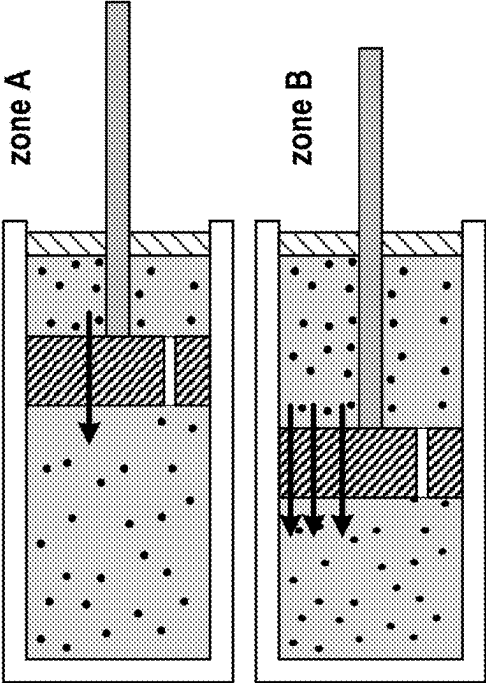
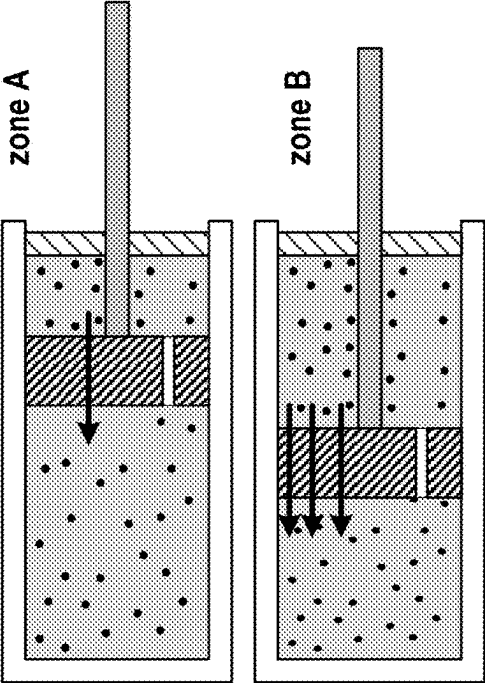


FIG. 1B

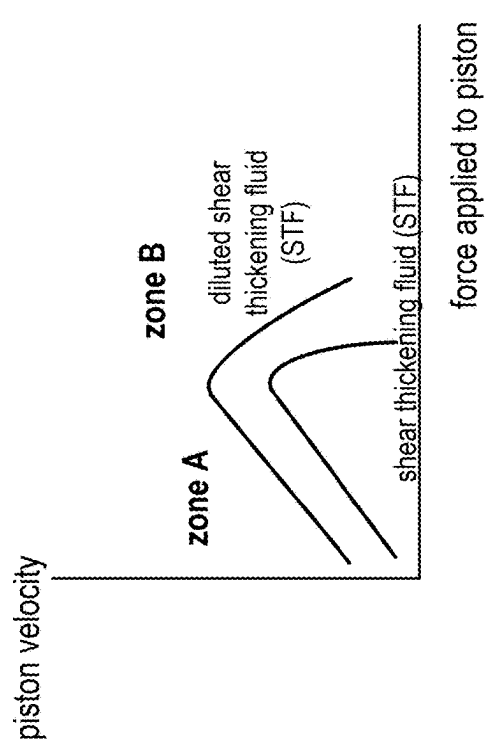


FIG. 1C

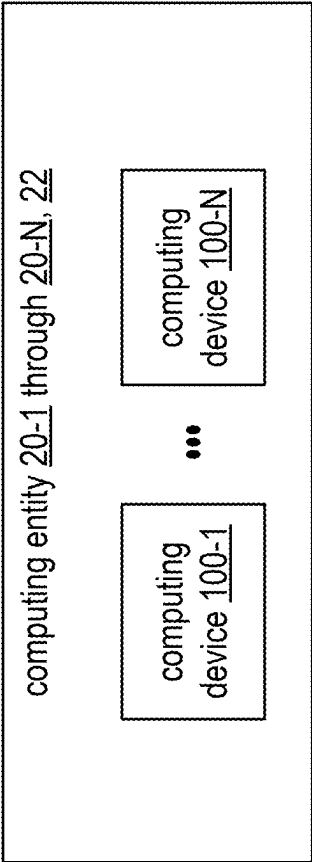


FIG. 2A

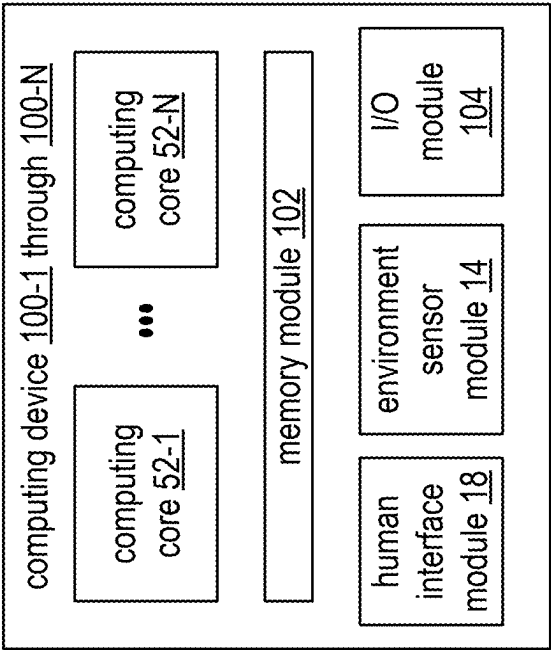
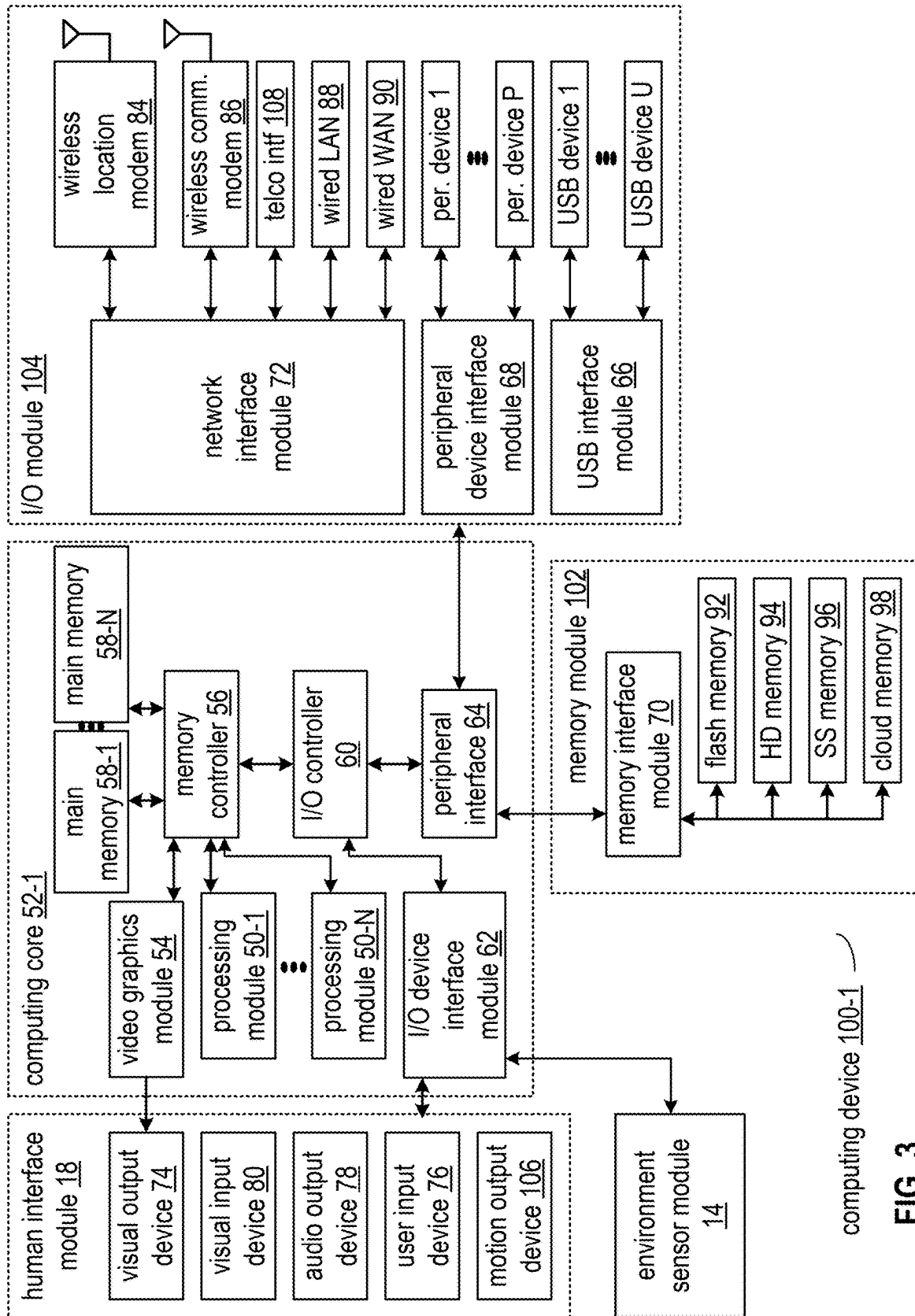


FIG. 2B



computing device 100-1

FIG. 3

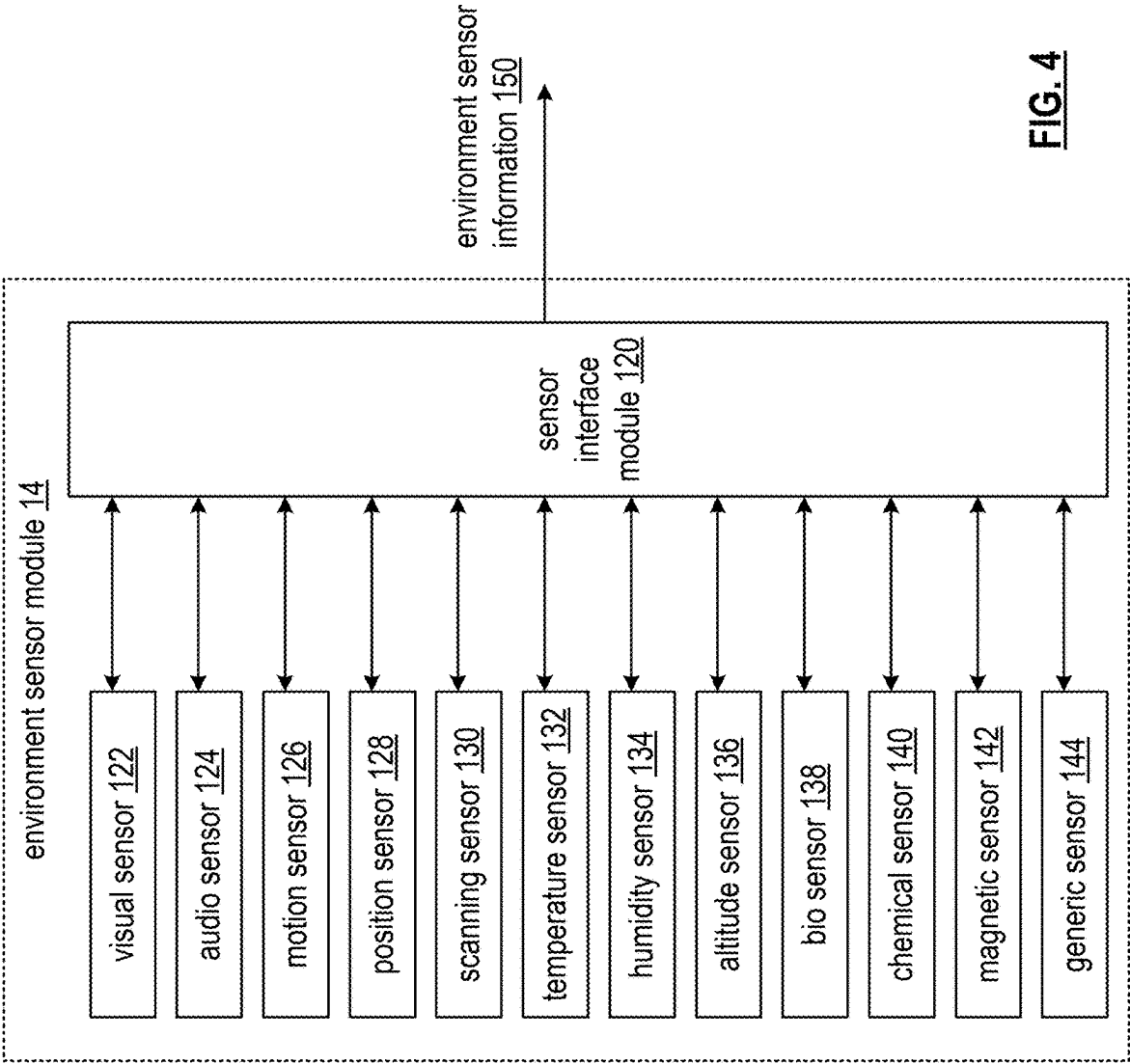
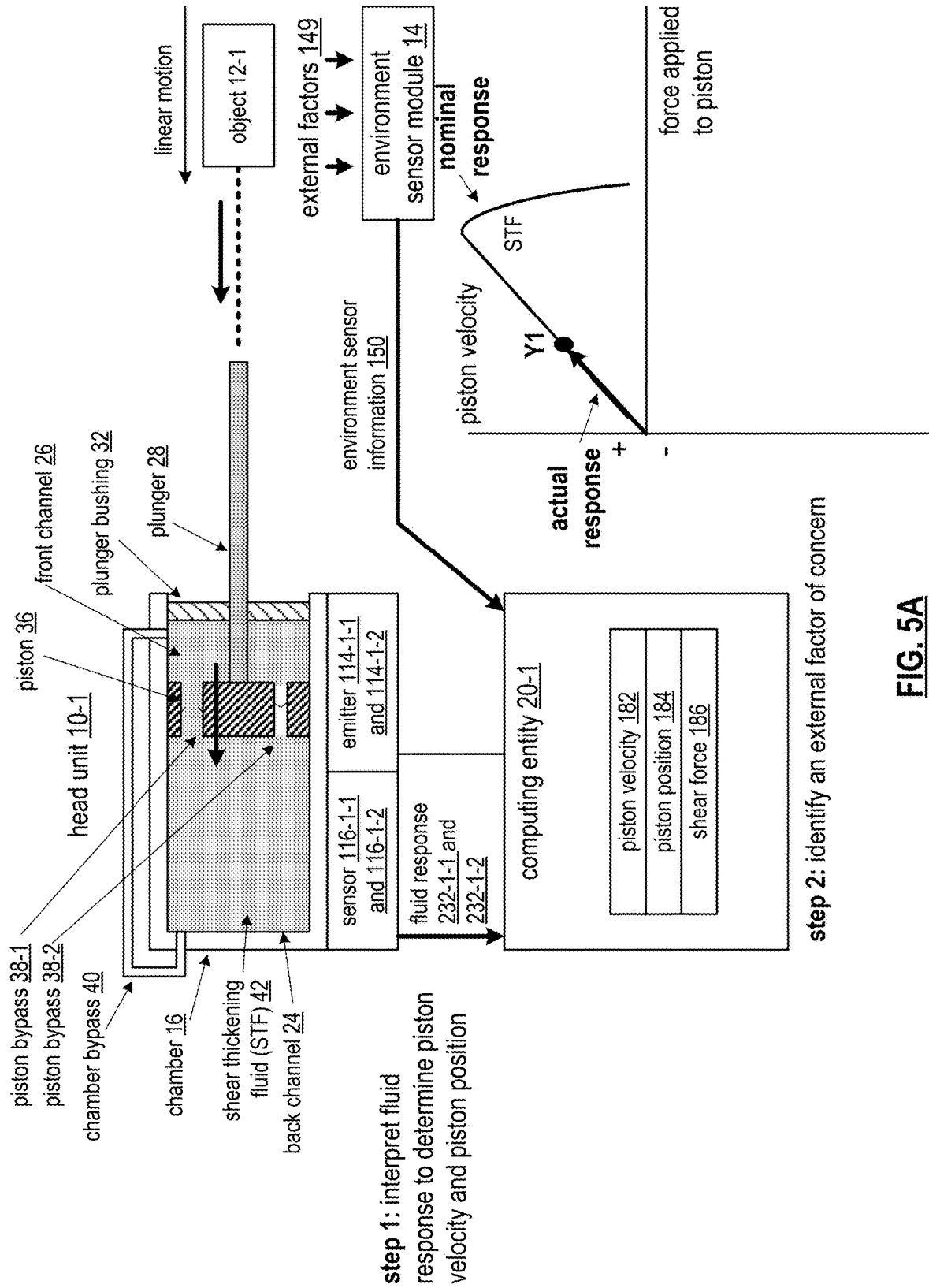
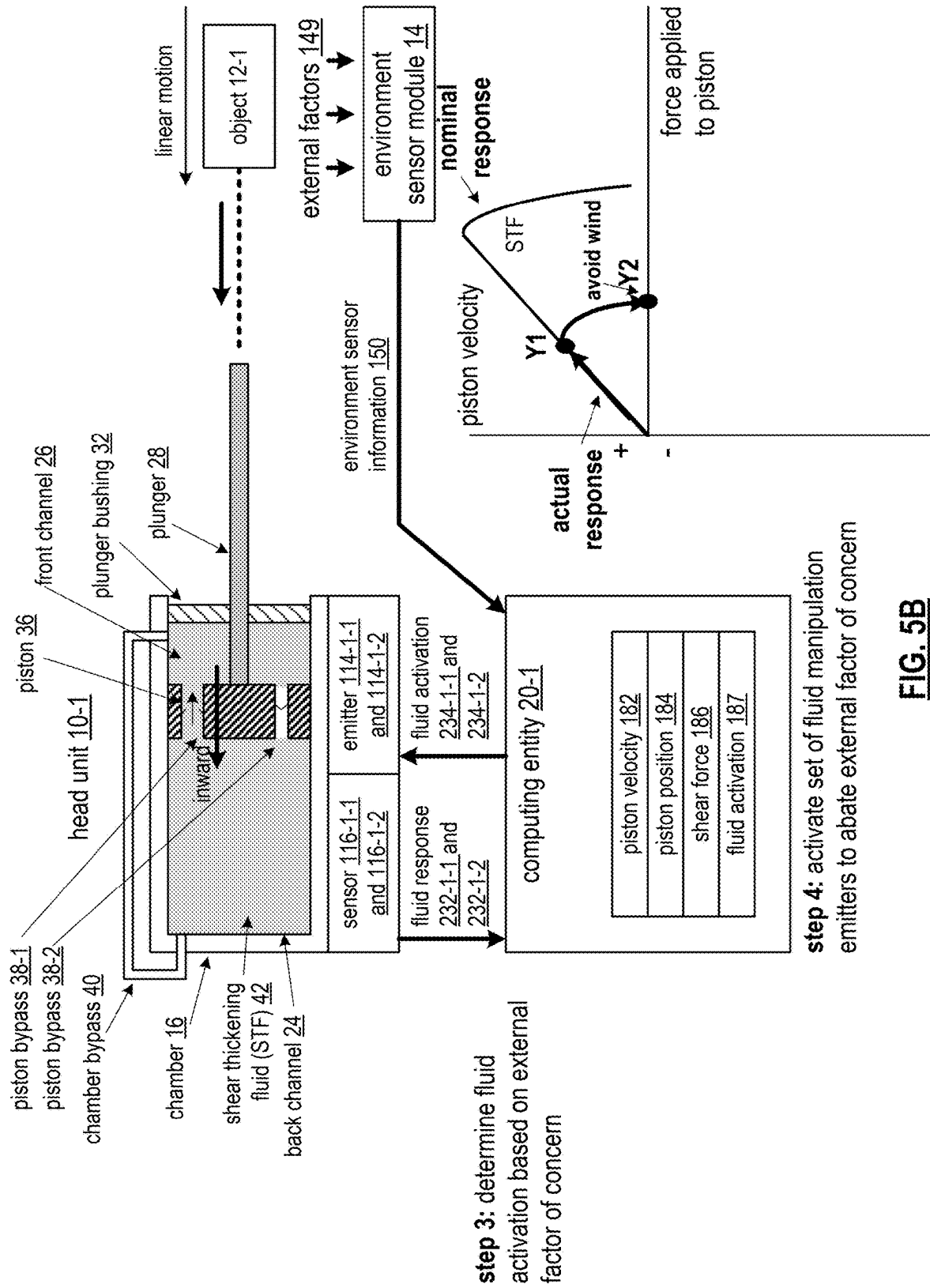
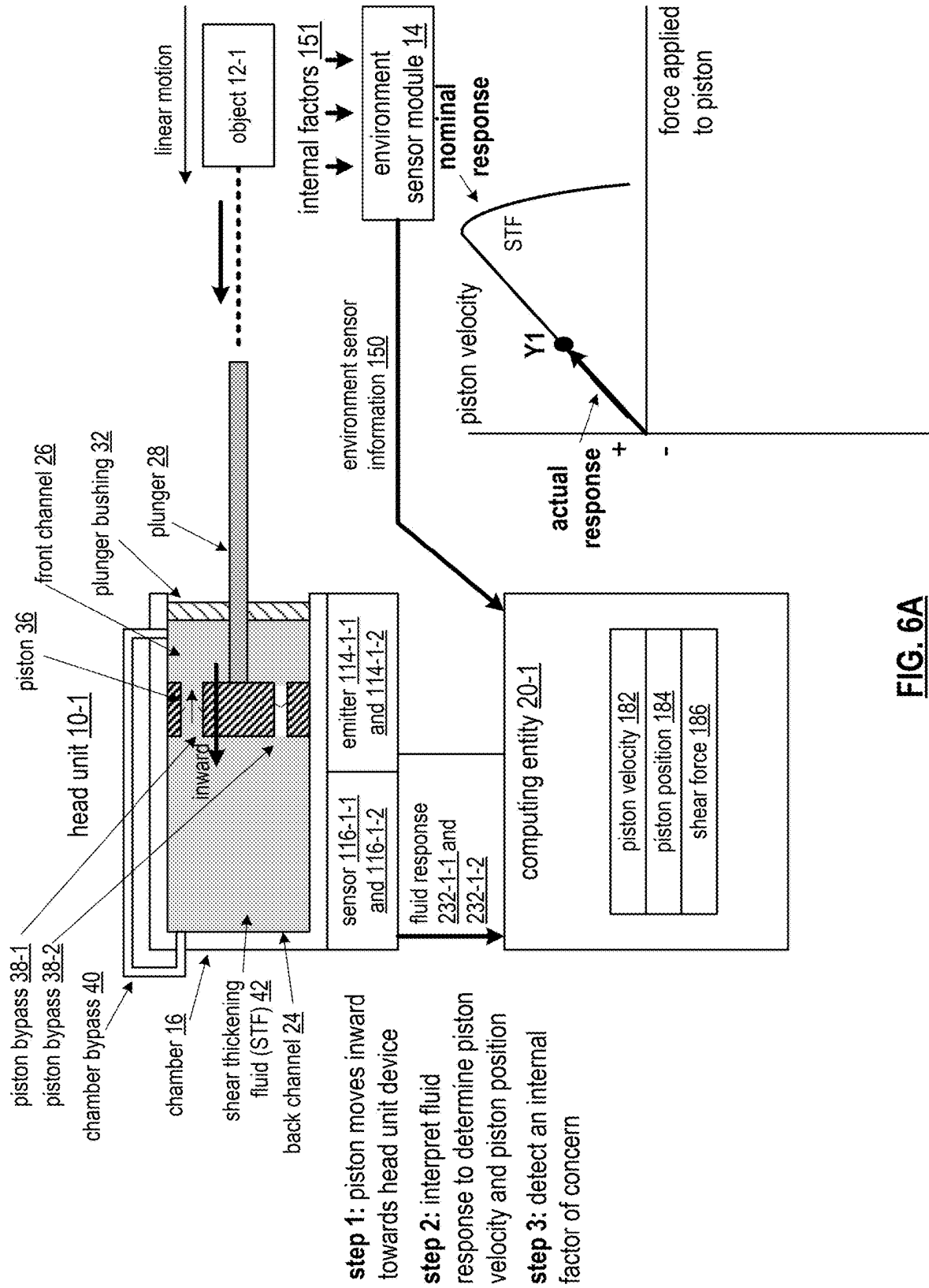
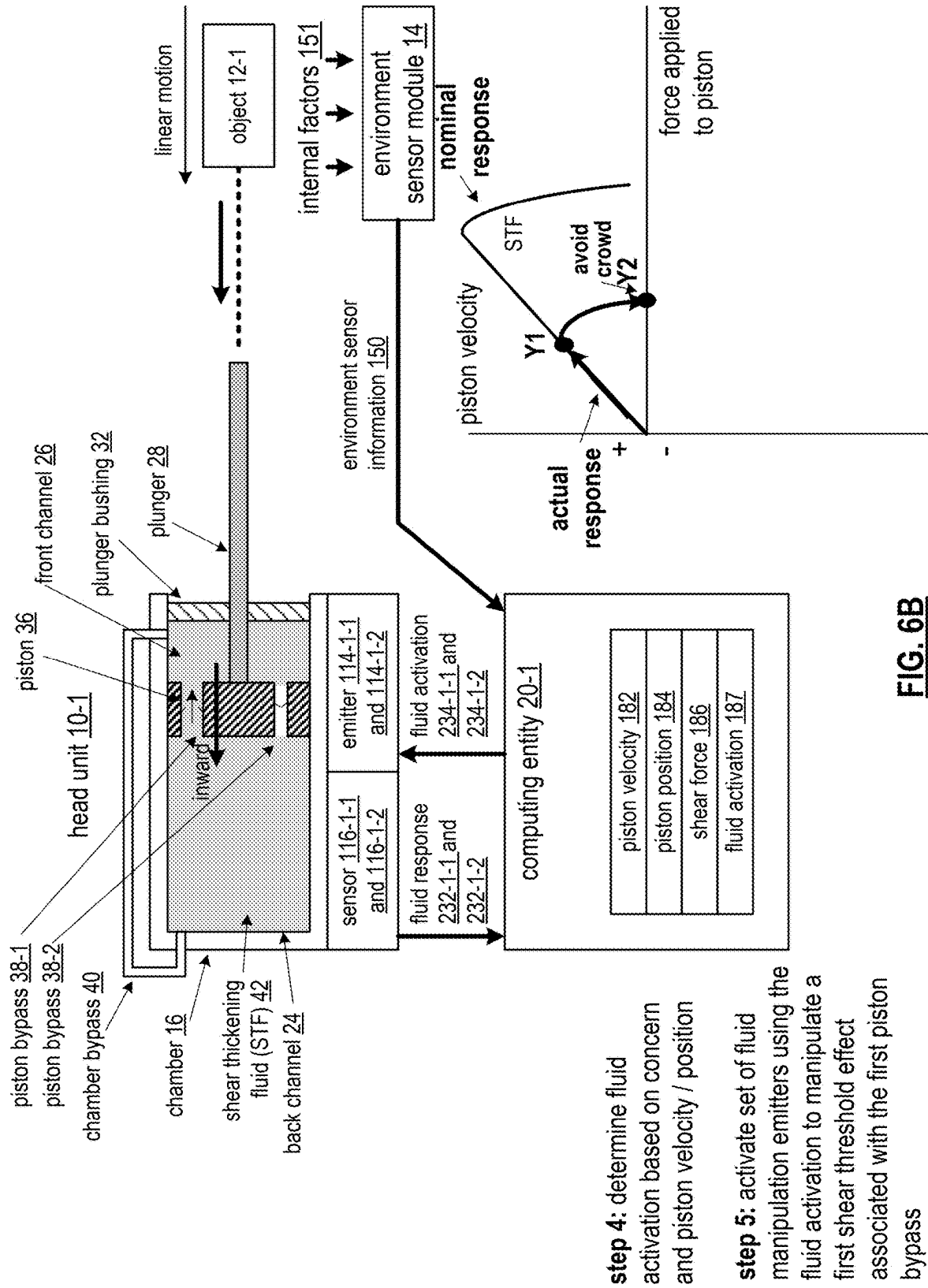


FIG. 4









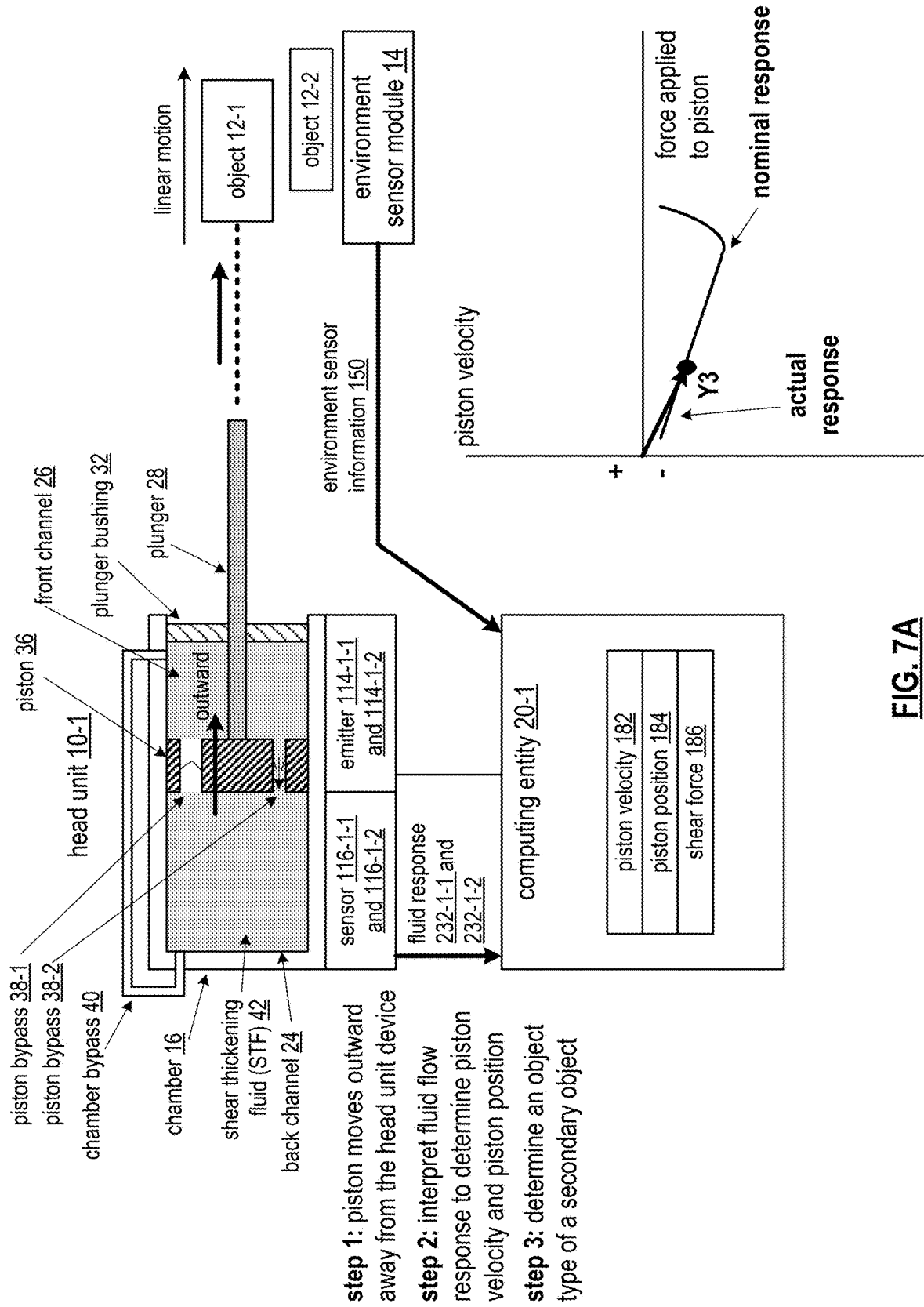
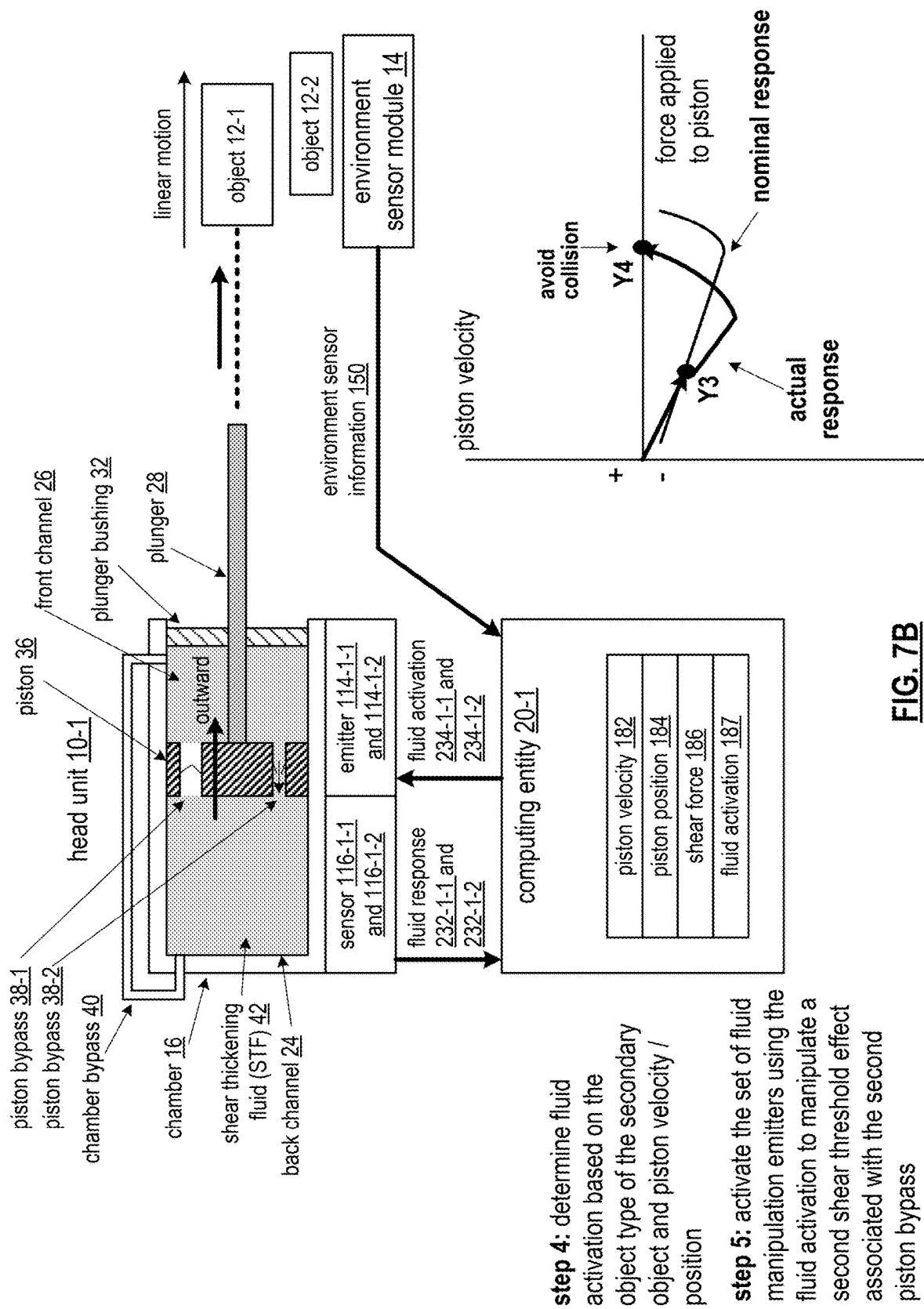
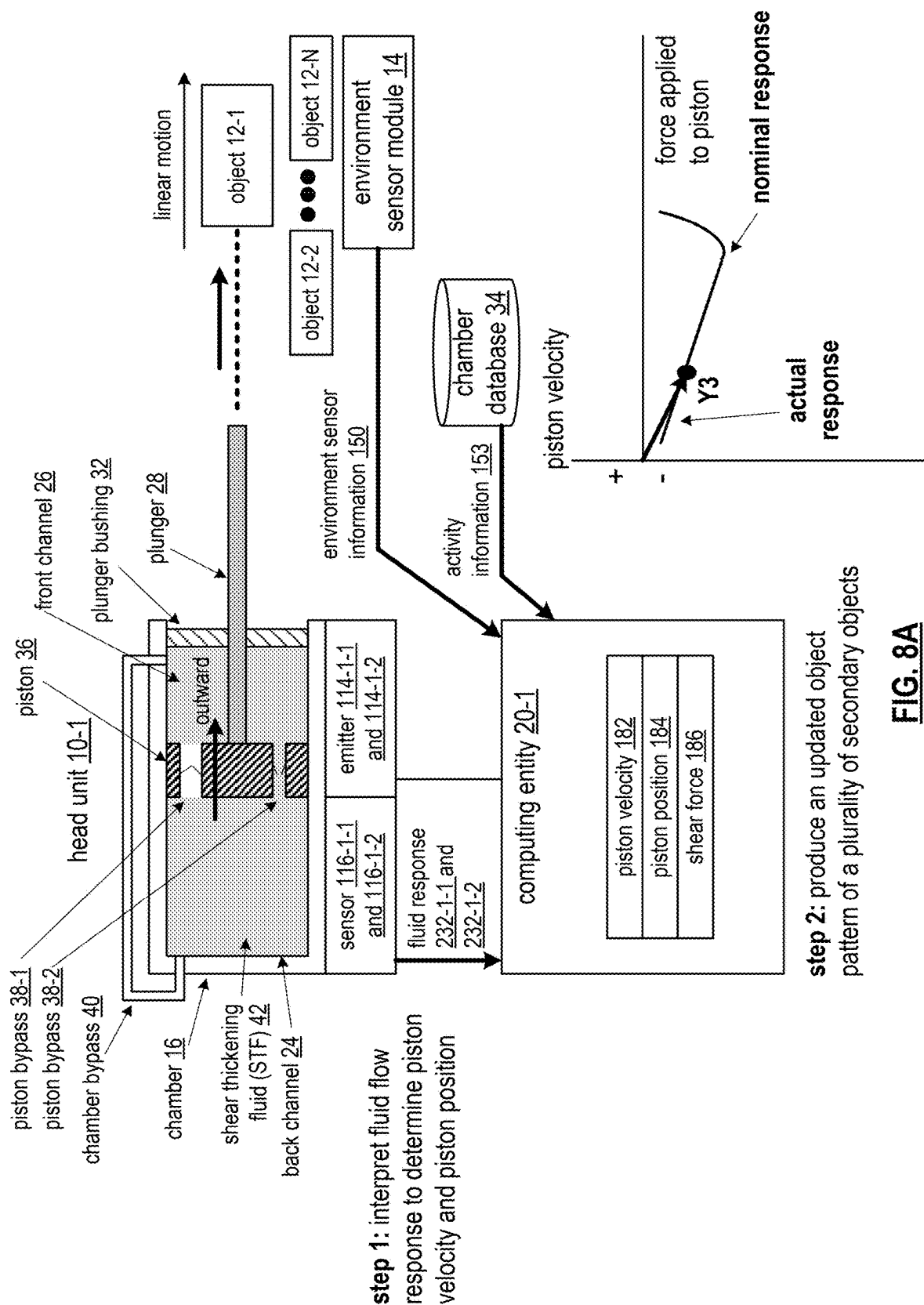
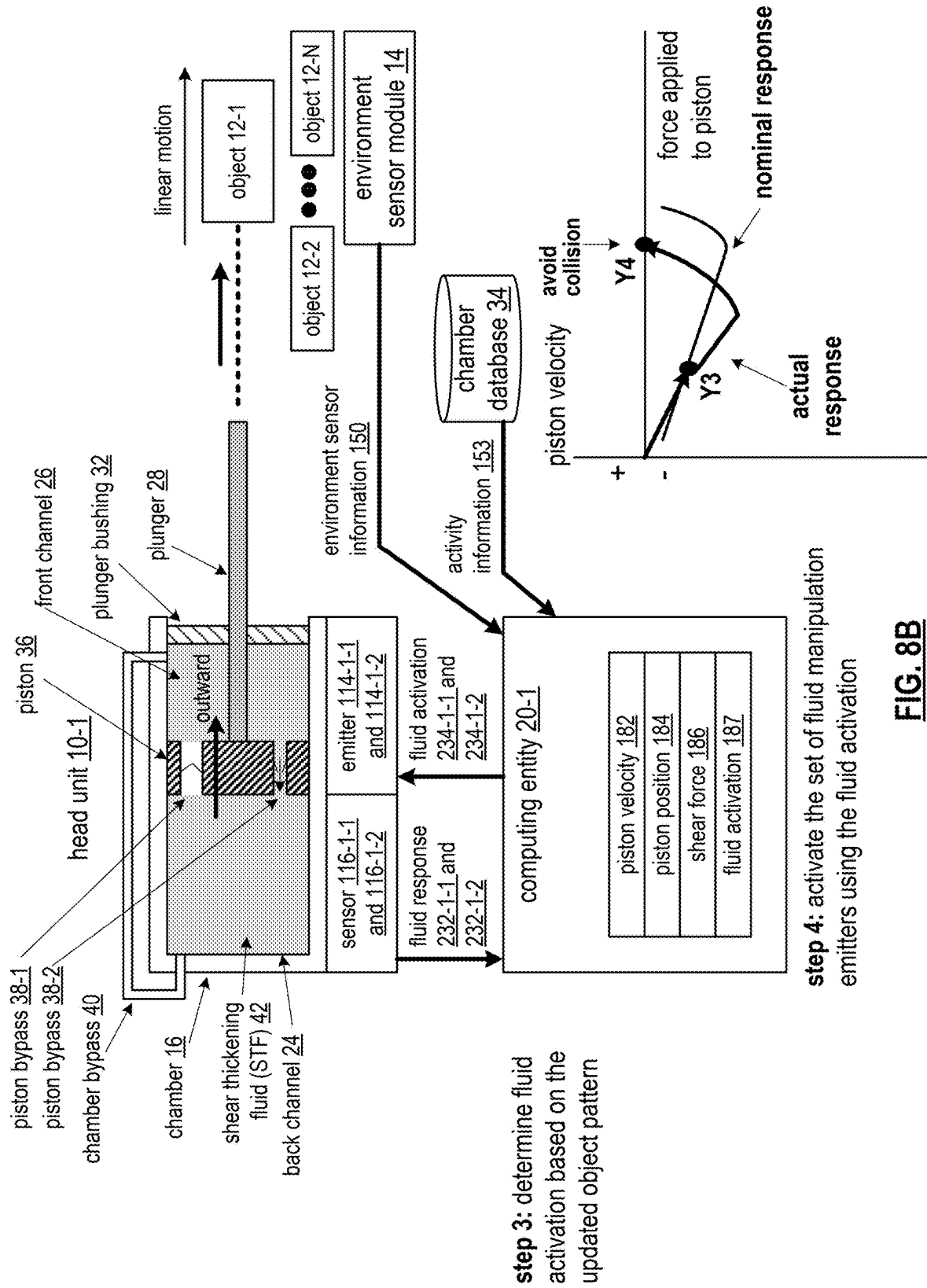
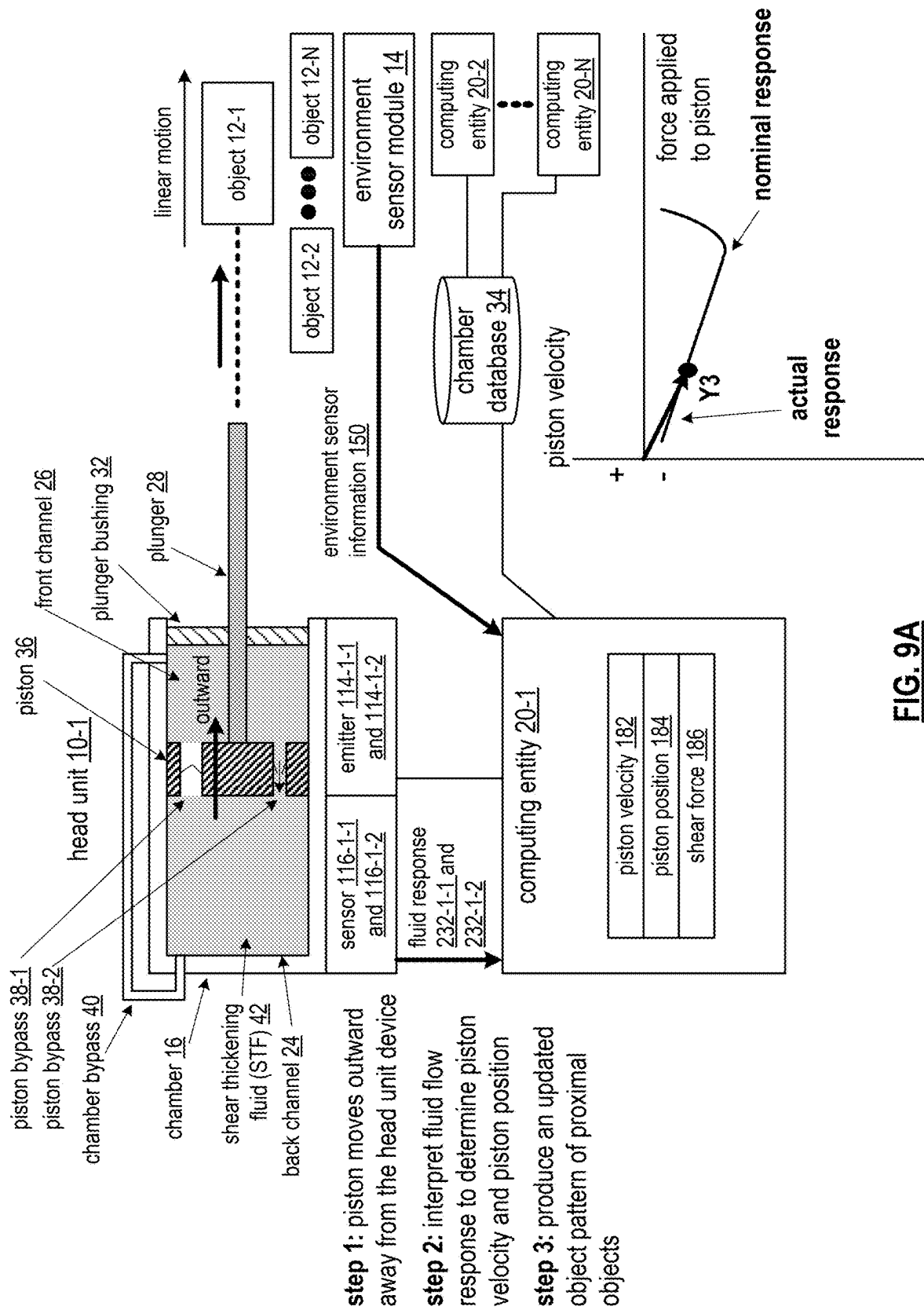


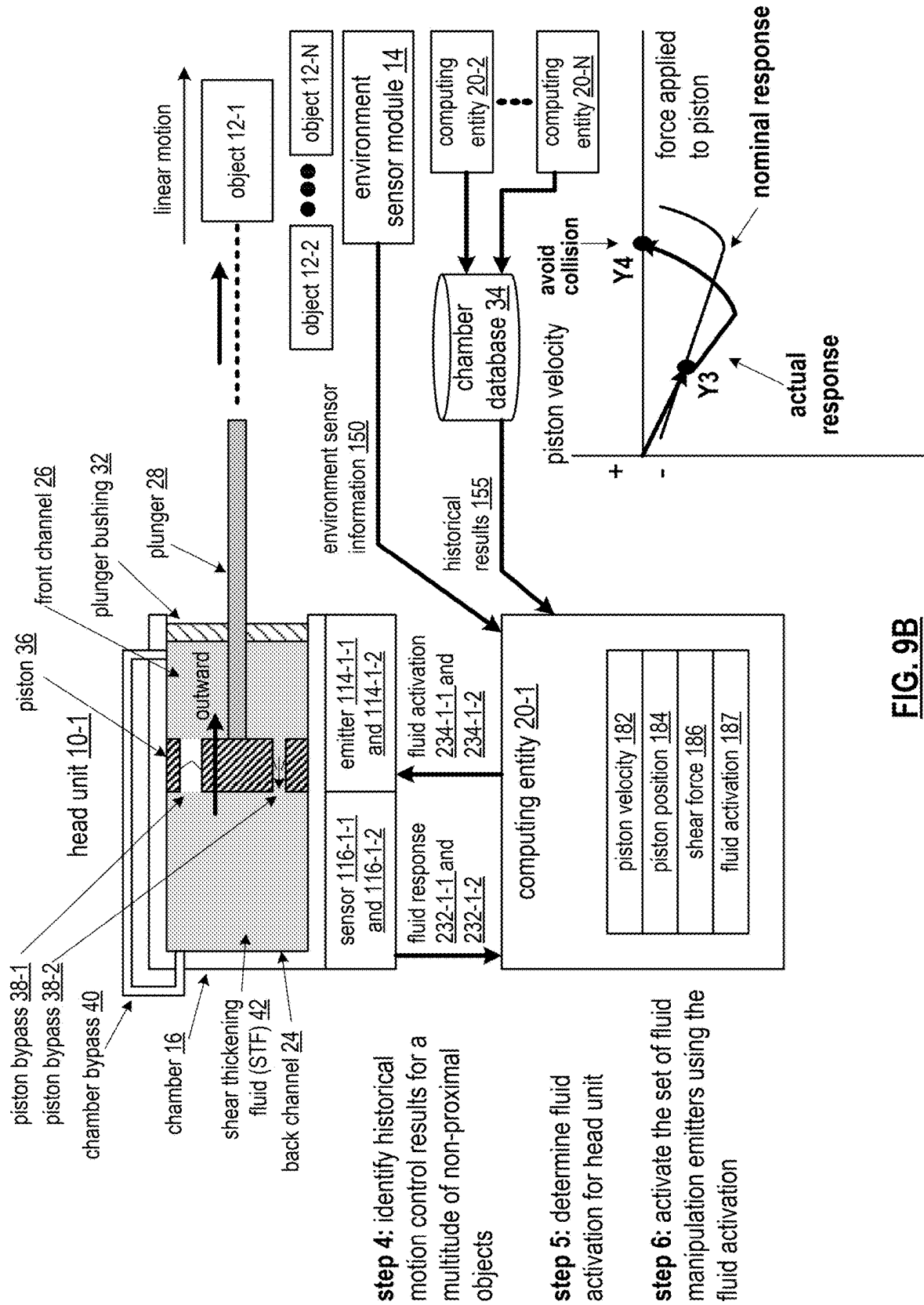
FIG. 7A

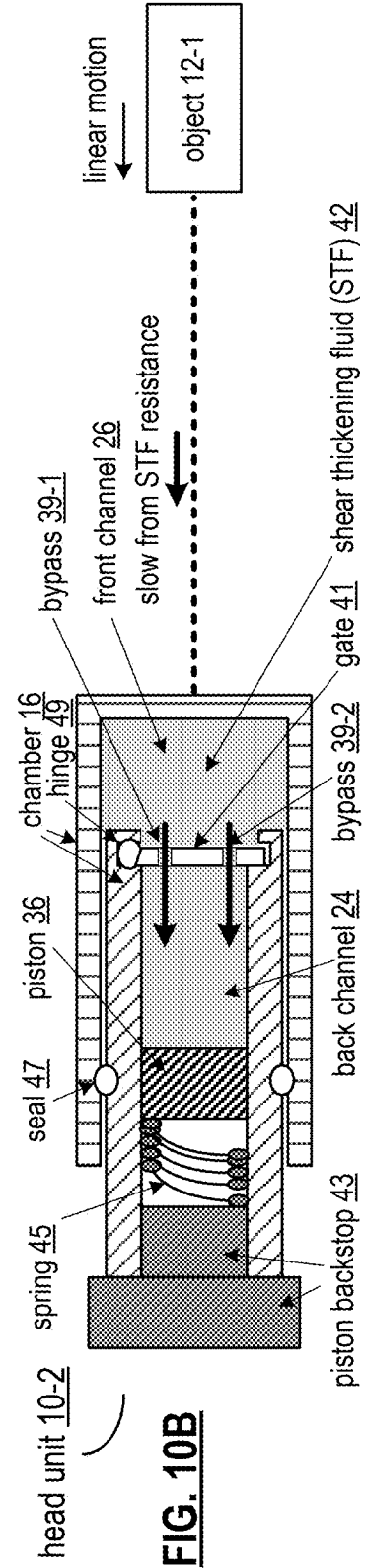
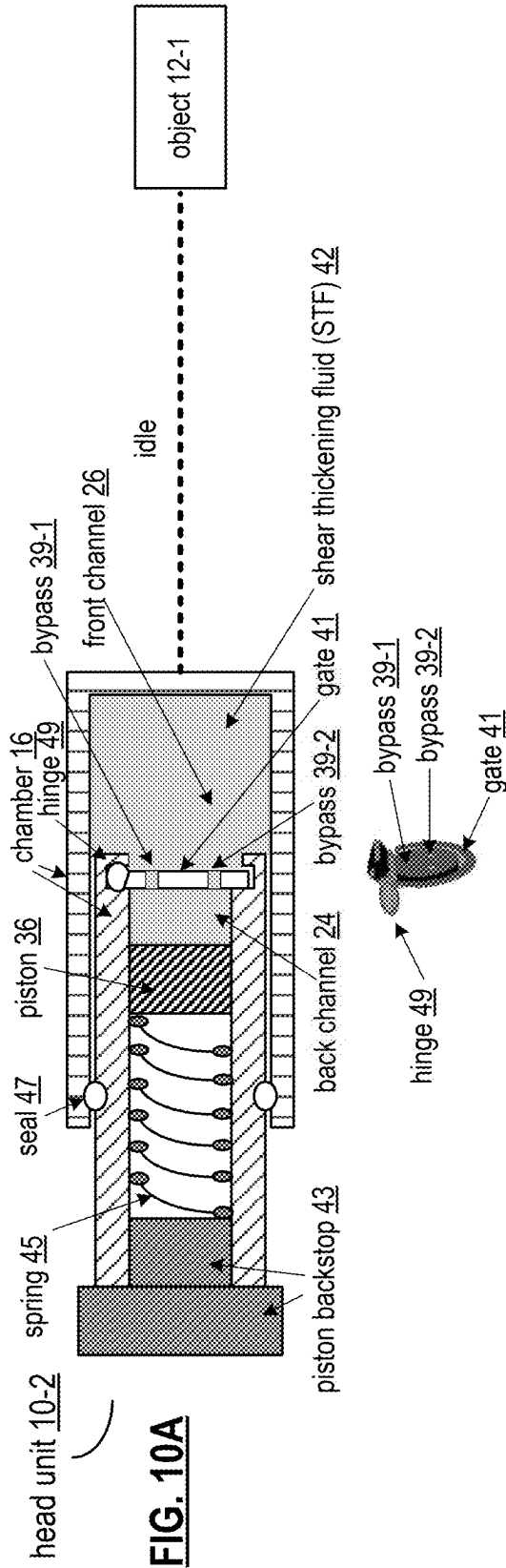


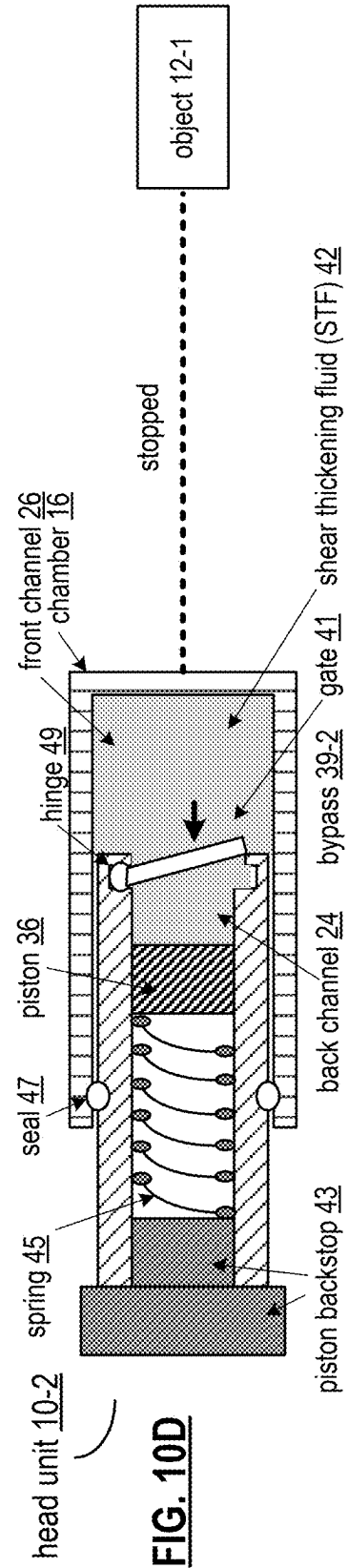
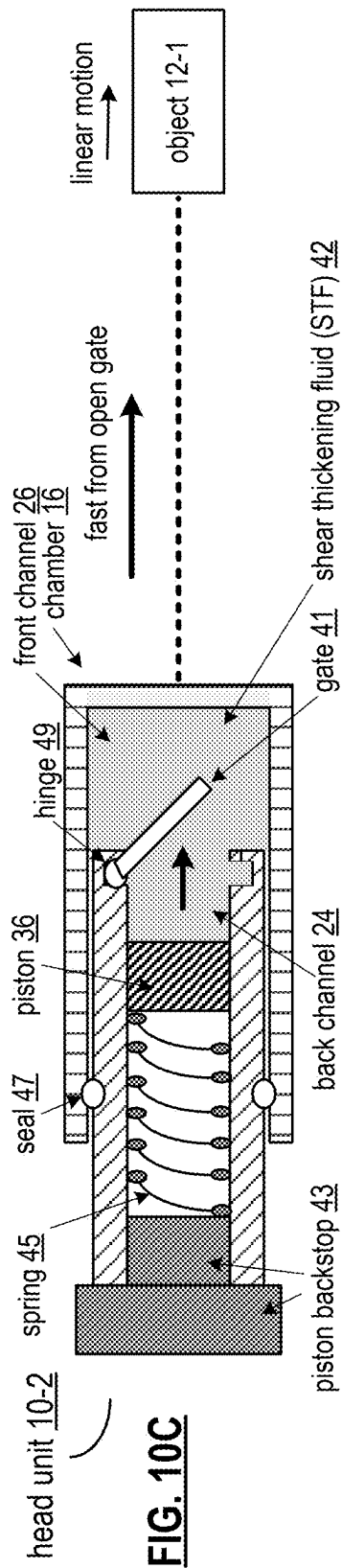


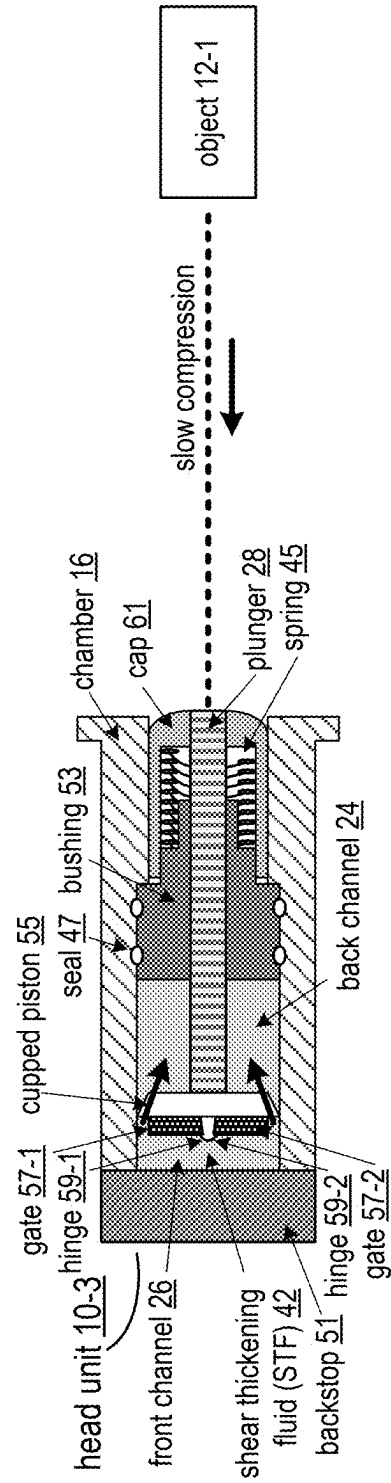
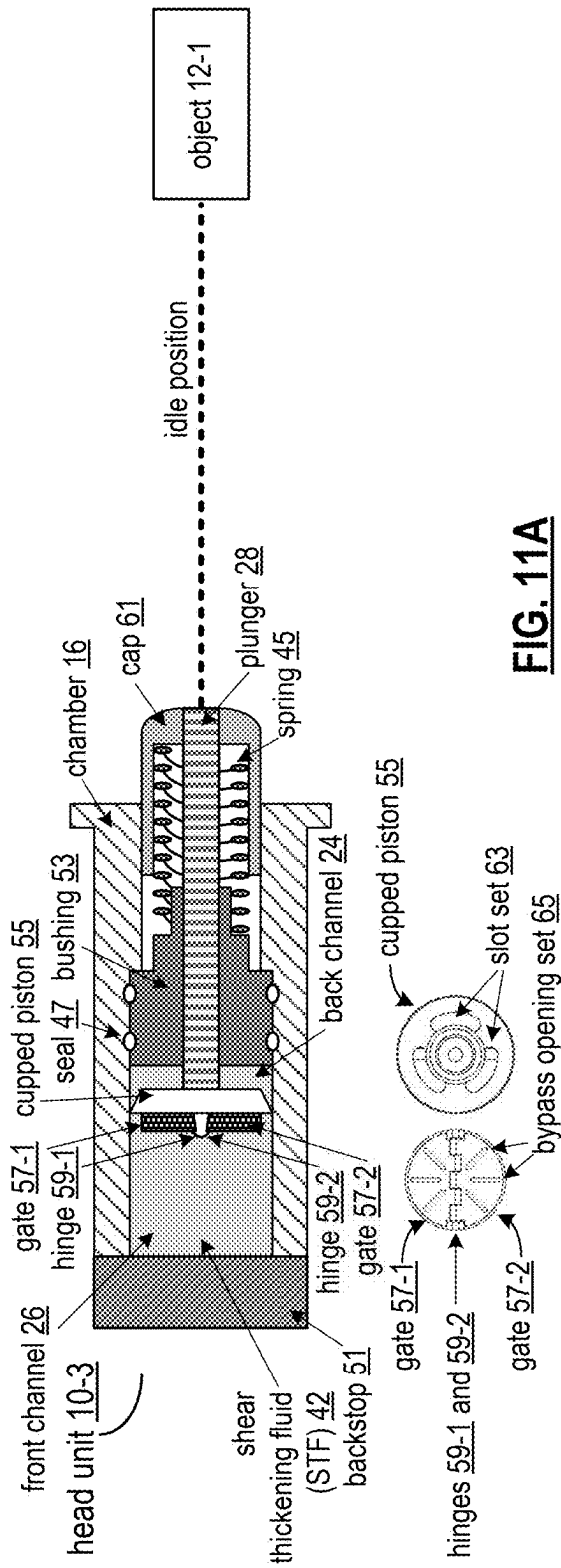












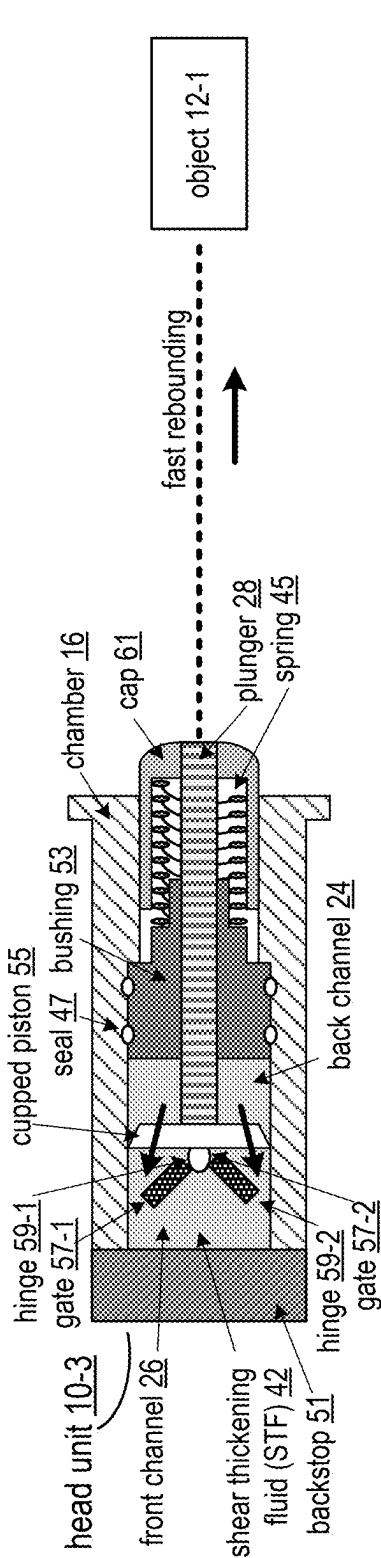


FIG. 11C

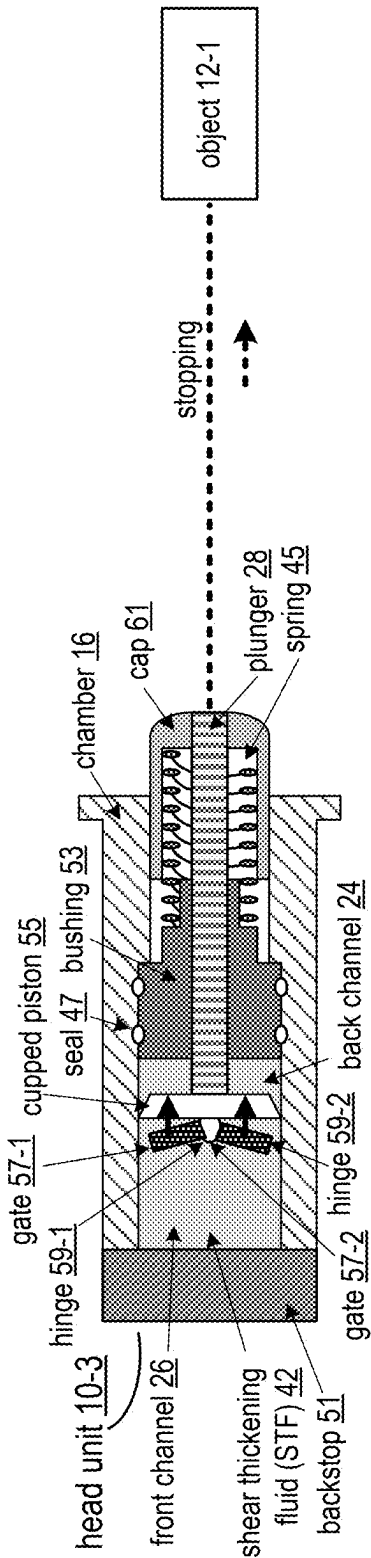


FIG. 11D

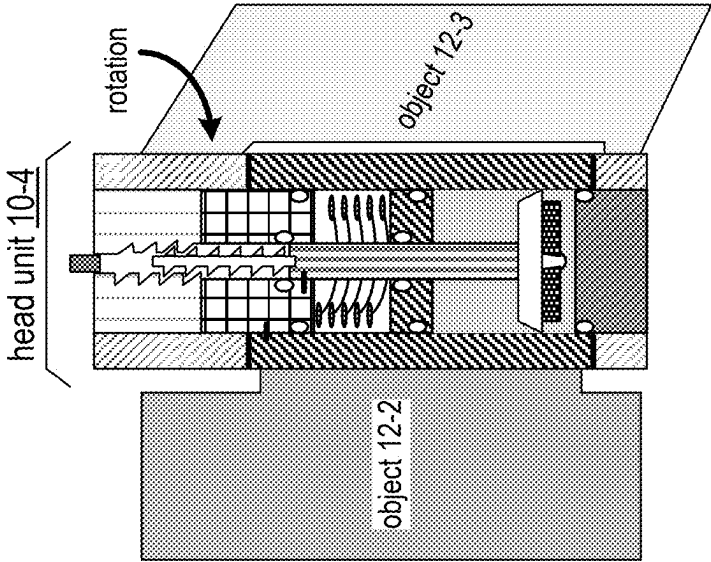


FIG. 12B

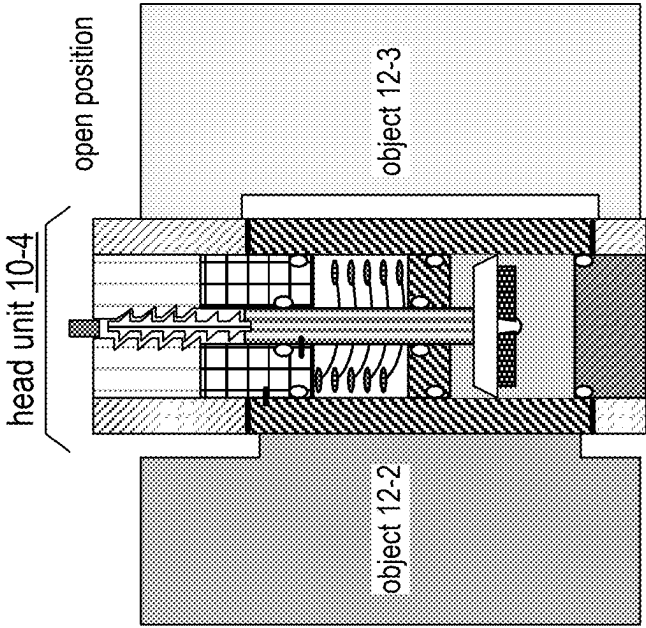


FIG. 12A

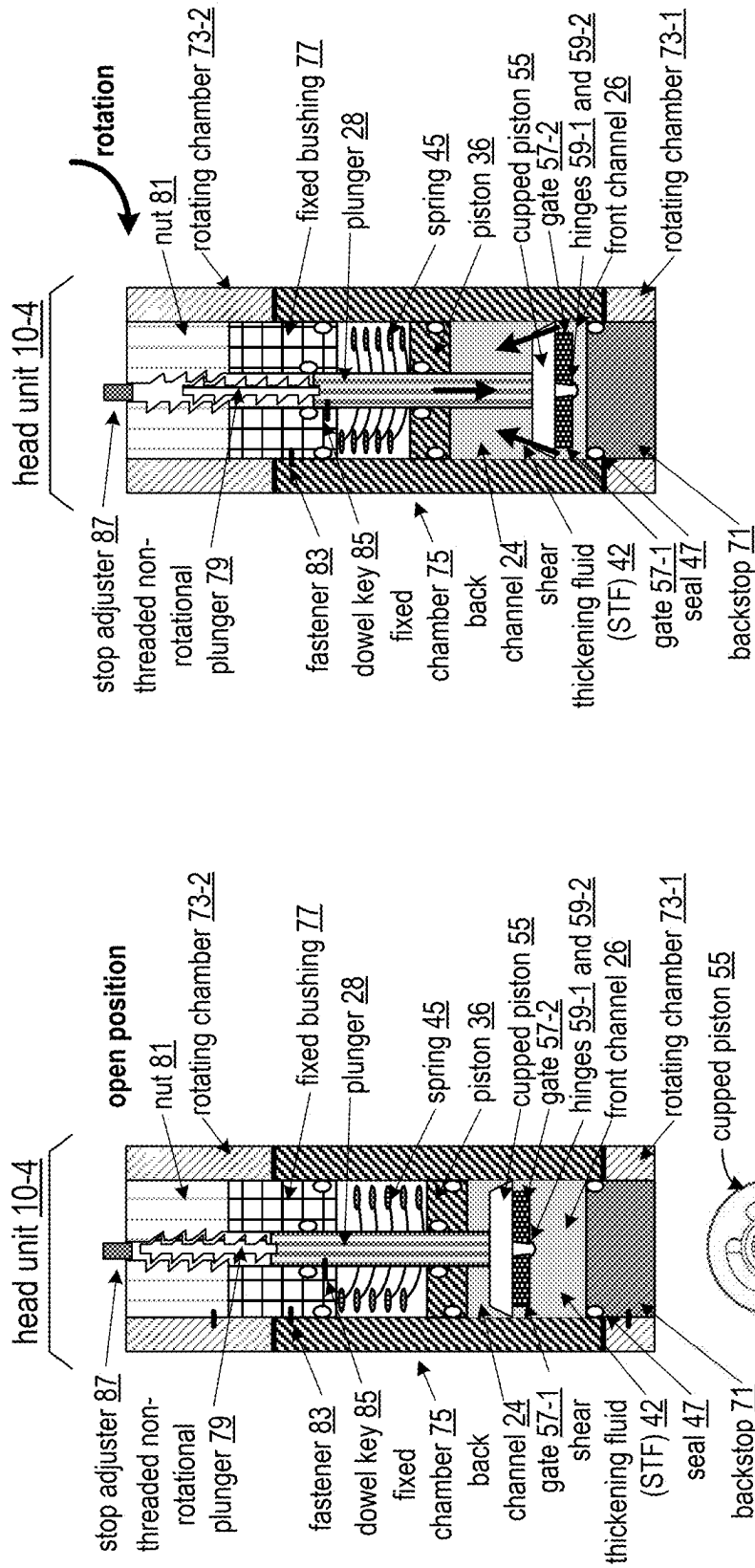


FIG. 12D

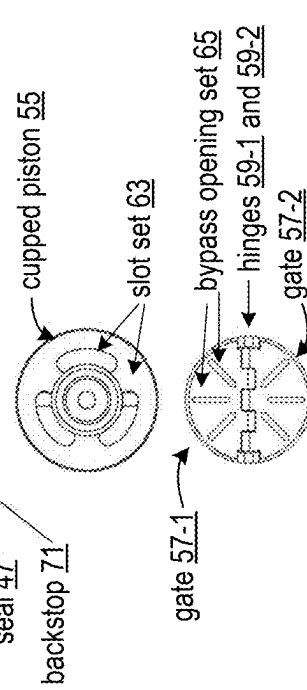


FIG. 12C

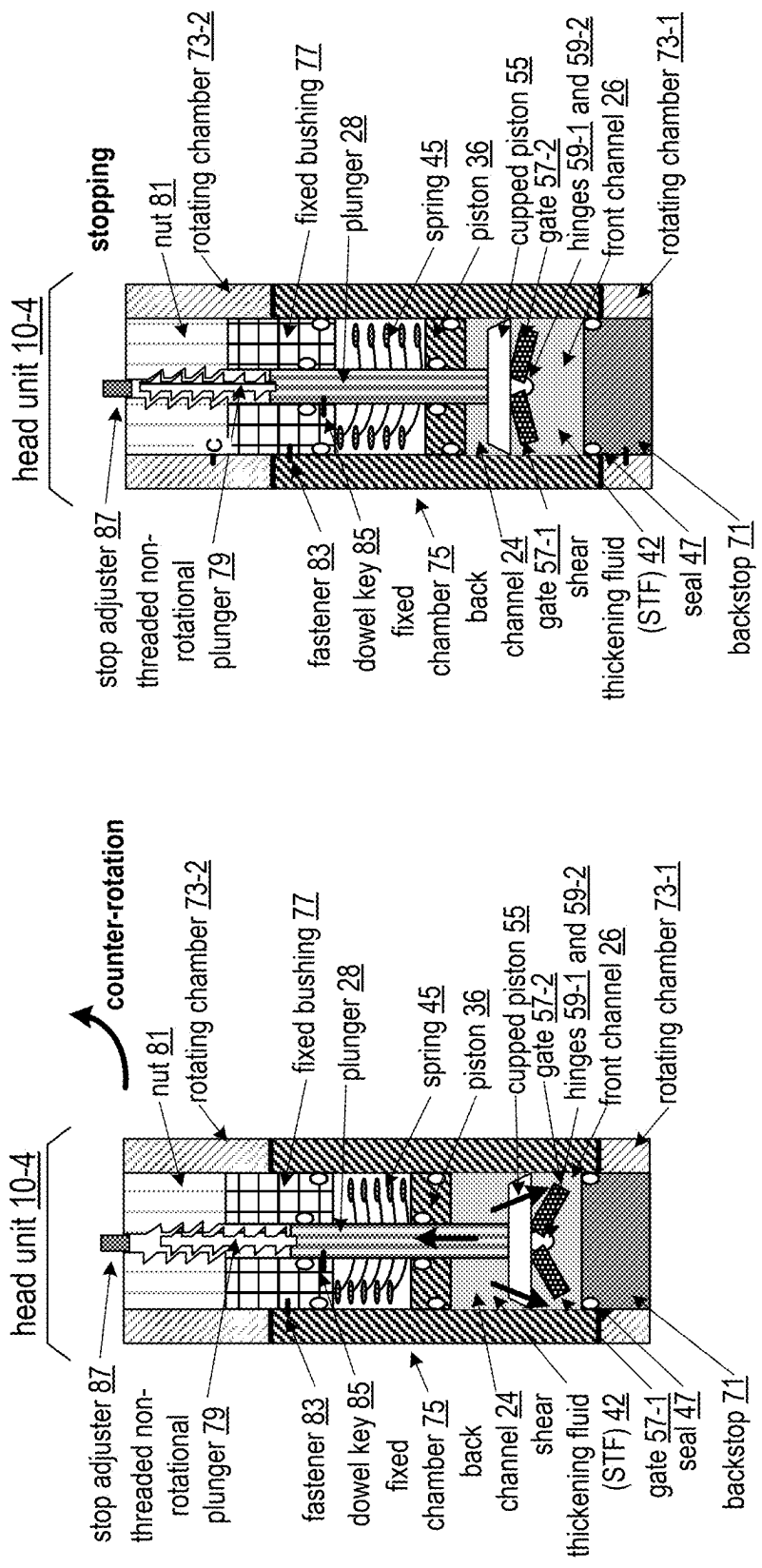
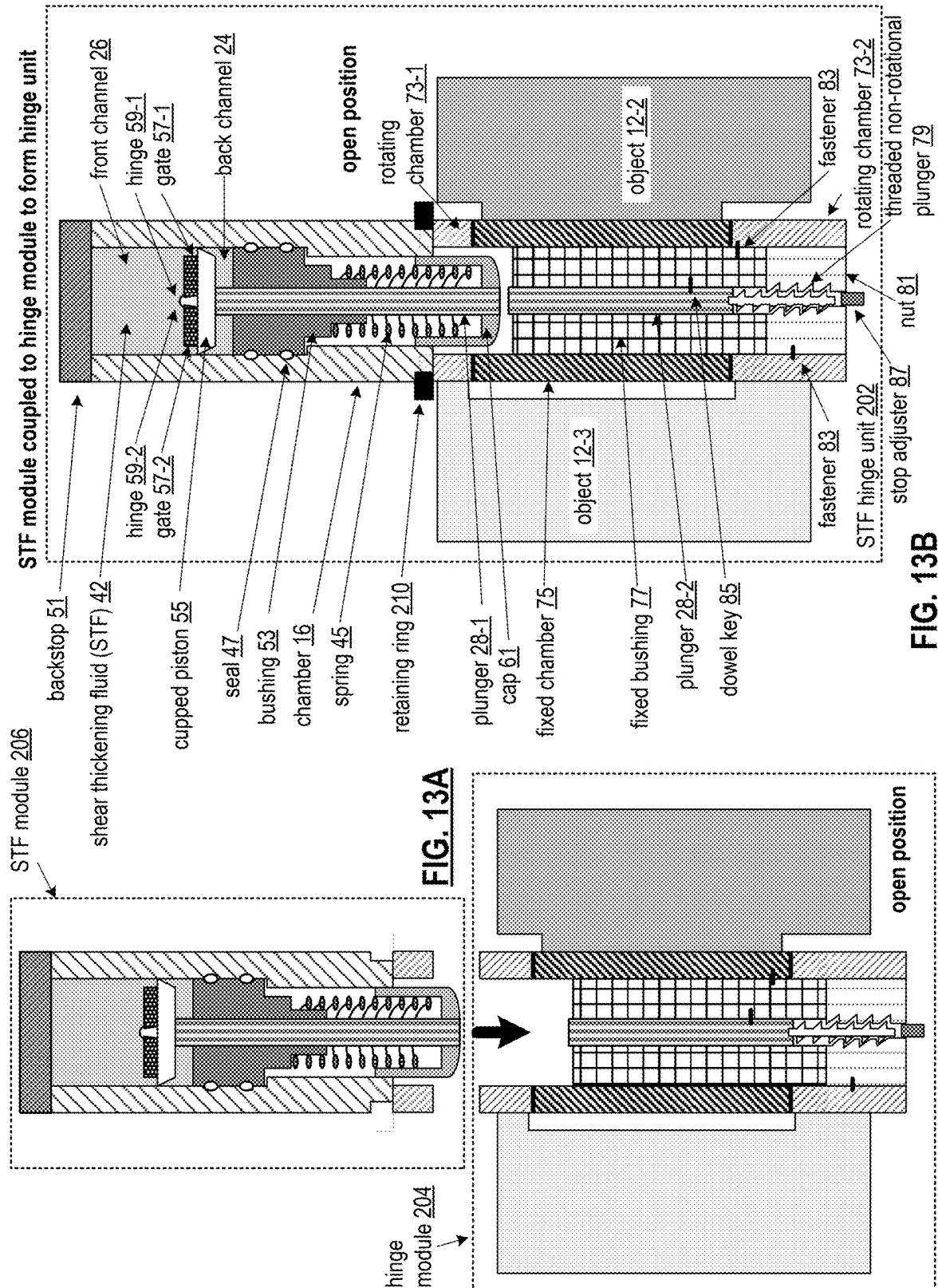
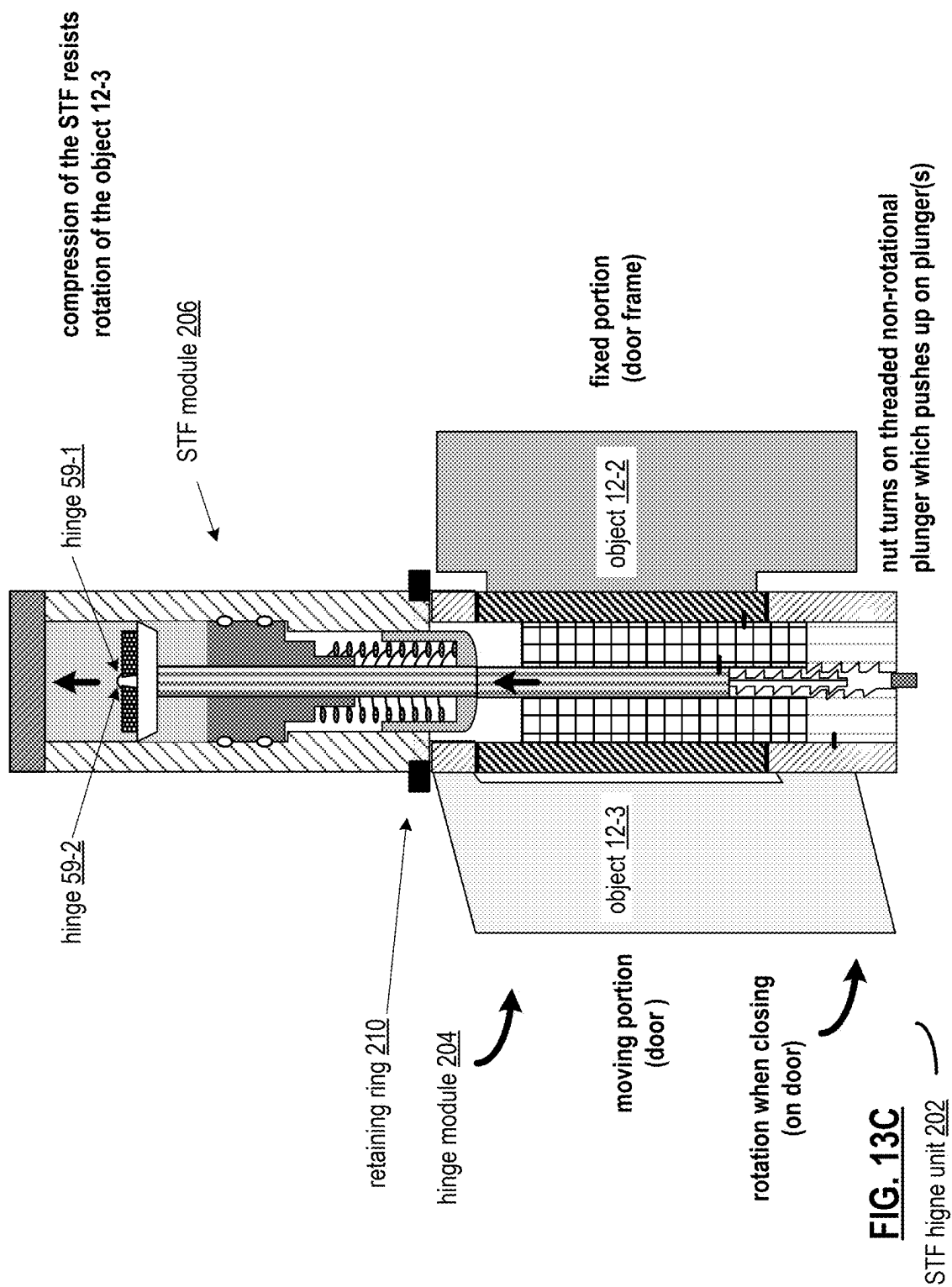
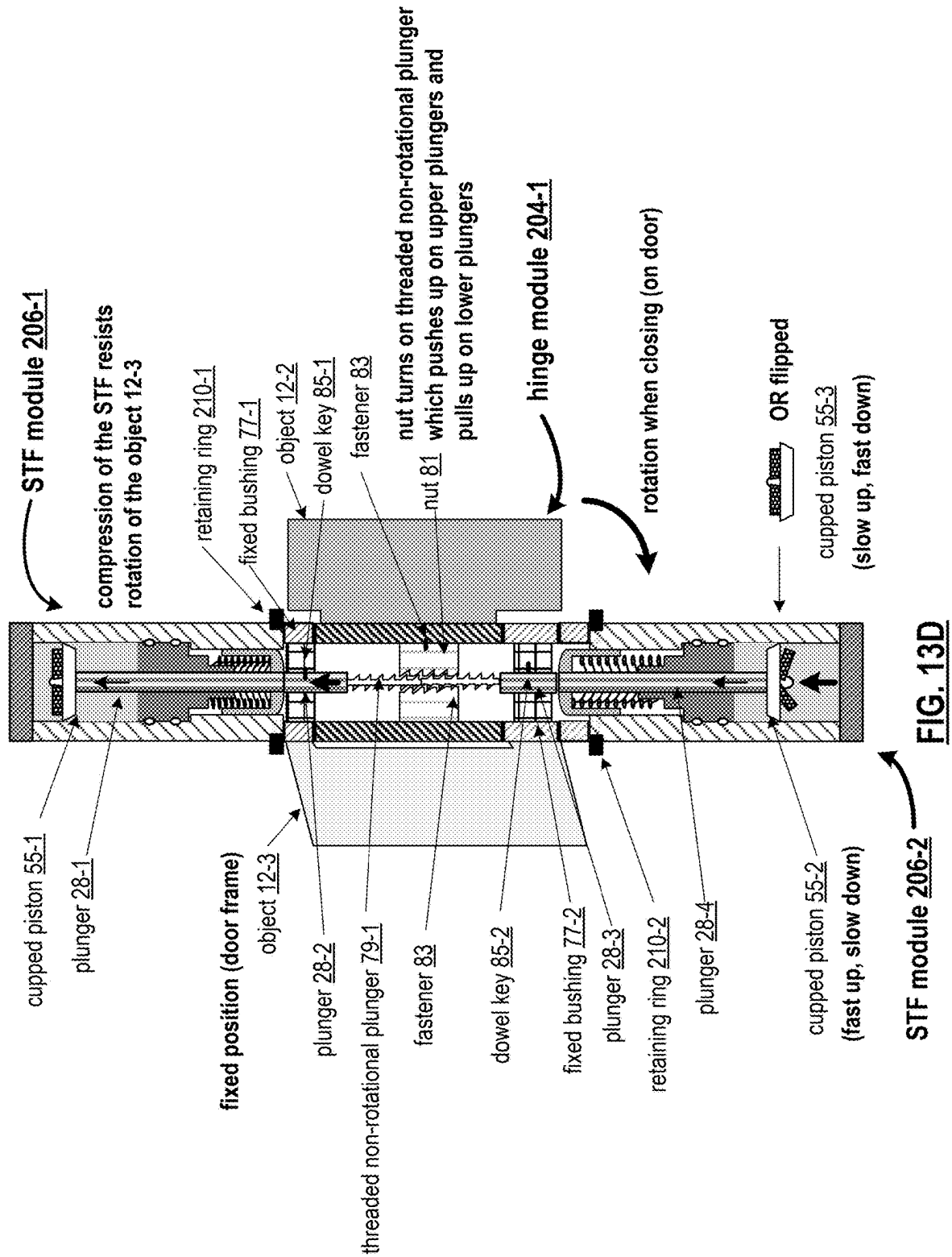


FIG. 12F

FIG. 12E







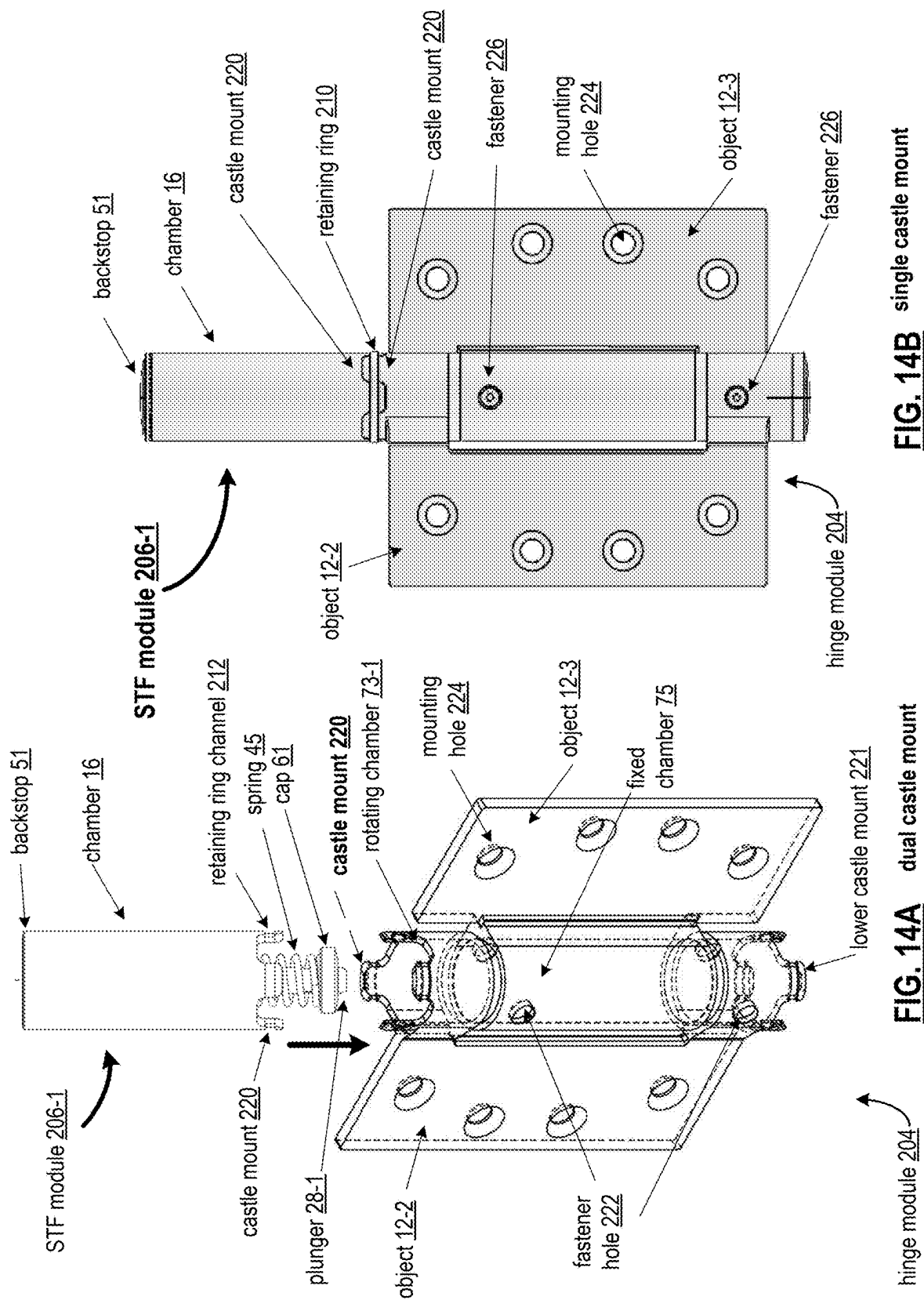


FIG. 14A dual castle mount

FIG. 14B single castle mount

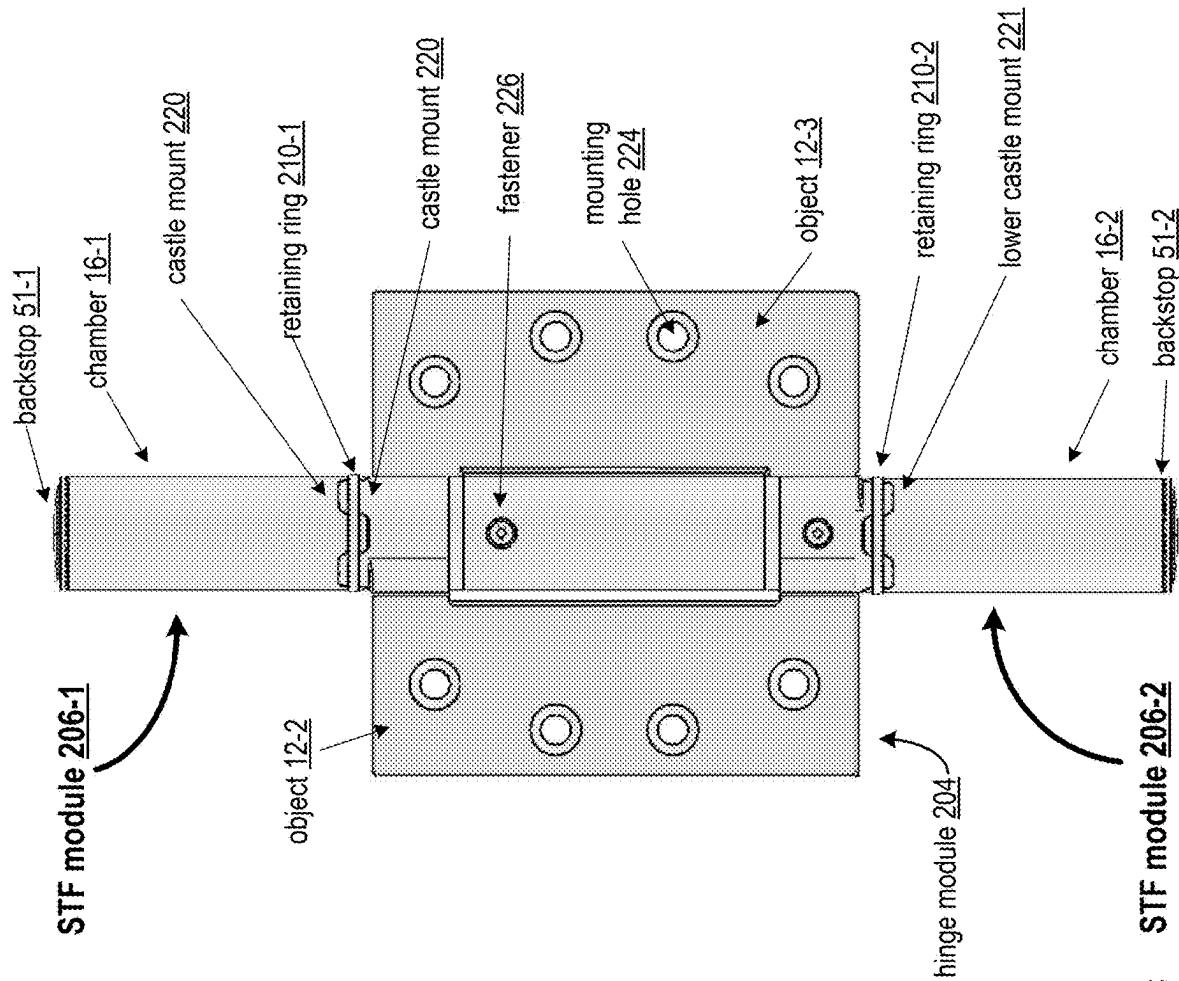


FIG. 14C

double castle mount

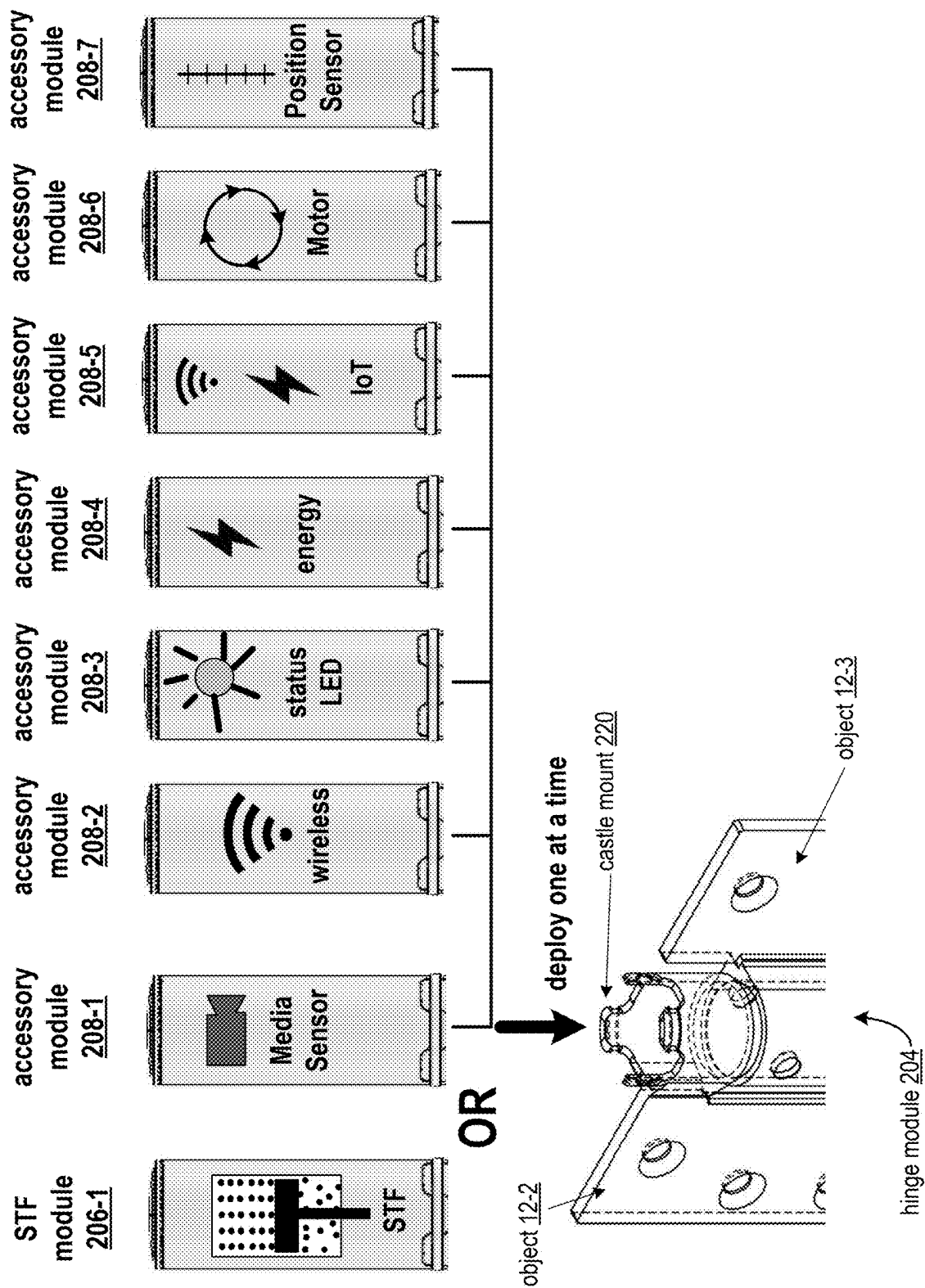


FIG. 15A

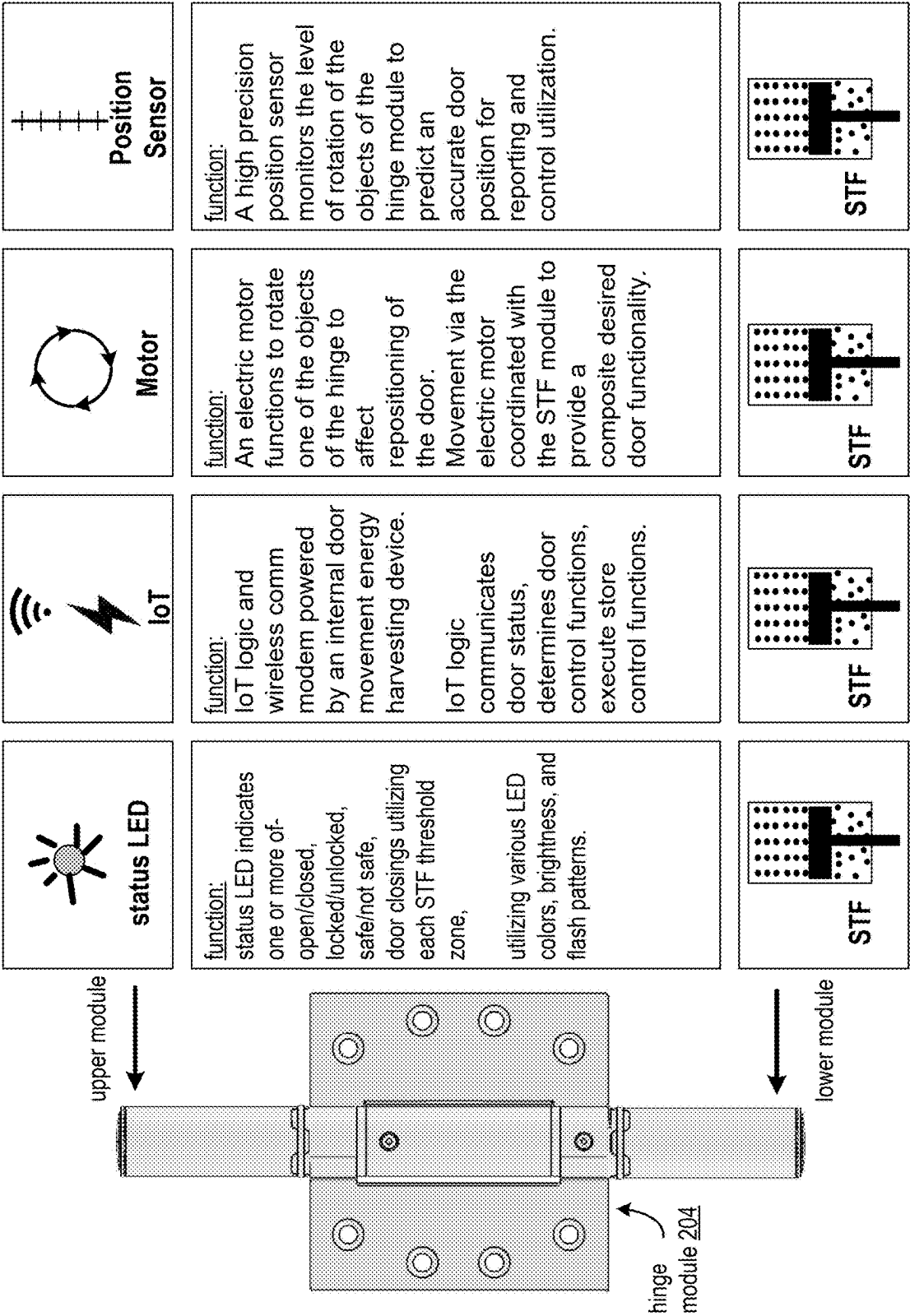


FIG. 15B

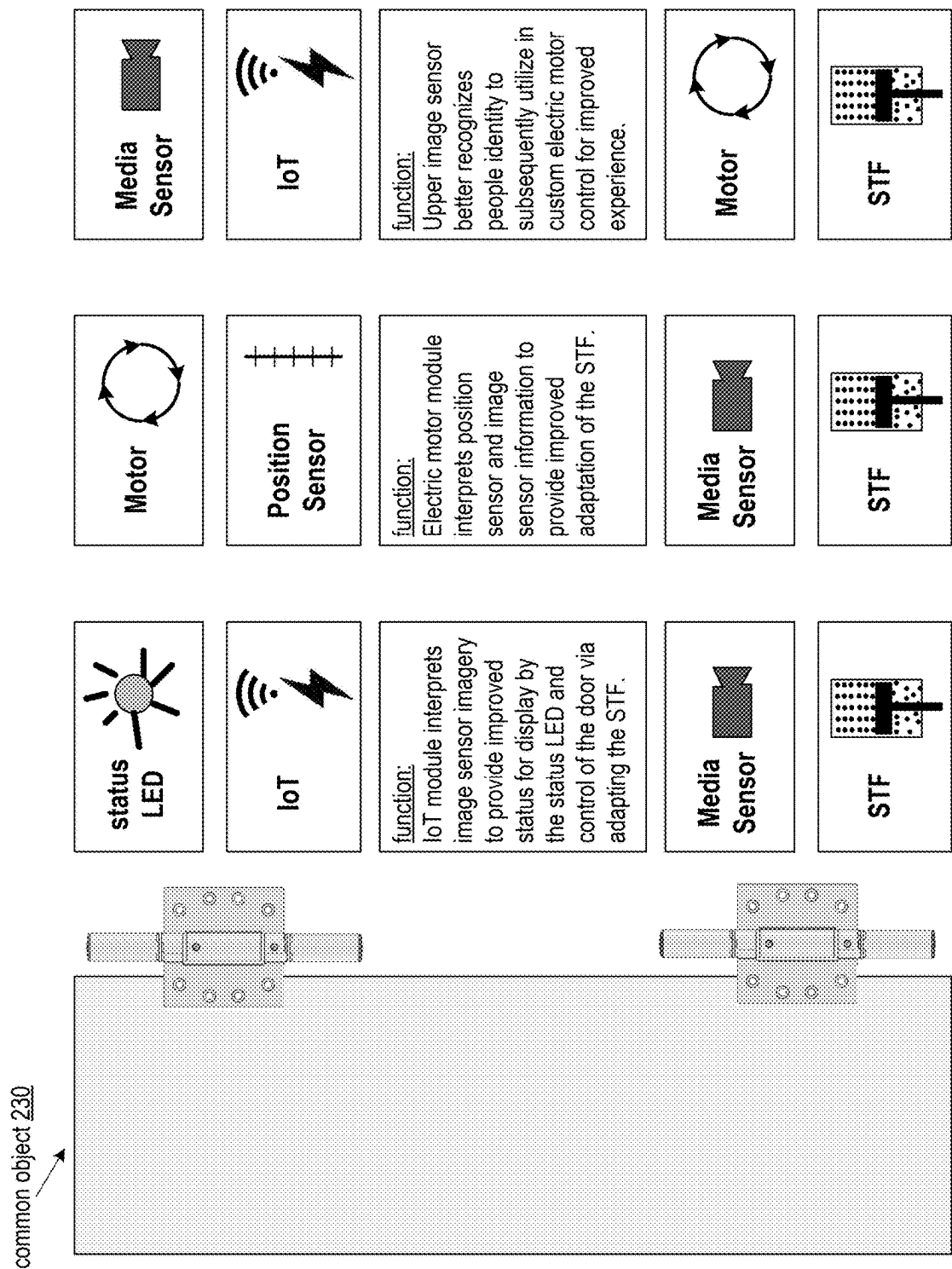
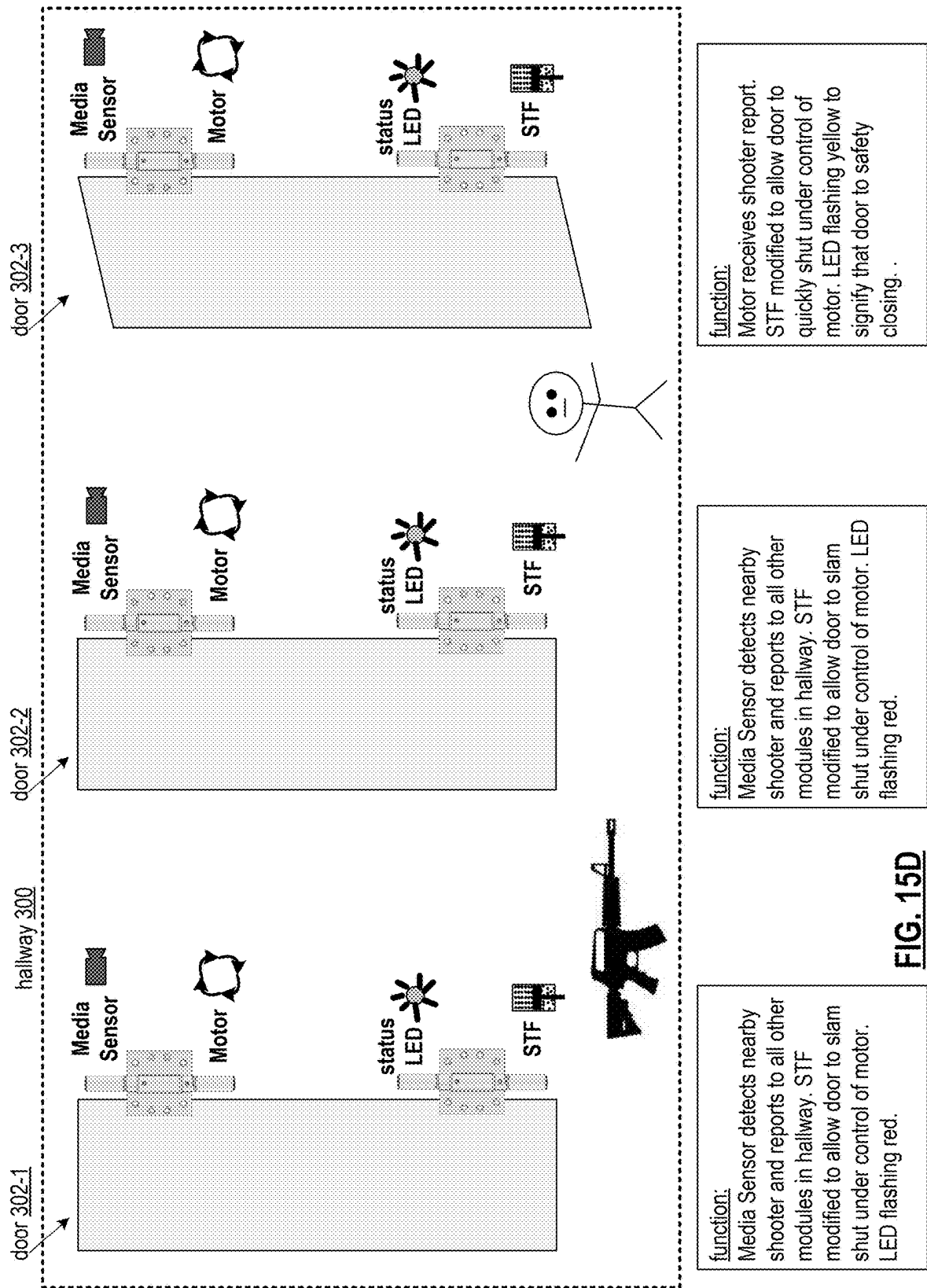


FIG. 15C



MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present U.S. Utility patent application claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 18/646,943, entitled “MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Apr. 26, 2024, issuing Apr. 29, 2025 as U.S. Pat. No. 12,286,060, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 18/586,061, entitled “MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Feb. 23, 2024, issuing Apr. 29, 2025 as U.S. Pat. No. 12,286,829, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 18/515,620, entitled “MODULAR ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Nov. 21, 2023, issuing Apr. 29, 2025 as U.S. Pat. No. 12,287,024, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 18/071,680, entitled “ROTATING SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM,” filed Nov. 30, 2022, issued Nov. 28, 2023 as U.S. Pat. No. 11,828,309, which claims priority pursuant to 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/427,949, entitled “SHEAR THICKENING FLUID BASED OBJECT CONTROL MECHANISM”, filed Nov. 25, 2022, expired, all of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility patent application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC

[0003] Not Applicable.

BACKGROUND OF THE INVENTION

Technical Field of the Invention

[0004] This invention relates generally to systems that measure and control mechanical movement and more particularly to sensing and controlling of a linear and/or rotary movement mechanism that includes a chamber with dilatant fluid (e.g., a shear thickening fluid).

Description of Related Art

[0005] Many mechanical mechanisms are subject to undesired movement that can lead to annoying sounds, property damage and/or loss, and personal injury and even death. Desired and undesired movements of the mechanical mechanisms may involve a wide range of forces. A need exists to control the wide range of forces to solve these problems.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0006] FIG. 1A is a schematic block diagram of an embodiment of a mechanical and computing system in accordance with the present invention;

[0007] FIG. 1B is a graph of viscosity vs. shear rate for an aspect of an embodiment of a mechanical and computing system in accordance with the present invention;

[0008] FIG. 1C is a graph of plunger velocity vs. force applied to the plunger for an aspect of an embodiment of a mechanical and computing system in accordance with the present invention;

[0009] FIG. 2A is a schematic block diagram of an embodiment of a computing entity of a computing system in accordance with the present invention;

[0010] FIG. 2B is a schematic block diagram of an embodiment of a computing device of a computing system in accordance with the present invention;

[0011] FIG. 3 is a schematic block diagram of another embodiment of a computing device of a computing system in accordance with the present invention;

[0012] FIG. 4 is a schematic block diagram of an embodiment of an environment sensor module of a computing system in accordance with the present invention;

[0013] FIGS. 5A-5B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0014] FIGS. 6A-6B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0015] FIGS. 7A-7B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0016] FIGS. 8A-8B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0017] FIGS. 9A-9B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects in accordance with the present invention;

[0018] FIGS. 10A-10D are schematic block diagrams of an embodiment of a mechanical system to control an object in accordance with the present invention;

[0019] FIGS. 11A-11D are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention;

[0020] FIGS. 12A-12F are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention;

[0021] FIGS. 13A-13D are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention;

[0022] FIGS. 14A-14C are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention; and

[0023] FIGS. 15A-15D are schematic block diagrams of another embodiment of a mechanical system to control an object in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0024] FIG. 1A is a schematic block diagram of an embodiment of a mechanical and computing system that includes a set of head units 10-1 through 10-N, objects 12-1 through 12-3, computing entities 20-1 through 20-N associated with the head units 10-1 through 10-N, and a computing entity 22. The objects include any object that has mass and moves. Examples of an object include a door, an aircraft wing, a portion of a building support mechanism, and a particular drivetrain, etc.

[0025] The cross-sectional view of FIG. 1A illustrates a head unit that includes a chamber 16, a piston 36, a plunger 28, a plunger bushing 32, and a chamber bypass 40. The chamber 16 contains a shear thickening fluid (STF) 42. The chamber 16 includes a back channel 24 and a front channel 26, where the piston partitions the back channel 24 and the front channel 26. The piston 36 travels axially within the chamber 16. The chamber 16 may be a cylinder or any other shape that enables movement of the piston 36 and compression of the STF 42. The STF 42 is discussed in greater detail with reference to FIGS. 1B and 1C.

[0026] The plunger bushing 32 guides the plunger 28 into the chamber 16 in response to force from the object 12-1. The plunger bushing 32 facilitates containment of the STF within the chamber 16. The plunger bushing 32 remains in a fixed position relative to the chamber 16 when the force from the object moves the piston 36 within the chamber 16. In an embodiment the plunger bushing 32 includes an O-ring between the plunger bushing 32 and the chamber 16. In another embodiment the plunger bushing 32 includes an O-ring between the plunger bushing 32 and the plunger 28.

[0027] The piston 36 includes a piston bypass 38 between opposite sides of the piston to facilitate flow of a portion of the STF between the opposite sides of the piston (e.g., between the back channel 24 and the front channel 26) when the piston travels through the chamber in an inward or an outward direction.

[0028] Alternatively, or in addition to, the chamber bypass 40 is configured between opposite ends of the chamber 16, wherein the chamber bypass 40 facilitates flow of a portion of the STF between the opposite ends of the chamber when the piston travels through the chamber in the inward or outward direction (e.g., between the back channel 24 and the front channel 26).

[0029] In alternative embodiments, the piston bypass 38 and the chamber bypass 40 includes mechanisms to enable STF flow in one direction and not an opposite direction. In further alternative embodiments, a control valve within the piston bypass 38 and/or the chamber bypass 40 controls the STF flow between the back channel 24 and the front channel 26. Each bypass includes one or more of a one-way check valve and a variable flow valve.

[0030] The plunger 28 is operably coupled to a corresponding object by one of a variety of approaches. A first approach includes a direct connection of the plunger 28 to the object 12-1 such that linear motion in any direction couples from the object 12-1 to the plunger 28. A second approach includes the plunger 28 coupled to a cap 44 which receives a one way force from a strike 48 attached to the object 12-2. A third approach includes a pushcap 46 that receives a force from a rotary-to-linear motion conversion component that is attached to the object 12-3. In an example,

the object 12-3 is connected to a camshaft 110 which turns a cam 109 to strike the pushcap 46.

[0031] In an embodiment, two or more of the head units are coupled by a head unit connector 112. When so connected, actuation of a piston in a first head unit is essentially replicated in a piston of a second head unit. The head unit connector 112 includes a mechanical element between plungers of the two or more head units and/or direct connection of two or more plungers to a common object. For example, plunger 28 of head unit 10-1 and plunger 28 of head unit 10-2 are directly connected to object 12-1 when utilizing a direct connection.

[0032] Further associated with each head unit is a set of emitters and a set of sensors. For example, head unit 10-N includes a set of emitters 114-N-1 through 114-N-M and a set of sensors 116-N-1 through 116-N-M. Emitters includes any type of energy and or field emitting device to affect the STF, either directly or indirectly via other nanoparticles suspended in the STF. Examples of emitter categories include light, audio, electric field, magnetic field, wireless field, etc. Specific examples of fluid manipulation emitters include a variable flow valve associated with a bypass or injector or similar, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter. Specific examples of magnetic field emitters include a Helmholtz coil, a Maxwell coil, a permanent magnet, a solenoid, a superconducting electromagnet, and a radio frequency transmitting coil.

[0033] Sensors include any type of energy and/or field sensing device to output a signal that represents a reaction, motion or position of the STF. Examples of sensor categories include bypass valve position, mechanical position, image, light, audio, electric field, magnetic field, wireless field, etc. Specific examples of fluid flow sensors include a valve opening detector associated with the chamber 16 or any type of bypass (e.g., piston bypass 38, chamber bypass 40, a reservoir injector, or similar), a mechanical position sensor, an image sensor, a light sensor, an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor. Specific examples of magnetic field sensors include a Hall effect sensor, a magnetic coil, a rotating coil magnetometer, an inductive pickup coil, an optical magnetometry sensor, a nuclear magnetic resonance sensor, and a caesium vapor magnetometer.

[0034] The computing entities 20-1 through 20-N are discussed in detail with reference to FIG. 2A. The computing entity 22 includes a control module 30 and a chamber database 34 to facilitate storage of history of operation, desired operations, and other aspects of the system.

[0035] In an example of operation, the head unit 10-1 controls motion of the object 12-1 and includes the chamber 16 filled at least in part with the shear thickening fluid 42, the piston 36 housed at least partially radially within the chamber 16, and the piston 36 is configured to exert pressure against the shear thickening fluid 42 in response to movement of the piston 36 from a force applied to the piston from the object 12-1. The movement of the piston 36 includes one of traveling through the chamber 16 in an inward direction or traveling through the chamber 16 in an outward direction. The STF is configured to have a decreasing viscosity in

response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates.

[0036] The shear thickening fluid 42 (e.g., dilatant non-Newtonian fluid) has nanoparticles of a specific dimension that are mixed in a carrier fluid or solvent. Force applied to the shear thickening fluid 42 results in these nanoparticles stacking up, thus stiffening and acting more like a solid than a flowable liquid when a shear threshold is reached. In particular, viscosity of the shear thickening fluid 42 rises significantly when shear rate is increased to a point of the shear threshold. The relationship between viscosity and shear rates is discussed in greater detail with reference to FIGS. 1A and 1B.

[0037] In another example of operation, the object 12-1 applies an inward motion force on the plunger 28 which moves the piston 36 in words within the chamber 16. As the piston moves inward, shear rate of the shear thickening fluid 42 changes. A sensor 116-1-1 associated with the chamber 16 of the head unit 10-1 outputs chamber I/O 160 to the computing entity 20-1, where the chamber I/O 160 includes a movement data associated with the STF 42 as a result of the piston 36 moving inwards. Having received the chamber I/O 160, the computing entity 20-1 interprets the chamber I/O 160 to reproduce the movement data.

[0038] The computing entity 20-1 outputs the movement data as a system message 162 to the computing entity 22. The control module 30 stores the movement data in the chamber database 34 and interprets the movement data to determine whether to dynamically adjust the viscosity of the shear thickening fluid. Dynamic adjustment of the viscosity results in dynamic control of the movement of the piston 36, the plunger 28, and ultimately the object 12-1. Adjustment of the viscosity affects velocity, acceleration, and position of the piston 36.

[0039] The control module 30 determines whether to adjust the viscosity based on one or more desired controls of the object 12-1. The desired controls include accelerating, decelerating, abruptly stopping, continuing on a current trajectory, continuing at a constant velocity, or any other movement control. For example, the control module 30 determines to abruptly stop the movement of the object 12-1 when the object 12-1 is a door and the door is detected to be closing at a rate above a maximum closing rate threshold level and when the expected shear rate versus viscosity of the shear thickening fluid 42 requires modification (e.g., boost the viscosity now to slow the door from closing too quickly).

[0040] When determining to modify the viscosity, the control module 30 outputs a system message 162 to the computing entity 20-1, where the system message 162 includes instructions to immediately boost the viscosity beyond the expected shear rate versus viscosity of the shear thickening fluid 42. Alternatively, the system message 162 includes specific information on the relationship of viscosity versus shear rate.

[0041] Having received the system message 162, the computing entity 20-1 determines a set of adjustments to make with regards to the shear thickening fluid 42 within the chamber 16. The set of adjustments includes one or more of adjusting STF 42 flow through the chamber bypass 40, adjusting STF 42 flow through the piston bypass 38, and activating an emitter of a set of emitters 114-1-1 through 114-N-1. The flow adjustments include regulating within a flow range, stopping, starting, and allowing in one particular

direction. For example, the computing entity 20-1 determines to activate emitter 114-1-1 to produce a magnetic field such as to interact with magnetic nanoparticles within the STF 42 to raise the viscosity. The computing entity 20-1 issues another chamber I/O 160 to the emitter 114-1-1 to initiate a magnetic influence process to boost the viscosity of the STF 42.

[0042] In an alternative embodiment, the computing entity 22 issues another system message 162 to two or more computing entities (e.g., 20-1 and 20-2) to boost the viscosity for corresponding head units 10-1 and 10-2 when the head unit connector 112 connects head units 10-1 and 10-2 and both head units are controlling the motion of the object 12-1. For instance, one of the head units informs the computing entity 22 that the object 12-1 is moving too quickly inward and the predicted stopping power of the expected viscosity versus shear rate of the STF 42 of the head unit, even when boosted, will not be enough to slow the object 12-1 to a desired velocity or position. When informed that one head unit, even with a modified viscosity, is not enough to control the object 12-1, the control module 30 determines how many other head units (e.g., connected via the head unit connector 112) to apply and to dynamically modify the viscosity.

[0043] In yet another alternative embodiment, the computing entity 22 issues a series of system messages 162 to a set of computing entities associated with a corresponding set of head units to produce a cascading effect of altering of the viscosity of the STF 42 of each of the chambers 16 associated with the set of head units. For example, 3 head units are controlled by 3 corresponding computing entities to adjust viscosity in a time cascaded manner. For instance, head unit 10-1 abruptly changes the viscosity to attempt to slow the object 12-1 followed seconds later by head unit 10-2 abruptly changing the viscosity to attempt to further slow the object 12-1, followed seconds later by head unit 10-1 abruptly changing the viscosity to attempt to further slow the object 12-1.

[0044] In a still further alternative embodiment, the computing entity 22 conditionally issues each message of the series of system messages 162 to the set of computing entities associated with the corresponding set of head units to produce the cascading effect of altering of the viscosity of the STF 42 of each of the chambers 16 associated with the set of head units only when a most recent adaptation of viscosity is not enough to slow the object 12-1 with desired results. For example, the 3 head units are controlled by the 3 corresponding computing entities to adjust viscosity in a conditional time cascaded manner. For instance, head unit 10-1 abruptly changes the viscosity to attempt to slow the object 12-1 followed seconds later by head unit 10-2 abruptly changing the viscosity if head unit 10-1 was unsuccessful to attempt to further slow the object 12-1, followed seconds later by head unit 10-2 abruptly changing the viscosity if head unit 10-2 was unsuccessful to attempt to further slow the object 12-1.

[0045] FIG. 1B is a graph of viscosity vs. shear rate for an aspect of an embodiment of a mechanical and computing system that includes a chamber, a shear thickening fluid, and a piston that moves through the chamber applying forces on the shear thickening fluid. The shear thickening fluid includes a non-Newtonian fluid since the relationship between shear rate and viscosity is nonlinear.

[0046] A relationship between compressive impulse (e.g., shear rate) and the viscosity of the shear thickening fluid is nonlinear and may comprise one or more inflection points as the piston travels within the chamber in response to different magnitudes of forces and different accelerations. The viscosity of the STF may also be a function of other influences, such as electric fields, acoustical waves, magnetic fields, and other similar influences. As a first example of a response of a shear thickening fluid, a first range of shear rates in zone A has a decreasing viscosity as the shear rate increases and then in a second range of shear rates in zone B the viscosity increases abruptly. As a second example of a response of a diluted shear thickening fluid, the first range of shear rates in zone A extends to a higher level of shear rates with the decreasing viscosity and then in the still higher second range of shear rates in zone B the viscosity increases abruptly similar to that of the shear thickening include.

[0047] The shear thickening fluid includes particles within a solvent. Examples of particles of the shear thickening fluid include oxides, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, or a mixture thereof. Further examples of the particles of the shear thickening fluid include SiO₂, polystyrene, or polymethyl-methacrylate.

[0048] The particles are suspended in a solvent. Example components of the solvent include water, a salt, a surfactant, and a polymer. Further example components of the solvent include ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone or a mixture thereof. Example particle diameters range from less than 100 μm to less than 1 millimeter. In an instance, the shear thickening fluid is made of silica particles suspended in polyethylene glycol at a volume fraction of approximately 0.57 with the silica particles having an average particle diameter of approximately 446 nm. As a result, the shear thickening fluid exhibits a shear thickening transition at a shear rate of approximately 102-103 s⁻¹.

[0049] A volume fraction of particles dispersed within the solvent distinguishes the viscosity versus shear rate of different shear thickening fluids. The viscosity of the STF changes in response to the applied shear stress. At rest and under weak applied shear stress, a STF may have a fairly constant or even slightly decreasing viscosity because the random distribution of particles causes the particles to frequently collide. However, as a greater shear stress is applied so that the shear rate increases, the particles flow in a more streamlined manner. However, as an even greater shear stress is applied so that the shear rate increases further, a hydrodynamic coupling between the particles may overcome the interparticle forces responsible for Brownian motion. The particles may be driven closer together, and the microstructure of the colloidal dispersion may change, so that particles cluster together in hydroclusters.

[0050] The viscosity curve of the STF can be fine-tuned through changes in the characteristics of the particles suspended in the solvent. For example, the particles shape, surface chemistry, ionic strength, and size affect the various interparticle forces involved, as does the properties of the solvent. However, in general, hydrodynamic forces dominate at a high shear stress, which also makes the addition of a polymer attached to the particle surface effective in limiting clumping in hydroclusters. Various factors influence this clumping behavior, including, fluid slip, adsorbed ions, surfactants, polymers, surface roughness, graft density (e.g.,

of a grafted polymer), molecular weight, and solvent, so that the onset of shear thickening can be modified. In general, the onset of shear thickening can be slowed by the introduction of techniques to prevent the clumping of particles. For example, influencing the STF with emissions from an emitter in proximal location to the chamber.

[0051] FIG. 1C is a graph of piston velocity vs. force applied to the piston for an aspect of an embodiment of a mechanical and computing system that includes a chamber, a shear thickening fluid, and a piston that moves through the chamber applying forces on the shear thickening fluid. The shear thickening fluid includes a non-Newtonian fluid since the relationship between shear rate and viscosity is nonlinear.

[0052] An example curve for a shear thickening fluid indicates that as more force is applied to the piston in zone A, a higher piston velocity is realized until the corresponding transition to zone B occurs where the shear threshold affect takes hold and the viscosity abruptly increases significantly. When the viscosity increases abruptly, the piston velocity slows back down and may even stop.

[0053] Another example curve for a diluted shear thickening fluid indicates that as more force is applied to the piston in zone A, an even higher piston velocity is realized until the corresponding transition to zone B occurs where the shear threshold affect takes hold and the viscosity abruptly increases significantly. When the viscosity increases abruptly, the piston velocity slows back down and may even stop.

[0054] FIG. 2A is a schematic block diagram of an embodiment of the computing entity (e.g., 20-1 through 20-N; and 22) of the mechanical and computing system of FIG. 1. The computing entity includes one or more computing devices 100-1 through 100-N. A computing device is any electronic device that communicates data, processes data, represents data (e.g., user interface) and/or stores data.

[0055] Computing devices include portable computing devices and fixed computing devices. Examples of portable computing devices include an embedded controller, a smart sensor, a social networking device, a gaming device, a smart phone, a laptop computer, a tablet computer, a video game controller, and/or any other portable device that includes a computing core. Examples of fixed computing devices includes a personal computer, a computer server, a cable set-top box, a fixed display device, an appliance, and industrial controller, a video game counsel, a home entertainment controller, a critical infrastructure controller, and/or any type of home, office or cloud computing equipment that includes a computing core.

[0056] FIG. 2B is a schematic block diagram of an embodiment of a computing device (e.g., 100-1 through 100-N) of the computing entity of FIG. 2A that includes one or more computing cores 52-1 through 52-N, a memory module 102, a human interface module 18, an environment sensor module 14, and an input/output (I/O) module 104. In alternative embodiments, the human interface module 18, the environment sensor module 14, the I/O module 104, and the memory module 102 may be standalone (e.g., external to the computing device). An embodiment of the computing device is discussed in greater detail with reference to FIG. 3.

[0057] FIG. 3 is a schematic block diagram of another embodiment of the computing device 100-1 of the mechanical and computing system of FIG. 1 that includes the human

interface module **18**, the environment sensor module **14**, the computing core **52-1**, the memory module **102**, and the I/O module **104**. The human interface module **18** includes one or more visual output devices **74** (e.g., video graphics display, 3-D viewer, touchscreen, LED, etc.), one or more visual input devices **80** (e.g., a still image camera, a video camera, a 3-D video camera, photocell, etc.), and one or more audio output devices **78** (e.g., speaker(s), headphone jack, a motor, etc.). The human interface module **18** further includes one or more user input devices **76** (e.g., keypad, keyboard, touchscreen, voice to text, a push button, a microphone, a card reader, a door position switch, a biometric input device, etc.) and one or more motion output devices **106** (e.g., servos, motors, lifts, pumps, actuators, anything to get real-world objects to move).

[0058] The computing core **52-1** includes a video graphics module **54**, one or more processing modules **50-1** through **50-N**, a memory controller **56**, one or more main memories **58-1** through **58-N** (e.g., RAM), one or more input/output (I/O) device interface modules **62**, an input/output (I/O) controller **60**, and a peripheral interface **64**. A processing module is as defined at the end of the detailed description.

[0059] The memory module **102** includes a memory interface module **70** and one or more memory devices, including flash memory devices **92**, hard drive (HD) memory **94**, solid state (SS) memory **96**, and cloud memory **98**. The cloud memory **98** includes an on-line storage system and an on-line backup system.

[0060] The I/O module **104** includes a network interface module **72**, a peripheral device interface module **68**, and a universal serial bus (USB) interface module **66**. Each of the I/O device interface module **62**, the peripheral interface **64**, the memory interface module **70**, the network interface module **72**, the peripheral device interface module **68**, and the USB interface modules **66** includes a combination of hardware (e.g., connectors, wiring, etc.) and operational instructions stored on memory (e.g., driver software) that are executed by one or more of the processing modules **50-1** through **50-N** and/or a processing circuit within the particular module.

[0061] The I/O module **104** further includes one or more wireless location modems **84** (e.g., global positioning satellite (GPS), Wi-Fi, angle of arrival, time difference of arrival, signal strength, dedicated wireless location, etc.) and one or more wireless communication modems **86** (e.g., a cellular network transceiver, a wireless data network transceiver, a Wi-Fi transceiver, a Bluetooth transceiver, a 315 MHz transceiver, a zig bee transceiver, a 60 GHz transceiver, etc.). The I/O module **104** further includes a telco interface **108** (e.g., to interface to a public switched telephone network), a wired local area network (LAN) **88** (e.g., optical, electrical), and a wired wide area network (WAN) **90** (e.g., optical, electrical). The I/O module **104** further includes one or more peripheral devices (e.g., peripheral devices **1-P**) and one or more universal serial bus (USB) devices (USB devices **1-U**). In other embodiments, the computing device **100-1** may include more or less devices and modules than shown in this example embodiment.

[0062] FIG. 4 is a schematic block diagram of an embodiment of the environment sensor module **14** of the computing device of FIG. 2B that includes a sensor interface module **120** to output environment sensor information **150** based on information communicated with a set of sensors. The set of sensors includes a visual sensor **122** (e.g., to the camera, 3-D

camera, 360° view camera, a camera array, an optical spectrometer, etc.) and an audio sensor **124** (e.g., a microphone, a microphone array). The set of sensors further includes a motion sensor **126** (e.g., a solid-state Gyro, a vibration detector, a laser motion detector) and a position sensor **128** (e.g., a Hall effect sensor, an image detector, a GPS receiver, a radar system).

[0063] The set of sensors further includes a scanning sensor **130** (e.g., CAT scan, MRI, x-ray, ultrasound, radio scatter, particle detector, laser measure, further radar) and a temperature sensor **132** (e.g., thermometer, thermal coupler). The set of sensors further includes a humidity sensor **134** (resistance based, capacitance based) and an altitude sensor **136** (e.g., pressure based, GPS-based, laser-based).

[0064] The set of sensors further includes a biosensor **138** (e.g., enzyme, microbial) and a chemical sensor **140** (e.g., mass spectrometer, gas, polymer). The set of sensors further includes a magnetic sensor **142** (e.g., Hall effect, piezo electric, coil, magnetic tunnel junction) and any generic sensor **144** (e.g., including a hybrid combination of two or more of the other sensors).

[0065] FIGS. 5A-5B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system includes a head unit system that includes the head unit **10-1** of FIG. 1, the object **12-1** of FIG. 1, the environment sensor module **14** of FIG. 2B, and the computing entity **20-1** of FIG. 1.

[0066] The head unit system further includes an environment sensor (e.g., environment sensor module **14** of FIG. 2B). The environment sensor is associated with an external environment that is external (e.g., outdoors) to an internal environment associated with the object. For example, the internal environment includes facilities inside a building and the external environment includes the environment outside the building.

[0067] The head unit **10-1** includes a shear thickening fluid (STF) **42**. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. 1B. The second range of shear rates are greater than the first range of shear rates.

[0068] The head unit further includes a chamber **16**. The chamber is configured to contain a portion of the STF and includes a front channel **26** and a back channel **24**.

[0069] The head unit further includes a piston **36** housed at least partially radially within the chamber **16** and separating the back channel **24** and the front channel **26**. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object **12-1**. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0070] The piston **36** includes a first piston bypass **38-1** between opposite sides of the piston that controls flow of the STF **42** between the opposite sides of the piston from the back channel to the front channel when the piston is trav-

eling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0071] The piston 36 further includes a second piston bypass 38-2 between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0072] The head unit 10-1 further includes a set of fluid flow sensors 116-1-1 and 116-1-2 positioned proximal to the chamber 16. The set of fluid flow sensors provide the fluid response 232-1-1 and 232-1-2 respectively from the STF 42.

[0073] The head unit 10-1 further includes set of fluid manipulation emitters 114-1-1 and 114-1-2 positioned proximal to the chamber 16. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42 (e.g., shifting the shear rate versus viscosity curve), the first piston bypass 38-1 (e.g., to block or allow flow of the STF), and the second piston bypass 38-2 to control the motion of the object 12-1.

[0074] FIG. 5A illustrates an example of operation of a method for the controlling the operational aspects that includes the piston 36 moving inward towards the head unit 10-1 when the object 12-1 exerts the force on the plunger 28 that transfers the force to the piston 36. As a result, the piston 36 exerts the force on the STF 42 within the back channel 24.

[0075] A first step of the example of operation includes the computing entity 20-1 interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity 182 and a piston position 184 of the piston 36 associated with the head unit device of the head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object 12-1 within the internal environment. For example, the computing entity 20-1 interprets fluid responses 232-1-1 and 232-1-2 from the STF 42 in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0076] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity 20-1 receives fluid responses 232-1-1 and 232-1-2 over the time range, where the fluid responses include the fluid flow signals.

[0077] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity 20-1 interprets the fluid flow signals to produce the fluid flow response.

[0078] A third sub-step includes determining the piston velocity based on the fluid flow response of the set of fluid flow sensors over the time range. For example, the computing entity 20-1 calculates piston velocity based on changes in the fluid flow response over the time range.

[0079] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity 20-1 calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0080] As yet another example of interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 compares the fluid response 232-1-1 and 232-1-2 to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity 182 and piston position 184. As a still further example of the interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 extracts the piston velocity 182 and the piston position 184 directly from the fluid response 232-1-1 and/or 232-1-2 when the sensors 116-1-1 and 116-1-2 generate the piston velocity and piston position directly.

[0081] The first step of the example of operation further includes the computing entity 20-1 determining a shear force 186 based on the piston velocity 182 and the piston position 184. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid flow response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity 20-1 extracts the shear force 186 directly from the fluid responses 232-1-1 and 232-1-2. In an instance, the shear force 186 reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. 5A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y1.

[0082] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42.

[0083] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42 based on an actual valve opening status of the first piston bypass 38-1 (e.g., which allows flow of the STF from the back channel 24 to the front channel 26 when the piston is moving in the inward direction of the example).

[0084] Having determined the piston velocity and the piston position, a second step of the example of operation includes the computing entity 20-1 interpreting an output of the environment sensor to identify an external factor of concern associated with the external environment that is likely to affect the internal environment. The external factor 149 of concern is associated with one or more of a wind gust, wind from a particular direction, rain, snow, humidity, air pressure, temperature, an air pollutant, and/or any other factor of concern. The interpreting the output of the environment sensor to identify the external factor of concern associated with the external environment that is likely to affect the internal environment includes a series of sub-steps.

[0085] A first sub-step includes the computing entity 20-1 obtaining environment sensor information 150 from the environment sensor 14 for a set of timeframes (e.g., every second for 5 minutes). A second sub-step includes the computing entity 20-1 identifying an external factor of external factors 149 from the environment sensor information 150 for the set of timeframes. For example, periodic wind directions and/or periodic external temperatures.

[0086] A third sub-step includes the computing entity 20-1 determining whether an amplitude of the external factor for the set of timeframes is greater than a maximum amplitude threshold level. For example, the computing entity 20-1 interprets environment sensor information 150 to identify external wind direction from the northwest with a wind speed of 75 miles an hour and determining that the wind speed is greater than a maximum wind speed threshold level (e.g., 40 mph).

[0087] A fourth sub-step includes the computing entity 20-1 indicating the external factor of concern associated with the external environment that is likely to affect the internal environment when the amplitude of the external factor for the set of timeframes is greater than the maximum amplitude threshold level. For example, the computing entity 20-1 indicates a low external temperature as external factor of concern when interpreting the environment sensor information 150 to identify an external temperature of 20° below zero Fahrenheit and then determining that the external temperature is less than a minimum external temperature threshold level (e.g., 10° above zero Fahrenheit). The FIG. 5B further illustrates the example of operation, where having detected the external factor of concern, in a third step the computing entity 20-1 determines the fluid activation 187 for the head unit based on the external factor of concern and one or more of the piston velocity, the piston position, and the shear force 186. The determining the fluid activation for the head unit device based on the external factor of concern and one or more of the piston velocity and the piston position includes one or more of a variety of approaches.

[0088] A first approach includes the computing entity 20-1 interpreting a request associated with modifying one or more of object velocity and object position. For example, the computing entity 20-1 receives the request from another computing entity with superior information with regards to the external environment and effects on the internal environment.

[0089] A second approach includes the computing entity 20-1 interpreting guidance from a chamber database based on the external factor of concern. For example, the computing entity 20-1 accesses the chamber database 34 FIG. 1A to recover the guidance for high winds and low temperatures with regards to controlling internal environment objects (e.g., doors).

[0090] A third approach includes the computing entity 20-1 establishing the fluid activation to include facilitating the second range of shear rates to slow down the object when detecting that the external factor of concern includes an external factor that indicates an abatement of the effect on the internal environment from the external factor of concern includes slowing down the object. For example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of an external door more slowly when the external factor indicates an abundance of desirable fresh air.

[0091] A fourth approach includes the computing entity 20-1 establishing the fluid activation to include facilitating the first range of shear rates to speed up the object when detecting that the external factor of concern includes another external factor that indicates the abatement of the effect on the internal environment from the external factor of concern includes speeding up the object. For example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of an external door more quickly, yet safely, when the external factor is the wind speed and direction and the

object 12-1 includes the external door. As another example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of the external door more quickly, yet safely, when external factor is low ambient temperature.

[0092] The third step further includes the computing entity 20-1 interpreting the output of the environment sensor to update the external factor of concern associated with the external environment to produce updated external factor of concern associated with the external environment. For example, the computing entity 20-1 reestablishes the external factor of concern 30 minutes later. Having updated the external factor of concern, the computing entity 20-1 updates the fluid activation for the head unit device based on the updated external factor of concern associated with the external environment and one or more of the piston velocity and the piston position. For example, the computing entity 20-1 reestablishes the fluid activation to adapt to the most recent external factor of concern.

[0093] Having determined the fluid activation, a fourth step of the example method of operation includes the computing entity 20-1 activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the object to abate the effect on the internal environment from the external factor of concern. The control of the motion includes one or more of direct manipulation of the STF, facilitation of the first shear threshold effect associated with the first piston bypass, and facilitation of the second shear threshold effect associated with the second piston bypass.

[0094] The activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the object to abate the effect on the internal environment from the external factor of concern includes a variety of approaches. When the piston is traveling through the chamber in the inward direction and when the STF is to have the decreasing viscosity, a first sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-1 to the set of fluid manipulation emitters 114-1-1 to cause one of the first piston bypass 38-1 to facilitate the first shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF 42 to facilitate the first range of shear rates (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0095] When the piston is traveling through the chamber in the inward direction and when the STF is to have the increasing viscosity, a second sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-2 to the set of fluid manipulation emitters 114-1-2 to cause one of the first piston bypass 38-1 to facilitate the first shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF 42 to facilitate the second range of shear rates. For example, when the object 12-1 includes the external door closing (e.g., piston moving in the inward direction) in a high wind situation, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-1 to facilitate closing down the one-way check valve to prevent STF from moving from the back channel 24 to the front channel 26 thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y1 to the point Y2 (e.g., at a dead stop when closed) as illustrated in FIG. 5B.

[0096] When the piston is traveling through the chamber in the outward direction and when the STF is to have the decreasing viscosity a third sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-1 to the set of fluid manipulation emitters 114-1-1 to cause one of the second piston bypass 38-2 to facilitate the second shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF 42 to facilitate the first range of shear rates.

[0097] When the piston is traveling through the chamber in the outward direction and when the STF is to have the increasing viscosity a fourth sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-2 to the set of fluid manipulation emitters 114-1-2 to cause one of the second piston bypass 38-2 to facilitate the second shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF 42 to facilitate the second range of shear rates.

[0098] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. 1 can alternatively be performed by other modules of the system of FIG. 1 or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system 10, cause one or more computing devices of the mechanical and computing system of FIG. 1 to perform any or all of the method steps described above.

[0099] FIGS. 6A-6B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system includes the head unit 10-1 of FIG. 1, the object 12-1 of FIG. 1, and the computing entity 20-1 of FIG. 1.

[0100] The head unit system includes an environment sensor (e.g., environment sensor module 14 of FIG. 2B). The environment sensor is associated with an internal environment that is internal to an external environment associated with the object. For example, the internal environment includes facilities inside a building and the external environment includes the environment outside the building.

[0101] The head unit 10-1 includes a shear thickening fluid (STF) 42. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. 1B. The second range of shear rates are greater than the first range of shear rates.

[0102] The head unit further includes a chamber 16. The chamber is configured to contain a portion of the STF and includes a front channel 26 and a back channel 24.

[0103] The head unit further includes a piston 36 housed at least partially radially within the chamber 16 and separating the back channel 24 and the front channel 26. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object 12-1. The movement of the piston includes one of traveling through

the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0104] The piston 36 includes a first piston bypass 38-1 between opposite sides of the piston that controls flow of the STF 42 between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0105] The piston 36 further includes a second piston bypass 38-2 between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0106] The head unit 10-1 further includes a set of fluid flow sensors 116-1-1 and 116-1-2 positioned proximal to the chamber 16. The set of fluid flow sensors provide the fluid response 232-1-1 and 232-1-2 respectively from the STF 42.

[0107] The head unit 10-1 further includes set of fluid manipulation emitters 114-1-1 and 114-1-2 positioned proximal to the chamber 16. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42 (e.g., shifting the shear rate versus viscosity curve), the first piston bypass 38-1 (e.g., to block or allow flow of the STF), and the second piston bypass 38-2 to control the motion of the object 12-1.

[0108] FIG. 6A illustrates an example of operation of a method for the controlling the operational aspects. A first step of the example of operation includes the piston 36 moving inward towards the head unit 10-1 when the object 12-1 exerts the force on the plunger 28 that transfers the force to the piston 36. As a result, the piston 36 exerts the force on the STF 42 within the back channel 24.

[0109] A second step of the example of operation includes the computing entity 20-1 interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity 182 and a piston position 184 of the piston 36 associated with a head unit device of a head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object 12-1 within the internal environment. For example, the computing entity 20-1 interprets fluid responses 232-1-1 and 232-1-2 from the STF 42 in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0110] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity 20-1 receives fluid responses 232-1-1 and 232-1-2 over the time range, where the fluid responses include the fluid flow signals.

[0111] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity 20-1 interprets the fluid flow signals to produce the fluid flow response.

[0112] A third sub-step includes determining the piston velocity based on the fluid flow response of the set of fluid flow sensors over the time range. For example, the computing entity 20-1 calculates piston velocity based on changes in the fluid flow response over the time range.

[0113] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity 20-1 calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0114] As yet another example of interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 compares the fluid response 232-1-1 and 232-1-2 to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity 182 and piston position 184. As a still further example of the interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 extracts the piston velocity 182 and the piston position 184 directly from the fluid response 232-1-1 and/or 232-1-2 when the sensors 116-1-1 and 116-1-2 generate the piston velocity and piston position directly.

[0115] The second step of the example of operation further includes the computing entity 20-1 determining a shear force 186 based on the piston velocity 182 and the piston position 184. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid flow response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity 20-1 extracts the shear force 186 directly from the fluid responses 232-1-1 and 232-1-2. In an instance, the shear force 186 reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. 6A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y1.

[0116] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42.

[0117] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42 based on an actual valve opening status of the first piston bypass 38-1 (e.g., which allows flow of the STF from the back channel 24 to the front channel 26 when the piston is moving in the inward direction of the example).

[0118] A third step of the example of operation includes the computing entity 20-1 interpreting an output of the environment sensor to identify an internal factor 151 of concern associated with the internal environment. The internal factor 151 of concern is associated with one or more of internal air quality, an internal smoke level, date, time of day, a building activity schedule, internal humidity, internal air pressure, internal temperature, fire alarm status, intrusion alarm status, building lockdown status, a building occupancy level, a maximum building occupancy threshold level, building power status, a solar array status, a heating-ventilation-and-cooling (HVAC) status, and/or any other factor of

concern. For example, the computing entity 20-1 interprets environment sensor information 150 to identify smoke in a storage room. As another example, the computing entity 20-1 interprets the environment sensor information 150 to identify a scheduled lunch time building activity (e.g., where more than average number of people are moving about within the building).

[0119] FIG. 6B further illustrates the example of operation, where having detected the internal factor of concern, in a fourth step the computing entity 20-1 determines the fluid activation 187 for the head unit based on the internal factor of concern and one or more of the piston velocity, the piston position, and the shear force 186. For example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of an internal door associated with the storage room more quickly when the internal factor is the smoke detected in the storage room and the object 12-1 includes the internal door to the storage room. As another example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of the external door slower than average, and safely, when the scheduled lunch time building activity is detected as the internal factor of concern.

[0120] In particular, the computing entity 20-1 determines the fluid activation 187 to adjust the viscosity of the STF to facilitate movement of the piston and hence door in a more desirable fashion based on the identified internal factor of concern. The determining the fluid activation 187 includes a variety of approaches. A first approach includes opening of either of the piston bypass 38-1 and piston bypass 38-2 allow the STF to move between the back channel 24 and the front channel 26 to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second approach includes opening of the chamber bypass 40 to lower the viscosity of the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0121] A fifth step of the example method of operation includes the computing entity 20-1 activating the set of fluid manipulation emitters 114-1-1 and 114-1-2 in accordance with the fluid activation 187 to manipulate one of the first shear threshold effect associated with the first piston bypass 38-1 and the second shear threshold effect associated with the second piston bypass 38-2 to control the motion of the object 12-1 to abate the effect on the internal environment from the internal factor of concern. For example, when the object 12-1 includes the internal door to the storage room and is closing when moving in the inward direction, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-1 to facilitate further opening of a one-way check valve to allow more of the STF to move from the back channel 24 to the front channel 26 thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to close when the internal factor of concern is the detected smoke in the storage room.

[0122] As another example, when the object 12-1 includes the external door closing (e.g., piston moving in the inward direction) in a high-traffic lunchtime situation, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-1 to facilitate closing down the one-way check valve to prevent STF from moving from the back channel 24 to the front channel 26 thusly selecting the

second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y1 to the point Y2 (e.g., at a dead stop when closed and no one is coming through the door) as illustrated in FIG. 6B.

[0123] In yet another embodiment, the computing entity 20-1 detects both an external factor of concern and the internal factor of concern to produce the fluid activation 187. For example, even when the external temperature is very low, the computing entity 20-1 controls the head unit system to close the door slowly when a higher than normal traffic pattern of people going through that door is detected.

[0124] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. 1 can alternatively be performed by other modules of the system of FIG. 1 or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system 10, cause one or more computing devices of the mechanical and computing system of FIG. 1 to perform any or all of the method steps described above.

[0125] FIGS. 7A-7B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system includes the head unit 10-1 of FIG. 1, the object 12-1 of FIG. 1 (e.g., a door), a secondary object 12-2 (e.g., a person), and the computing entity 20-1 of FIG. 1.

[0126] The head unit system includes a secondary object sensor (e.g., environment sensor module 14 of FIG. 2B). The secondary object is associated with the object (e.g., the person goes through the door).

[0127] The head unit 10-1 includes a shear thickening fluid (STF) 42. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. 1B. The second range of shear rates are greater than the first range of shear rates.

[0128] The head unit further includes a chamber 16. The chamber is configured to contain a portion of the STF and includes a front channel 26 and a back channel 24.

[0129] The head unit further includes a piston 36 housed at least partially radially within the chamber 16 and separating the back channel 24 and the front channel 26. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object 12-1. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0130] The piston 36 includes a first piston bypass 38-1 between opposite sides of the piston that controls flow of the STF 42 between the opposite sides of the piston from the

back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0131] The piston 36 further includes a second piston bypass 38-2 between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0132] The head unit 10-1 further includes a set of fluid flow sensors 116-1-1 and 116-1-2 positioned proximal to the chamber 16. The set of fluid flow sensors provide the fluid response 232-1-1 and 232-1-2 respectively from the STF 42.

[0133] The head unit 10-1 further includes set of fluid manipulation emitters 114-1-1 and 114-1-2 positioned proximal to the chamber 16. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42 (e.g., shifting the shear rate versus viscosity curve), the first piston bypass 38-1 (e.g., to block or allow flow of the STF), and the second piston bypass 38-2 to control the motion of the object 12-1 with regards to the secondary object (e.g., avoiding a collision between the door and the person).

[0134] FIG. 7A illustrates an example of operation of a method for the controlling the operational aspects. A first step of the example of operation includes the piston 36 moving outward away from the head unit 10-1 when the object 12-1 exerts a pulling force on the plunger 28 that transfers the force to the piston 36. As a result, the piston 36 exerts the force on the STF 42 within the front channel 26.

[0135] A second step of the example of operation includes the computing entity 20-1 interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity 182 and a piston position 184 of the piston 36 associated with a head unit device of a head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object 12-1 within the internal environment. For example, the computing entity 20-1 interprets fluid responses 232-1-1 and 232-1-2 from the STF 42 in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0136] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity 20-1 receives fluid responses 232-1-1 and 232-1-2 over the time range, where the fluid responses include the fluid flow signals.

[0137] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity 20-1 interprets the fluid flow signals to produce the fluid response.

[0138] A third sub-step includes determining the piston velocity based on the fluid response of the set of fluid flow sensors over the time range. For example, the computing entity 20-1 calculates piston velocity based on changes in the fluid response over the time range.

[0139] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity 20-1 calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0140] As yet another example of interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 compares the fluid response 232-1-1 and 232-1-2 to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity 182 and piston position 184. As a still further example of the interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 extracts the piston velocity 182 and the piston position 184 directly from the fluid response 232-1-1 and/or 232-1-2 when the sensors 116-1-1 and 116-1-2 generate the piston velocity and piston position directly.

[0141] The second step of the example of operation further includes the computing entity 20-1 determining a shear force 186 based on the piston velocity 182 and the piston position 184. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity 20-1 extracts the shear force 186 directly from the fluid responses 232-1-1 and 232-1-2. In an instance, the shear force 186 reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. 7A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y3 (e.g., a negative velocity since moving in the outward direction).

[0142] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42.

[0143] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42 based on an actual valve opening status of the first piston bypass 38-1 (e.g., which allows flow of the STF from the front channel 26 to the back channel 24 when the piston is moving in the outward direction of the example).

[0144] A third step of the example of operation includes the computing entity 20-1 interpreting an output of the secondary object sensor to produce an object type of the secondary object. The object type includes a person, a child, an elderly person, a group of people, a patient transport gurney, a cart, a short cart, a long cart, a cart train, a motor scooter, a vehicle, a jack truck, a pallet hauler, a 2 wheel hauler, a set of animals, and/or type of anything that can move through a door. The interpreting includes interpreting environment sensor information 150 from the secondary object sensor (e.g., from the environment sensor module 14), comparing the environment sensor information 150 to previously stored information for each of the object types, and selecting the object type when a match is detected. For example, the computing entity 20-1 compares an image of the environment sensor information 152 a stored image of a cart and identifies the secondary object as the cart when matching the image of the environment sensor information 150 to the stored image.

[0145] FIG. 7B further illustrates the example of operation, where having determined the object type, in a fourth

step the computing entity 20-1 determines the fluid activation 187 for the head unit based on the object type and one or more of the piston velocity, the piston position, and the shear force 186. For example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of an external door more quickly, yet safely, when the object type is a short cart. As another example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of the external door more slowly when the object type is an elderly person.

[0146] In particular, the computing entity 20-1 determines the fluid activation 187 to adjust the viscosity of the STF to facilitate movement of the piston and hence door in a more desirable fashion based on the identified object type. The determining the fluid activation 187 includes a variety of approaches. A first approach includes opening of either of the piston bypass 38-1 and piston bypass 38-2 allow the STF to move between the back channel 24 and the front channel 26 to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second approach includes opening of the chamber bypass 40 to lower the viscosity of the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0147] A fifth step of the example method of operation includes the computing entity 20-1 activating the set of fluid manipulation emitters 114-1-1 and 114-1-2 in accordance with the fluid activation 187 to manipulate one of the first shear threshold effect associated with the first piston bypass 38-1 and the second shear threshold effect associated with the second piston bypass 38-2 to control the motion of the object 12-1 to control the motion of the object with regards to the secondary object. For example, when the object 12-1 includes an operating room door and is opening when moving in the outward direction, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-2 to facilitate further opening of a one-way check valve to allow more of the STF to move from the front channel 26 to the back channel 24 thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to open when the detected secondary object is a patient transport gurney entering the operating room.

[0148] As another example, when the object 12-2 includes the group of people, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-1 to facilitate closing down the one-way check valve to prevent STF from moving from the back channel 24 to the front channel 26 thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y3 to the point Y4 (e.g., when the piston is moving in the outward direction) as illustrated in FIG. 7B.

[0149] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. 1 can alternatively be performed by other modules of the system of FIG. 1 or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element

section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system 10, cause one or more computing devices of the mechanical and computing system of FIG. 1 to perform any or all of the method steps described above.

[0150] FIGS. 8A-8B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system provides a head unit system that includes the head unit 10-1 of FIG. 1, the object 12-1 of FIG. 1 (e.g., a door), a plurality of secondary objects 12-2 through 12-N, a secondary object sensor (e.g., the environment sensor module 14 of FIG. 2B to detect the secondary objects), and the computing entity 20-1 of FIG. 1. The secondary object sensor is associated with the object 12-1 and the secondary objects are associated with the object.

[0151] The head unit system includes the secondary object sensor. The plurality of secondary objects is associated with the object (e.g., a group of people go through the door).

[0152] The head unit 10-1 includes a shear thickening fluid (STF) 42. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. 1B. The second range of shear rates are greater than the first range of shear rates.

[0153] The head unit further includes a chamber 16. The chamber is configured to contain a portion of the STF and includes a front channel 26 and a back channel 24.

[0154] The head unit further includes a piston 36 housed at least partially radially within the chamber 16 and separating the back channel 24 and the front channel 26. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from a force applied to the piston from the object 12-1. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0155] The piston 36 includes a first piston bypass 38-1 between opposite sides of the piston that controls flow of the STF 42 between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0156] The piston 36 further includes a second piston bypass 38-2 between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0157] The head unit 10-1 further includes a set of fluid flow sensors 116-1-1 and 116-1-2 positioned proximal to the chamber 16. The set of fluid flow sensors provide the fluid response 232-1-1 and 232-1-2 respectively from the STF 42.

[0158] The head unit 10-1 further includes set of fluid manipulation emitters 114-1-1 and 114-1-2 positioned proximal

to the chamber 16. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42 (e.g., shifting the shear rate versus viscosity curve), the first piston bypass 38-1 (e.g., to block or allow flow of the STF), and the second piston bypass 38-2 to control the motion of the object 12-1 with regards to the set of secondary objects (e.g., avoiding a collision between the door and the group people).

[0159] FIG. 8A illustrates an example of operation of a method for the controlling the operational aspects. In the example of operation, the piston 36 moves outward away from the head unit 10-1 when the object 12-1 exerts a pulling force on the plunger 28 that transfers the force to the piston 36. As a result, the piston 36 exerts the force on the STF 42 within the front channel 26.

[0160] A first step of the example of operation includes the computing entity 20-1 interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity 182 and a piston position 184 of the piston 36 associated with a head unit device of a head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object 12-1. For example, the computing entity 20-1 interprets fluid responses 232-1-1 and 232-1-2 from the STF 42 in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0161] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity 20-1 receives fluid responses 232-1-1 and 232-1-2 over the time range, where the fluid responses include the fluid flow signals.

[0162] A second sub-step includes determining the fluid flow response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity 20-1 interprets the fluid flow signals to produce the fluid response.

[0163] A third sub-step includes determining the piston velocity based on the fluid response of the set of fluid flow sensors over the time range. For example, the computing entity 20-1 calculates piston velocity based on changes in the fluid response over the time range.

[0164] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity 20-1 calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0165] As yet another example of interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 compares the fluid response 232-1-1 and 232-1-2 to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity 182 and piston position 184. As a still further example of the interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 extracts the piston velocity 182 and the piston position 184 directly from the fluid response 232-1-1 and/or 232-1-2 when the sensors 116-1-1 and 116-1-2 generate the piston velocity and piston position directly.

[0166] The first step of the example of operation further includes the computing entity 20-1 determining a shear force 186 based on the piston velocity 182 and the piston position 184. The determining the shear force based on the piston velocity and the piston position includes one approach of a

variety of approaches. A first approach includes extracting the shear force directly from the fluid response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity **20-1** extracts the shear force **186** directly from the fluid responses **232-1-1** and **232-1-2**. In an instance, the shear force **186** reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. **8A**, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point **Y3** (e.g., a negative velocity since moving in the outward direction).

[**0167**] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42**.

[**0168**] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity **20-1** compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF **42** based on an actual valve opening status of the second piston bypass **38-2** (e.g., which allows flow of the STF from the front channel **26** to the back channel **24** when the piston is moving in the outward direction of the example).

[**0169**] A second step of the example of operation includes the computing entity **20-1** interpreting an output of the secondary object sensor to update an object pattern of the plurality of secondary objects. The object pattern includes the object type as previously discussed and activity parameters associated with objects of the object type. The activity parameters includes a person walking, a person running, a group of people walking, a group of people running, a set of secondary objects moving towards the object, the set of second area objects moving away from the object, and/or any other possible activity type associated with one or more of the secondary objects.

[**0170**] The interpreting the output of the secondary object sensor to update the object pattern of the plurality of secondary objects to produce the updated object pattern of the plurality of secondary objects includes a series of sub-steps. A first sub-step includes the computing entity **20-1** obtaining environment sensor information **150** from the secondary object sensor (e.g., environment sensor module **14**) for a subset of secondary objects of the plurality of secondary objects (e.g., **12-2** through **12-N**).

[**0171**] A second sub-step includes the computing entity **20-1** identifying activity parameters from the environment sensor information for the subset of secondary objects. For example, the computing entity **20-1** compares a video clip of the environment sensor information **150** to a stored video clip of a group of people walking towards a door and identifies the object pattern as a group of people walking when matching the video clip of the environment sensor information **150** to the stored video clip of the person walking. As another example, the computing entity **20-1** further determines walking velocities for the group of people from the video clip from the video clip.

[**0172**] A third sub-step includes the computing entity **20-1** obtaining the object pattern of the plurality of secondary objects. For example, the computing entity **20-1** recovers the object pattern from a local memory of the computing entity

20-1. As another example, the computing entity **20-1** extracts the object pattern from activity information **153** recovered from the chamber database **34**.

[**0173**] A fourth sub-step includes the computing entity **20-1** modifying the object pattern of the plurality of secondary objects based on the activity parameters for the subset of secondary objects to produce the updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** modifies the walking velocities for the group of people based on the recently extracted walking velocities for the group of people.

[**0174**] FIG. **8B** further illustrates the example of operation, where having determined the updated object pattern for the plurality of secondary objects, a third step includes the computing entity **20-1** determining fluid activation **187** for the head unit based on the updated object pattern of the plurality of secondary objects and one or more of activity information **153**, the piston velocity **182**, the piston position **184**, and the shear force **186**. The activity information **153** includes historical records and schedules for future activities associated with the set of secondary objects. For example, the activity information **153** includes a work schedule, a class schedule, timeclock punching information indicating which employees are present at an area of employment, invoicing information associated with product flow, manufacturing process information, inventory control information, and/or any other information that assists in identifying the object pattern for the set of secondary objects.

[**0175**] The determining the fluid activation for the head unit device is based on the updated object pattern of the plurality of secondary objects and one or more of the piston velocity and the piston position includes one or more sub-steps. A first sub-step includes the computing entity **20-1** interpreting a request associated with modifying one or more of object velocity and object position. For example, the computing entity **20-1** receives the request from another computing entity. A second sub-step includes the computing entity interpreting fluid activation guidance from the chamber database **34** based on the updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** accesses the chamber database **34** based on the updated object pattern to locate the fluid activation guidance.

[**0176**] A third sub step includes the computing entity **20-1** interpreting activity information **153** from the chamber database **34** based on the updated object pattern of the plurality of secondary objects to produce an object movement recommendation. A correlation between the activity information and the updated object pattern of the plurality of secondary objects suggests an expected movement behavior of the secondary object. For example, analysis of **100** plus instances of groups of people walking through doors is summarized in the chamber database as the activity information **153** such that one more instance of the group of people walking through a particular door is expected to be similar.

[**0177**] A fourth sub-step includes the computing entity **20-1** determining a position for the secondary object based on the updated object pattern of the plurality of secondary objects. For example, the computing entity **20-1** estimates a next position for a person walking through the door based on the pattern of the groups of people walking through the doors.

[0178] A fifth sub-step includes the computing entity 20-1 determining an object position for the object based on the piston velocity and the piston position. For example, the computing entity 20-1 estimates the object position for the object (e.g., a door with regards to the secondary object, a person) based on historical data of piston velocity and piston position verses position of the object (e.g., the door).

[0179] A sixth sub-step includes the computing entity 20-1 establishing the fluid activation to include facilitating the first range of shear rates when the object movement recommendation includes increasing velocity of the object. For example, when the piston is traveling in the outward direction to open the door, the first range of shear rates is selected when the object movement recommendation includes opening the door more quickly to avoid a collision between the person and the door.

[0180] A seventh sub-step includes the computing entity 20-1 establishing the fluid activation to include facilitating the second range of shear rates when the object movement recommendation includes decreasing the velocity of the object. For example, when the piston is traveling in the outward direction to close the door, the second range of shear rates is selected when the object movement recommendation includes closing the door more slowly to avoid the collision between the person and the door. As another example of the determining of the fluid activation 187, the computing entity 20-1 establishes the fluid activation 187 to facilitate opening of a large door more quickly, yet safely, when the object pattern is the group of people approaching the door and the activity information 153 indicates that a current time frame is a lunch. When large groups of people are expected to be moving through the door. As a still further example, the computing entity 20-1 establishes the fluid activation 187 to facilitate closing of the large door more slowly when the object pattern is the group of people passing through the door.

[0181] In particular, the computing entity 20-1 determines the fluid activation 187 to produce a mechanism to adjust the viscosity of the STF to facilitate movement of the piston and hence door in a more desirable fashion based on the identified object type. The mechanism to adjust the viscosity includes a variety of approaches. A first approach includes opening of either of the piston bypass 38-1 and piston bypass 38-2 allow the STF to move between the back channel 24 and the front channel 26 to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second approach includes opening of the chamber bypass 40 to lower the viscosity of the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0182] The determining the fluid activation further includes interpreting the output of the secondary object sensor to further update the object pattern of the plurality of secondary objects to produce a further updated object pattern of the plurality of secondary objects. For example, the computing entity 20-1 further analyzes environment sensor information 150 from the environment sensor module 14 to update the object pattern. Having produced the further updated object pattern, the computing entity 20-1 updates the fluid activation for the head unit device based on the

further updated object pattern of the plurality of secondary objects and one or more of the piston velocity and the piston position as previously discussed.

[0183] A fourth step of the example method of operation includes the computing entity 20-1 activating the set of fluid manipulation emitters 114-1-1 and 114-1-2 in accordance with the fluid activation 187 to manipulate one of the first shear threshold effect associated with the first piston bypass 38-1 and the second shear threshold effect associated with the second piston bypass 38-2 to control the motion of the object 12-1 to control the motion of the object with regards to the secondary object. For example, when the object 12-1 includes a large lunchroom door and is opening when moving in the outward direction, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-2 to facilitate further opening of a one-way check valve to allow more of the STF to move from the front channel 26 to the back channel 24 thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to open when the detected set of secondary objects is a group of people passing through the lunch room door during the lunch period.

[0184] As another example, when the set of secondary objects 12-2 through 12-N includes the group of people, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-2 to facilitate closing down the one-way check valve to prevent STF from moving from the front channel 26 to the back channel 24 thusly selecting the second range of shear rates and a higher viscosity the STF to slow down the door to close safely moving from the point Y3 to the point Y4 (e.g., when the piston is moving in the outward direction) as illustrated in FIG. 8B.

[0185] The activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the object includes a variety of approaches. When the piston is traveling through the chamber in the inward direction and when the STF is to have the decreasing viscosity, a first sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-1 to the set of fluid manipulation emitters 114-1-1 to cause one of the first piston bypass 38-1 to facilitate the first shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF 42 to facilitate the first range of shear rates (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0186] When the piston is traveling through the chamber in the inward direction and when the STF is to have the increasing viscosity, a second sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-2 to the set of fluid manipulation emitters 114-1-2 to cause one of the first piston bypass 38-1 to facilitate the first shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF 42 to facilitate the second range of shear rates.

[0187] When the piston is traveling through the chamber in the outward direction and when the STF is to have the decreasing viscosity a third sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-1 to the set of fluid manipulation emitters 114-1-1 to cause one of the second piston bypass 38-2 to facilitate the second shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF 42 to facilitate the first range of shear rates.

[0188] When the piston is traveling through the chamber in the outward direction and when the STF is to have the increasing viscosity a fourth sub-approach includes the computing entity 20-1 issuing the fluid activation 234-1-2 to the set of fluid manipulation emitters 114-1-2 to cause one of the second piston bypass 38-2 to facilitate the second shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF 42 to facilitate the second range of shear rates.

[0189] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. 1 can alternatively be performed by other modules of the system of FIG. 1 or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system 10, cause one or more computing devices of the mechanical and computing system of FIG. 1 to perform any or all of the method steps described above.

[0190] FIGS. 9A-9B are schematic block diagrams of another embodiment of a mechanical and computing system illustrating an example of controlling operational aspects. The mechanical and computing system provides a head unit system that includes the head unit 10-1 of FIG. 1, the object 12-1 of FIG. 1 (e.g., a first door, interchangeable referred to as a first object), a set of objects 12-2 through 12-N (e.g., people proximal to the first door of a first building passing through the first door), a secondary object sensor (e.g., the environment sensor module 14 of FIG. 2B to detect the people), the chamber database 34 FIG. 1, and the computing entities 20-1 through 20-N (e.g., associated with multiple other non-proximal areas of a large building, associated with a multitude of non-proximal buildings of a large geographic area such as worldwide).

[0191] The head unit 10-1 (e.g., interchangeably referred to as the head unit device) is configured to control motion of the first object (e.g., object 12-1) with regards to a second object of a plurality of proximal objects (e.g., the objects 12-2 through 12-N). The plurality of proximal objects are line-of-sight spatially associated with the first object from time to time. For example, a group of people or other objects that are in an immediate vicinity of the first door.

[0192] The head unit system includes the secondary object sensor for sensing the plurality of proximal objects. The head unit 10-1 includes a shear thickening fluid (STF) 42. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates as discussed with reference to FIG. 1B. The second range of shear rates are greater than the first range of shear rates.

[0193] The head unit further includes a chamber 16. The chamber is configured to contain a portion of the STF and includes a front channel 26 and a back channel 24.

[0194] The head unit further includes a piston 36 housed at least partially radially within the chamber 16 and separating the back channel 24 and the front channel 26. The piston is configured to exert pressure against the shear thickening fluid in response to movement of the piston from

a force applied to the piston from the object 12-1. The movement of the piston includes one of traveling through the chamber in an inward direction or traveling through the chamber in an outward direction. The piston travels toward the back channel and away from the front channel when traveling in the inward direction. The piston travels toward the front channel and away from the back channel when traveling in the outward direction.

[0195] The piston 36 includes a first piston bypass 38-1 between opposite sides of the piston that controls flow of the STF 42 between the opposite sides of the piston from the back channel to the front channel when the piston is traveling through the chamber in the inward direction to cause the STF to react with a first shear threshold effect.

[0196] The piston 36 further includes a second piston bypass 38-2 between the opposite sides of the piston that controls flow of the STF between the opposite sides of the piston from the front channel to the back channel when the piston is traveling through the chamber in the outward direction to cause the STF to react with a second shear threshold effect.

[0197] The head unit 10-1 further includes a set of fluid flow sensors 116-1-1 and 116-1-2 positioned proximal to the chamber 16. The set of fluid flow sensors provide the fluid response 232-1-1 and 232-1-2 respectively from the STF 42.

[0198] The head unit 10-1 further includes set of fluid manipulation emitters 114-1-1 and 114-1-2 positioned proximal to the chamber 16. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42 (e.g., shifting the shear rate versus viscosity curve), the first piston bypass 38-1 (e.g., to block or allow flow of the STF), and the second piston bypass 38-2 to control the motion of the object 12-1 with regards to the set of secondary objects (e.g., avoiding a collision between the door and the group people).

[0199] FIG. 9A illustrates an example of operation of a method for the controlling the operational aspects. A first step of the example of operation includes the piston 36 moving outward away from the head unit 10-1 when the object 12-1 exerts a pulling force on the plunger 28 that transfers the force to the piston 36. As a result, the piston 36 exerts the force on the STF 42 within the front channel 26.

[0200] A second step of the example of operation includes the computing entity 20-1 interpreting a fluid response from the set of fluid flow sensors to produce a piston velocity 182 and a piston position 184 of the piston 36 associated with the head unit device of the head unit system. The set of fluid flow sensors are positioned proximal to the head unit device for controlling motion of the object 12-1 with regards to the second object of the plurality of proximal objects (e.g., any one of objects 12-2 through 12-N). For example, the computing entity 20-1 interprets fluid responses 232-1-1 and 232-1-2 from the STF 42 in response to varying responsiveness of particles of the STF to produce the piston velocity and the piston position.

[0201] The interpreting the fluid response from the set of fluid flow sensors to produce the piston velocity and the piston position of the piston includes a series of sub-steps. A first sub-step includes inputting, from one or more fluid flow sensors of the set of fluid flow sensors, a set of fluid flow signals over a time range. For example, the computing entity 20-1 receives fluid responses 232-1-1 and 232-1-2 over the time range, where the fluid responses include the fluid flow signals.

[0202] A second sub-step includes determining the fluid response of the set of fluid flow sensors based on the set of fluid flow signals. For example, the computing entity 20-1 interprets the fluid flow signals to produce the fluid response.

[0203] A third sub-step includes determining the piston velocity based on the fluid response of the set of fluid flow sensors over the time range. For example, the computing entity 20-1 calculates piston velocity based on changes in the fluid response over the time range.

[0204] A fourth sub-step includes determining the piston position based on the piston velocity and a real-time reference. For example, the computing entity 20-1 calculates the piston position based on time in the piston velocity as the piston moves through the chamber.

[0205] As yet another example of interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 compares the fluid response 232-1-1 and 232-1-2 to previous measurements of fluid flow versus piston velocity and piston position to produce the piston velocity 182 and piston position 184. As a still further example of the interpreting the fluid response 232-1-1 and 232-1-2, the computing entity 20-1 extracts the piston velocity 182 and the piston position 184 directly from the fluid response 232-1-1 and/or 232-1-2 when the sensors 116-1-1 and 116-1-2 generate the piston velocity and piston position directly.

[0206] The second step of the example of operation further includes the computing entity 20-1 determining a shear force 186 based on the piston velocity 182 and the piston position 184. The determining the shear force based on the piston velocity and the piston position includes one approach of a variety of approaches. A first approach includes extracting the shear force directly from the fluid response when one or more fluid flow sensors of the set of fluid flow sensors outputs a shear force encoded signal. For example, the computing entity 20-1 extracts the shear force 186 directly from the fluid responses 232-1-1 and 232-1-2. In an instance, the shear force 186 reveals the piston velocity versus force applied to the piston curve as illustrated in FIG. 9A, where at a current time of interpreting the fluid flow response, the force and piston velocity are at a point Y3 (e.g., a negative velocity since moving in the outward direction).

[0207] A second approach includes determining the shear force utilizing the piston velocity and stored data for piston velocity versus shear force for the STF. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42.

[0208] A third approach includes determining the shear force utilizing the piston position and stored data for piston position and a piston bypass versus shear force for the STF within the chamber. For example, the computing entity 20-1 compares the velocity and position to stored data for instantaneous velocity and position versus shear force for the STF 42 based on an actual valve opening status of the second piston bypass 38-2 (e.g., which allows flow of the STF from the front channel 26 to the back channel 24 when the piston is moving in the outward direction of the example).

[0209] A third step of the example of operation includes the computing entity 20-1 interpreting an output of the secondary object sensor to produce an updated object pattern of the plurality of proximal objects. The object pattern includes the object type as previously discussed and activity parameters associated with objects of the object type. The activity parameters include a person walking, a person

running, a group of people walking, a group of people running, a set of secondary objects moving towards the object, the set of second area objects moving away from the object, and/or any other possible activity type associated with one or more of the secondary objects.

[0210] The interpreting the output of the secondary object sensor to produce the updated object pattern of the plurality of proximal objects includes a series of sub-steps. A first sub-step includes obtaining environment sensor information from the secondary object sensor for a subset of the plurality of proximal objects.

[0211] For example, the computing entity 20-1 receives environment sensor information 150 from the environment sensor module 14 with regards to the object 12-2.

[0212] A second sub-step includes identifying activity parameters from the environment sensor information for the subset of the plurality of proximal objects. For example, the computing entity 20-1 identifies the person walking when the object 12-2 is the person.

[0213] A third sub-step includes obtaining a previous object pattern of the plurality of the plurality of proximal objects. For example, the computing entity 20-1 recovers the previous object pattern for the person walking and for other people walking from the chamber database 34.

[0214] A fourth sub-step includes modifying the previous object pattern of the plurality of proximal objects based on the activity parameters for the subset of proximal objects to produce the updated object pattern of the plurality of proximal objects. For example, the computing entity 20-1 modifies the previous object pattern to include particulars associated with the person walking.

[0215] FIG. 9B further illustrates the example the method of operation, where having produced the updated object pattern of the plurality of proximal objects, a fourth step includes the computing entity 20-1 identifying historical motion control results for a multitude of non-proximal objects with regards to a set of comparative objects (e.g., other doors) based on the updated object pattern of the plurality of proximal objects. The multitude of non-proximal objects are non-line-of-sight spatially associated with the set of comparative objects (e.g., people in other buildings across the world). The set of comparative objects share a set of common attributes with the first object (e.g., other doors in the other buildings).

[0216] For example, the computing entity 20-1 compares the updated object pattern to historical results 155 from the chamber database 34. The other computing entities 20-2 through 20-1 contribute to the historical results by generating the historical results based on other object patterns associated with the other computing entities, corresponding utilized fluid activations for those other object patterns, and actual results from the utilization of the fluid activations (e.g., unfavorable results including late door openings and early door closings; a verbal results including door openings and closings matched with actual needs).

[0217] Having identified the historical motion control results, a fifth step of the example method of operation includes the computing entity 20-1 determining the fluid activation 187 for the head unit based on the historical motion control results for the multitude of non-proximal objects and one or more of the piston velocity, the piston position, and the shear force 186. The determining the fluid activation for the head unit device based on the historical motion control results for the multitude of non-proximal

objects and one or more of the piston velocity and the piston position includes a variety of one or more approaches.

[0218] A first approach includes interpreting a request associated with modifying one or more of object velocity and object position. For example, the computing entity **20-1** receives a request to slow down the door from the computing entity **20-2** when the computing entity **20-2** is associated with a similar door and has produced the request based on a desired optimization.

[0219] A second approach includes interpreting fluid activation guidance from the chamber database based on the historical motion control results for the multitude of non-proximal objects. For example, the computing entity **20-1** analyzes historical results **155** for similar patterns that lead to desired results of controlling a door similar to the first door.

[0220] A third approach includes interpreting activity information from the chamber database based on the updated object pattern of the plurality of proximal objects to produce an object movement recommendation. A correlation between the activity information and the updated object pattern of the plurality of proximal objects suggests an expected movement behavior of the second object. For example, the computing entity **20-1** produces the object movement recommendation to quickly move the door out of the way when sudden but movement behavior of the object **12-2** is expected based on the activity information.

[0221] A fourth approach includes determining a position for the second object based on the updated object pattern of the plurality of proximal objects. For example, the computing entity **20-1** includes the position for the second object within the updated object pattern for most of the proximal objects.

[0222] A fifth approach includes determining an object position for the first object based on the piston velocity and the piston position. The computing entity **20-1** continuously tracks the object position for the first object based on its velocity and position over time.

[0223] A sixth approach includes establishing the fluid activation to include facilitating the first range of shear rates when the object movement recommendation includes increasing velocity of the first object. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate closing of a large exterior door more quickly, yet safely, when the object pattern of the set of secondary objects is one person approaching the door and the historical results **155** indicates that a faster closing for one person is generally favorable.

[0224] A seventh approach includes establishing the fluid activation to include facilitating the second range of shear rates when the object movement recommendation includes decreasing the velocity of the first object. For example, the computing entity **20-1** establishes the fluid activation **187** to facilitate opening of the large exterior door more slowly when the object pattern of the set of secondary objects is a group of people mulling around that could be in the way of the door opening and the historical results **155** indicates that a slower opening when people may be in the way of the opening door is generally favorable.

[0225] In particular, the computing entity **20-1** determines the fluid activation **187** to adjust the viscosity of the STF to facilitate movement of the piston and hence the first door in a more desirable fashion based on the identified object pattern for the set of proximal objects and based on the

historical results. The fluid activation **187** itself includes a variety of approaches. A first approach includes opening of either of the piston bypass **38-1** and piston bypass **38-2** allow the STF to move between the back channel **24** and the front channel **26** to lower the shear rate and thus select a lower viscosity which in turn allows more rapid movement of the piston in the chamber and hence speeds up the door. A second fluid activation approach includes opening of the chamber bypass **40** to lower the viscosity the STF. A third approach includes activating the set of emitters to directly alter the viscosity of the STF in a desired fashion (e.g., lowering viscosity to speed up opening or closing of the door, raising viscosity to slow down the opening or the closing of the door).

[0226] Alternatively, or in addition to, the fifth step further includes the computing entity **20-1** interpreting the output of the secondary object sensor to further update the updated object pattern of the plurality of proximal objects to produce a further updated object pattern of the plurality of proximal objects (e.g., continued to track the group of people walking about). Having produced the further updated object pattern, the fifth step further includes the computing entity **20-1** updating the fluid activation for the head unit device based on the further updated object pattern of the plurality of proximal objects and one or more of the piston velocity and the piston position as previously discussed.

[0227] Having determined the fluid activation for the head unit, a sixth step of the example method of operation includes the activating the set of fluid manipulation emitters **114-1-1** and **114-1-2** in accordance with the fluid activation **187** to manipulate one of the first shear threshold effect associated with the first piston bypass **38-1** and the second shear threshold effect associated with the second piston bypass **38-2** to control the motion of the object **12-1** (e.g., the first object) to control the motion of the object with regards to the second object.

[0228] The activating the set of fluid manipulation emitters in accordance with the fluid activation to control the motion of the first object with regards to the second object includes a variety of approaches. A first approach includes, when the piston is traveling through the chamber in the inward direction and when the STF is to have the decreasing viscosity, issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the first piston bypass to facilitate the first shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF to facilitate the first range of shear rates.

[0229] Otherwise, when the STF is to have the increasing viscosity, the first approach includes issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the first piston bypass to facilitate the first shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF to facilitate the second range of shear rates.

[0230] A second approach includes, when the piston is traveling through the chamber in the outward direction and when the STF is to have the decreasing viscosity, issuing the fluid activation to the set of fluid manipulation emitters to cause one of: the second piston bypass to facilitate the second shear threshold effect to include the first range of shear rates, and the direct manipulation of the STF to facilitate the first range of shear rates.

[0231] Otherwise, when the STF is to have the increasing viscosity, the second approach includes issuing the fluid

activation to the set of fluid manipulation emitters to cause one of: the second piston bypass to facilitate the second shear threshold effect to include the second range of shear rates, and the direct manipulation of the STF to facilitate the second range of shear rates.

[0232] As another example, when the object 12-1 includes a large garage door and is opening when moving in the outward direction, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-2 to facilitate further opening of a one-way check valve to allow more of the STF to move from the front channel 26 to the back channel 24 thusly selecting the first range of shear rates and a lower viscosity of the STF to speed up the door to open when the detected set of secondary objects is a large vehicle passing through the large garage door when the historical results indicate that it is favorable to open the door quickly when the vehicle is very large.

[0233] As yet another example, when the set of secondary objects 12-2 through 12-N includes a group of people, the computing entity 20-1 outputs the fluid activation 234-1-1 to the piston bypass 38-2 to facilitate closing down the one-way check valve to prevent STF from moving from the front channel 26 to the back channel 24 thusly selecting the second range of shear rates and a higher viscosity the STF to slow down a heavy door to close safely moving from the point Y3 to the point Y4 (e.g., when the piston is moving in the outward direction) as illustrated in FIG. 9B when the historical results indicate that it is favorable to slow down the closing of the heavy door.

[0234] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. 1 can alternatively be performed by other modules of the system of FIG. 1 or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system 10, cause one or more computing devices of the mechanical and computing system of FIG. 1 to perform any or all of the method steps described above.

[0235] FIGS. 10A-10D are schematic block diagrams of an embodiment of a mechanical system to control an object. The mechanical system provides a head unit system that includes a head unit 10-2 and the object 12-1 of FIG. 1 (e.g., a first door, interchangeable referred to as a first object). In an embodiment, a computing system is operably coupled to the mechanical system, where the computing system includes one or more of the sensor 116-1-1 of FIG. 1A, the emitter 114-1-1 of FIG. 1A, the computing entity 20-1 of FIG. 1A, and the computing entity 22 of FIG. 1A.

[0236] The head unit 10-2 includes a shear thickening fluid (STF) 42, where the STF 42 is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0237] The STF 42 includes nanoparticles. The nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring miner-

als, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture. The STF 42 further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture.

[0238] The head unit 10-2 further includes a chamber 16 which includes two pieces that telescope. The chamber 16 is configured to contain a portion of the STF. The chamber 16 includes a front channel 26 and a back channel 24. The chamber 16 is further configured to volumetrically expand and contract (e.g., via telescoping) to provide the first and second ranges of shear rates for the STF.

[0239] The head unit 10-2 further includes a gate 41. The gate 41 is configured to separate the front channel 26 and the back channel 24 to control velocity of flow of the STF 42 between the front channel 26 and the back channel 24 by covering and uncovering an opening within the chamber 16 between the front channel 26 and the back channel 24. The gate 41 is further configured to enable the STF 42 to provide the first range of shear rates as the STF flows from the back channel 24 to the front channel 26 when the gate 41 is in an open position uncovering the opening as the chamber 16 volumetrically expands (e.g., telescopes outward). The gate 41 is further configured to enable the STF 42 to provide the second range of shear rates as the STF 42 flows from the front channel 26 to the back channel 24 when the gate 41 is in a closed position covering the opening as the chamber 16 volumetrically contracts (e.g., telescopes inward).

[0240] The head unit 10-2 further includes a piston 36 housed at least partially radially within the back channel 24 of the chamber 16. The piston 36 configured to load a spring 45 within the back channel 24 of the chamber 16 in response to the STF 42 flowing from the front channel 26 to back channel 24 when the chamber volumetrically contracts in response to a force applied to the chamber 16 from the object 12-1. The spring 45 pushes up against a piston backstop 43 at the end of the chamber 16.

[0241] The piston 36 is further configured to exert force on the STF 42 to cause the STF 42 to flow from the back channel 24 to front channel 26 in response to energy from the spring 45 unloading when the chamber 16 volumetrically expands absent the force applied to the chamber 16 from the object 12-1 (e.g., either the object 12-1 pulls the chamber outward as the object 12-1 moves away from the head unit 10-2 or the object 12-1 simply moves away from the chamber and the spring pushes the piston and the STF which telescopes the chamber outward).

[0242] The gate 41 further includes a set of bypass openings 39-1 and 39-2 between opposite sides of the gate 41. The set of bypass openings are configured to enable the STF 42 to provide the second range of shear rates as the STF 42 flows from the front channel 26 through the set of bypass openings 39-1 and/or 39-2 to the back channel 24 when the gate 41 is in the closed position covering the opening as the chamber 16 volumetrically contracts. In an embodiment, there are any number of openings of a set of openings.

[0243] The gate 41 further includes a hinge 49. The hinge 49 is configured to enable the gate 41 to swing to the open position from the closed position and to swing to the closed position from the open position. In an embodiment, a stop component of the hinge prevents the hinge from opening a full 90 degrees. For example, the stop component holds the gate open at a 45 degree maximum to help prevent the gate from sticking in the full open position.

[0244] The set of bypass openings 39-1 and 39-2 further includes one or more of a variety of configurations. A first configuration includes a first bypass opening configured with at least one cylindrical tube with substantially consistent diameter from one side of the gate to an opposite side of the gate. For example, a straight tube to enable even STF flow.

[0245] A second configuration includes a second bypass opening configured with at least one conical shaped tube with an increasing diameter from the one side of the gate to the opposite side of the gate. For example, a tapered tube.

[0246] A third configuration includes a third bypass opening configured with the at least one conical shaped tube with a decreasing diameter from the one side of the gate to the opposite side of the gate. For example, the tapered tube in an opposite direction.

[0247] A fourth configuration includes a fourth bypass opening configured with at least one venturi shaped tube from the one side of the gate to the opposite side of the gate. For example, the venturi to cause a venturi effect between sides of the gate.

[0248] In an embodiment, the head unit 10-2 further includes a chamber bypass 40 (e.g., as discussed with reference to FIG. 5A) between opposite ends of the chamber 16. The chamber bypass 40 facilitates flow of a portion of the STF between the opposite ends of the chamber 16 to facilitate equalization of pressure of the STF when the piston 36 travels through the chamber 16 in an inward or an outward direction.

[0249] In a further embodiment, the head unit system further includes the set of fluid flow sensors 116-1-1 and 116-1-2 positioned proximal to the chamber 16 as discussed with reference to FIG. 5A. The set of fluid flow sensors are configured to provide a fluid response from the STF 42. The set of fluid flow sensors includes one or more of a valve opening detector associated with a set of bypass openings, a mechanical position sensor, an image sensor, and a light sensor. The fluid flow sensors further includes an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0250] In a still further embodiment, the head unit system further includes the set of fluid manipulation emitters 114-1-1 and 114-1-2 positioned proximal to the chamber as discussed with reference to FIG. 5A. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42, the gate 41, the piston 36, and the set of bypass openings 39-1 and 39-2 to provide the control of the motion of the object 12-1. The set of fluid manipulation emitters includes one or more of a variable flow valve associated with the set of bypass openings, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, and a speaker. The set of fluid manipulation emitters further includes an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0251] FIG. 10A illustrates an example of operation of the head unit system where, in a first step, the head unit system is at idle. The object 12-1 is operably coupled to the chamber 16 but, the first step, exerts no force upon the chamber 16. The chamber 16 is telescoped to a full outward position and the STF 42 is contained within the chamber in both the front channel 26 and the back channel 24 up to the piston 36.

[0252] FIG. 10B further illustrates the example of operation of the head unit system where, in a second step, the head unit system moves from the idle position to a slow movement due to STF resistance. The gate 41 starts in the closed position such that when force is applied from the object 12-1 to the chamber 16 to compress the chamber 16, the STF 42 in the front channel 26 runs up against the closed gate 41 causing higher shear forces to trigger the second range of shear rates. Some of the STF flows through the bypass 39-1 and the bypass 39-2 from the front channel 26 to the back channel 24.

[0253] The bypass openings further include one or more of a one-way check valve and a variable flow valve. When the piston 36 is traveling through the chamber 16 in the inward direction away from the gate 41, a first setting of the variable flow valve facilitates the second range of shear rates when the STF is to have the increasing viscosity.

[0254] As the STF fills the back channel 24 the piston 36 is pushed deeper into the chamber 16 causing loading of the spring 45 up against the piston backstop 43. When the piston 36 is traveling through the chamber 16 in an inward direction away from the gate 41, a second shear threshold effect includes the second range of shear rates when the STF 42 is configured to have the increasing viscosity. As such, the FIG. 10C further illustrates the example of operation of the head unit system where, in a third step, the head unit system moves from the slow movement due to STF resistance position (e.g., compressed chamber) to a fast from open gate position (e.g., the object 12-1 moves back towards a starting position). As the object 12-1 moves outward (e.g., away from the chamber 16 such that the chamber 16 expands or the object 12-1 pulls the chamber 16) the STF in the back channel 24 is pushed up against the gate 41 by the piston 36 due to the spring 45 releasing its stored energy. The gate 41 opens in response to the STF 42 moving towards the gate 41 due to the pressure exerted on the STF 42 by the piston 36.

[0255] As the gate 41 opens, the STF moves from the back channel 24 to the front channel 26 with a higher velocity as compared to the opposite direction since the STF 42 is both pushed by the piston 36 and pulled by the suction of the STF 42 within the front channel 26 as the front channel volumetrically expands as the chamber expands. When the one-way check valve is utilized and the piston 36 is traveling through the chamber 16 in the outward direction towards the gate 41, the one-way check valve is configured to prevent STF flow through the one-way check valve. The STF 42 flows through the open gate into the front channel 26.

[0256] When the piston 36 is traveling through the chamber 16 in an outward direction towards the gate 41, a first shear threshold effect includes the first range of shear rates when the STF 42 is configured to have the decreasing viscosity. As such, a lower viscosity associated with this chamber expansion enables expansion of the chamber 16 at a faster rate than the compression of the chamber 16 as previously discussed. In an embodiment, the hinge 49 is configured to limit the opening of the gate 41 to an angle between 0° and 90°. For example, the hinge 49 is configured to limit the opening of the gate 41 to 45° such that the gate 41 is less likely to not remain stuck open when it is time to compress the chamber 16.

[0257] FIG. 10D further illustrates the example of operation of the head unit system where, in a fourth step, the head unit system moves from the fast from open gate position to the stopped and idle position. The spring 45 having released

all its energy, positions the piston 36 at a steady state position with regards to the force the spring applies up against the STF 42. The STF 42 stabilizes between the back channel 24 and the front channel 26 when the chamber 16 has reached its maximum expansion level. In response, the gate 41 tends towards closing over the opening between the back channel 24 and the front channel 26. In an embodiment, the hinge 49 includes a return spring to force the gate 41 closed when the pressure of the back channel 24 and the front channel 26 is substantially the same.

[0258] In another embodiment, when the sensors and emitters are utilized in conjunction with a computing entity, the rate at which the second shear threshold effect takes place is sensed and automatically adjusted for a desired rate such that the object 12-1 travels inward at a desired trajectory (e.g., a maximum speed, a maximum acceleration, a desired length of travel, etc.). In a similar fashion, the sensors and emitters in conjunction with the computing entity are utilized to automatically adjust for another desired rate with regards to outward travel of the object 12-1.

[0259] FIGS. 11A-11D are schematic block diagrams of another embodiment of a mechanical system to control an object. The mechanical system provides a head unit system that includes a head unit 10-3 and the object 12-1 of FIG. 1 (e.g., a first door, interchangeable referred to as a first object). In an embodiment, a computing system is operably coupled to the mechanical system, where the computing system includes one or more of the sensor 116-1-1 of FIG. 1A, the emitter 114-1-1 of FIG. 1A, the computing entity 20-1 of FIG. 1A, and the computing entity 22 of FIG. 1A.

[0260] The head unit 10-3 includes a shear thickening fluid (STF) 42, where the STF 42 is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0261] The STF 42 includes nanoparticles. The nanoparticles include one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture. The STF 42 further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture.

[0262] The head unit 10-3 further includes a chamber 16. The chamber 16 is configured to contain a portion of the STF 42. The chamber 16 includes a front channel 26 and a back channel 24.

[0263] The head unit 10-3 further includes a set of gates 57-1 and 57-2. Any number of gates may be utilized. The set of gates is configured to separate the front channel 26 and the back channel 24 to control velocity of flow of the STF 42 between the front channel 26 and the back channel 24. The set of gates 57-1 and 57-2 is further configured to enable the STF 42 to provide the first range of shear rates as the STF flows from the back channel 24 to the front channel 26 when the set of gates is in an open position. The set of gates 57-1 and 57-2 is further configured to enable the STF 42 to provide the second range of shear rates as the STF flows from the front channel 26 to the back channel 24 when the set of gates is in a closed position. In an embodiment, the set of gates open and close in unison (e.g., closed together or open substantially the same amount simultaneously).

[0264] The set of gates 57-1 and 57-2 further includes a set of hinges 59-1 and 59-2. The set of hinges is configured to

enable the set of gates 57-1 and 57-2 to swing to the open position from the closed position and to swing to the closed position from the open position. In an embodiment, the set of hinges is further configured to only allow the set of gates to open to a maximum open level. For example, the set of hinges only allow the gates to open to a maximum of 45° within a range of zero to 90°.

[0265] The head unit 10-3 further includes a cupped piston 55. The cupped piston is configured to facilitate flow of the STF 42 through a slot set 63 between the front channel 26 and the back channel 24. The slot set includes one or more varieties of slots. A first variety of the slot set is configured with at least one cylindrical tube with substantially consistent diameter from one side of the cupped piston to an opposite side of the cupped piston. For example, a uniform hole.

[0266] A second variety of slot of the slot set is configured with at least one conical shaped tube with an increasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, a tapered hole.

[0267] A third variety of slot of the slot set is configured with the at least one conical shaped tube with a decreasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, another tapered hole in another direction.

[0268] A fourth variety of slot of the slot set is configured with at least one venturi shaped tube from the one side of the cupped piston to the opposite side of the cupped piston. For example, the venturi shaped tube to produce a venturi effect between opposite sides of the cupped piston.

[0269] The head unit 10-3 further includes a bushing 53. The bushing is configured within the chamber 16 to cause containment of the STF 42 in the back channel 24. In an embodiment, the bushing 53 is further configured with one or more seals 47 provide further containment of the STF 42 within the back channel 24. The head unit 10-3 further includes a backstop 51. The backstop is configured with the chamber 16 to cause containment of the STF 42 in the front channel 26.

[0270] The head unit 10-3 further includes a plunger 28 housed at least partially radially within the back channel 24 of the chamber 16. The plunger is configured to facilitate the set of gates to operate in the closed position to cause the STF 42 within the front channel 26 to provide the second range of shear rates in response to an inward force applied to the plunger 28 from the object 12-1 as the STF 42 flows from the front channel 26 to the back channel 24. The plunger 28 is further configured to facilitate the set of gates 57-1 and 57-2 to operate in the open position to cause the STF 42 within the back channel 24 to provide the first range of shear rates in response to an outward force applied to the plunger as the STF 42 flows from the back channel 24 to the front channel 26.

[0271] The head unit 10-3 further includes a cap 61. The cap is configured to move in unison with the plunger 28 radially within the chamber 16. In an embodiment, the 61 is operably coupled to the plunger 28 such that a force applied by the object 12-1 to the plunger 28 and the 61 causes the 61 and the plunger 28 to move in unison in the inward direction in the chamber 16. The head unit 10-3 further includes a spring 45. The spring 45 is configured to store energy when the 61 moves in the inward direction in response to a force from the object 12-1. The spring 45 is

further configured to release the energy, absent the force applied by the object 12-1, such that the 61 and plunger 28 move in the outward direction quickly to reset the 61 to the idle position.

[0272] In an embodiment, the head unit 10-3 further includes a chamber bypass 40 as discussed with reference to FIG. 1A between opposite ends of the chamber 16. The chamber bypass 40 facilitates flow of a portion of the STF 42 between the opposite ends of the chamber 16 when the set of gates 57-1 and 57-2 travels through the chamber 16 in an inward or an outward direction.

[0273] In an embodiment where the head unit system further includes the set of fluid flow sensors, the set of fluid flow sensors are configured to provide a fluid response from the STF 42. The set of fluid flow sensors includes one or more of a valve opening detector associated with the slot set 63 of the cupped piston 55, a mechanical position sensor, an image sensor, a light sensor, and an audio sensor. The set of fluid flow sensors further includes a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0274] In an embodiment where the head unit system further includes the set of fluid manipulation emitters, the set of fluid manipulation emitters are positioned proximal to the chamber 16. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42, the set of gates 57-1 and 57-2, and the slot set 63 of the cupped piston 55 to provide the control of the motion of the object 12-1. The set of fluid manipulation emitters includes a variety of fluid manipulation emitters including one or more of a variable flow valve associated with the slot set 63, a mechanical vibration generator, an image generator, a light emitter, and an audio transducer. The set of fluid manipulation emitters further includes a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0275] FIG. 11A illustrates an example of operation of the head unit system where, in a first step, the head unit system is in an idle position. The object 12-1 is operably coupled to the plunger 28 and cap 61 ready to exert force upon the plunger and. At idle, the plunger and cap are at a full outward position such that the set of gates are in the closed position and are ready to exert force upon the STF 42 within the front channel 26.

[0276] FIG. 11B further illustrates the example of operation of the head unit system where, in a second step, the head unit system moves from the idle position to a slow compression operation due to STF resistance. For example, the object 12-1 applies the force to the cap 61 and the plunger 28 such that the set of gates 57-1 and 57-2 travels through the chamber 16 in the inward direction away from the back channel 24 to produce a second shear threshold effect as the STF 42 within the front channel 26 is compressed by the closed set of gates and the cupped piston. The second shear threshold effect includes the second range of shear rates when the STF is configured to have the increasing viscosity.

[0277] The set of gates 57-1 and 57-2 further include a bypass opening set 65. The bypass opening set 65 is configured to facilitate flow of the STF 42 from the front channel 26 through the bypass opening set 65 to the back channel 24 to cause the STF 42 within the front channel 26 to provide the second range of shear rates when the set of gates 57-1 and 57-2 is operating in the closed position. Movement of the set of gates and the cupped piston towards

the front channel 26 in response to the force applied from the plunger 28 serves to facilitate closure of the set of gates. As the set of gates closes, the STF 42 flows from the front channel 26 to the back channel 24 by one or more approaches. A first approach includes traveling around the set of gates when a diameter of the set of gates is less than a diameter of the chamber 16. A second approach includes traveling through the set of gates via one or more bypass openings of the bypass opening set 65.

[0278] The cupped piston 55 is further configured to facilitate flow of the STF 42 around the set of gates between the chamber and the set of gates from the front channel through the slot set 63 of the cupped piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position. For example, a smaller diameter slot set facilitates higher shear thresholds and a higher viscosity from the second range of shear rates such that a relatively small force from the object 12-1 causes an abrupt slow down of the movement of the object 12-1.

[0279] The cupped piston 55 is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the cupped piston to the back channel to cause the STF within the front channel to provide a sub-range of the second range of shear rates in accordance with a slot size configuration of the slot set when the set of gates is operating in the closed position. For example, a larger diameter slot set requires even more force from the object 12-1 to reach the second range of shear rates to cause the slowing down of the movement of the object 12-1.

[0280] The second step of the example of operation of the head unit 10-3 ends with the cap 61 and the plunger 28 reaching an end of travel (e.g., the cap 61 travels inward within the chamber 16 to rest when reaching the bushing 53). Soon after reaching the end of travel, the pressure of the STF within the front channel 26 equalizes with the pressure of the STF within the back channel 24. As long as a maintaining force from the object 12-1 is applied to the plunger 28 the head unit 10-3 remains in this position with resistance to the force from the object 12-1 coming from this stored energy of the spring 45 and no longer from the STF 42.

[0281] FIG. 11C further illustrates the example of operation of the head unit system where, in a third step, the head unit system moves from the slow compression operation to a fast rebounding operation absent the force applied from the object 12-1. The set of gates 57-1 and 57-2 is further configured such that, when the set of gates is traveling through the chamber in the outward direction towards the back channel, a first shear threshold effect is produced. The first shear threshold effect includes the first range of shear rates when the STF is configured to have the decreasing viscosity.

[0282] As the force from the object 12-1 is removed from the plunger 28 and the cap 61 or when the object 12-1 pulls the plunger 28 in the outward direction, the spring 45 is further configured within the chamber to have stored the energy as a result of the inward force applied to the plunger and to the cap from the object 12-1. The spring 45 is further configured to move the plunger 28, the cap 61, and set of gates 57-1 and 57-2 outward absent the inward force applied to the plunger and to the cap from the object 12-1. As the set

of gates is pulled by the plunger from the front channel to the back channel, the force of the STF serves to open the set of gates.

[0283] The cupped piston 55 is further configured to facilitate the flow of the STF from the back channel 24 through the slot set 63 of the cupped piston 55 to the front channel to cause the set of gates to operate in the open position and to cause the STF within the back channel to provide the first range of shear rates when the set of gates is operating in the open position. As such, an improvement to the operation of the head unit system is provided where the cap 61 quickly rebounds to the idle position absent the force applied by the object 12-1.

[0284] FIG. 11D further illustrates the example of operation of the head unit system where, in a fourth step, the head unit system transitions from the fast rebounding operation to a stopping operation where the system finally comes to a steady-state rest absent the force applied by the object 12-1. For example, the spring 45 releases all of its energy against the cap 61 and plunger 28 thus pulling the cupped piston 55 in the set of gates 57-1 and 57-2 to a fully outward resting position. The gates 57-1 and 57-2 begin to close when the pressure of the STF equalizes between the back channel 24 and the front channel 26. As such, the head unit 10-3 is resetting to be ready for another cycle where the object 12-1 applies the force to the head unit 10-3.

[0285] FIGS. 12A-12F are schematic block diagrams of another embodiment of a mechanical system to control an object. The mechanical system provides a head unit system that includes a head unit 10-4 and the objects 12-2 and 12-3 of FIG. 1A. In an embodiment, a computing system is operably coupled to the mechanical system, where the computing system includes one or more of the sensor 116-1-1 of FIG. 1A, the emitter 114-1-1 of FIG. 1A, the computing entity 20-1 of FIG. 1A, and the computing entity 22 of FIG. 1A.

[0286] FIGS. 12A and 12B illustrates an example of operation of the mechanical system to control the object. In an embodiment, the object 12-2 includes a first portion of a hinge assembly that is associated with a doorjamb and the object 12-3 includes a remaining portion of the hinge assembly that is associated with a door that swings utilizing the hinge assembly. In particular, the object 12-3 includes both the remaining portion of the hinge assembly and the door and the object 12-2 includes the first portion of the hinge assembly and the doorjamb.

[0287] FIG. 12A illustrates a relationship between the objects 12-2 and 12-3 with regards to the head unit 10-4 where the hinge assembly is in the open position. FIG. 12B further illustrates the relationship between objects 12-2 and 12-3 with regards to the head unit 10-4 where the hinge assembly is rotating towards a closed position (e.g., as the door closes). As will be discussed below, rotation of the object 12-3 causes a portion of the head unit 10-4 to rotate which causes compression of a shear thickening fluid within the head unit 10-4 to achieve a desired viscosity of the shear thickening fluid which in turn limits acceleration and/or velocity of the closing of the door.

[0288] FIG. 12C illustrates an embodiment of the head unit 10-4 and a first step of an example of operation of the head unit system that includes the head unit 10-4 at a starting open position. The head unit 10-4 includes a shear thickening fluid (STF) 42, where the STF 42 is configured to have a decreasing viscosity in response to a first range of shear

rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0289] The STF 42 includes nanoparticles. The nanoparticles include one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture. The STF 42 further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture.

[0290] The head unit 10-4 further includes a chamber 75. The chamber 75 is configured to contain a portion of the STF 42. The chamber 75 includes a front channel 26 and a back channel 24. In an embodiment, the chamber 75 is associated with the object 12-2 (e.g., the doorjamb) and is considered a fixed portion of the head unit 10-4. When the chamber 75 is considered the fixed portion, a rotating portion of the head unit 10-4 includes one or more rotating chambers 73-1 and 73-2. The one or more rotating chambers are associated with the object 12-3 (e.g., the door) and are considered a moving and/or rotational portion of the head unit 10-4 relative to the chamber 75 and the object 12-2.

[0291] The head unit 10-4 further includes a set of gates 57-1 and 57-2. Any number of gates may be utilized. The set of gates is configured to separate the front channel 26 and the back channel 24 to control velocity of flow of the STF 42 between the front channel 26 and the back channel 24. The set of gates 57-1 and 57-2 is further configured to enable the STF 42 to provide the first range of shear rates as the STF flows from the back channel 24 to the front channel 26 when the set of gates is in an open position. The set of gates 57-1 and 57-2 is further configured to enable the STF 42 to provide the second range of shear rates as the STF flows from the front channel 26 to the back channel 24 when the set of gates is in a closed position. In an embodiment, the set of gates open and close in unison (e.g., closed together or open substantially the same amount simultaneously). In another embodiment, the set of gates open within a first time frame and close within a second time frame. In yet another embodiment, the set of gates open sequentially. In a still further embodiment, the set of gates close sequentially.

[0292] The set of gates 57-1 and 57-2 further includes a set of hinges 59-1 and 59-2. The set of hinges is configured to enable the set of gates 57-1 and 57-2 to swing to the open position from the closed position and to swing to the closed position from the open position. In an embodiment, the set of hinges is further configured to only allow the set of gates to open to a maximum open level. For example, the set of hinges only allow the gates to open to a maximum of 45° within a range of zero to 90°.

[0293] The set of gates 57-1 and 57-2 further include a bypass opening set 65. The bypass opening set 65 is configured to facilitate flow of the STF 42 from the front channel 26 through the bypass opening set 65 to the back channel 24 to cause the STF 42 within the front channel 26 to provide the second range of shear rates when the set of gates 57-1 and 57-2 is operating in the closed position. Movement of the set of gates towards the front channel 26 in response to a force applied from the plunger 28 serves to facilitate closure of the set of gates. As the set of gates closes, the STF 42 flows from the front channel 26 to the back channel 24 by one or more approaches. A first approach includes traveling around the set of gates when a diameter of the set of gates is less than a diameter of the chamber 75. A

second approach includes traveling through the set of gates via one or more bypass openings of the bypass opening set 65.

[0294] The head unit 10-4 further includes a plunger 28 housed at least partially radially within the back channel 24 of the chamber 75. The plunger is configured to facilitate the set of gates to operate in the closed position to cause the STF 42 within the front channel 26 to provide the second range of shear rates inward force applied to the plunger 28 from a rotational force of the object 12-3 as the STF 42 flows from the front channel 26 to the back channel 24. The plunger 28 is further configured to facilitate the set of gates 57-1 and 57-2 to operate in the open position to cause the STF 42 within the back channel 24 to provide the first range of shear rates in response to an outward force applied to the plunger 28 from a counter-rotational force of the object 12-3 as the STF 42 flows from the back channel 24 to the front channel 26.

[0295] The head unit 10-4 further includes a threaded plunger 79. The threaded plunger 79 is configured to apply the inward force to the plunger 28 in response to the rotational force of the object 12-3. The threaded plunger 79 may also be referred to as a threaded non-rotational plunger 79. The threaded plunger 79 is further configured to generate the inward force when a nut 81 rotates on threads of the threaded plunger 79 of as a result of the rotational force of the object 12-3. In an embodiment, the nut 81 is configured within a rotating chamber 73-2 that is extended beyond the chamber 75. The nut 81 is configured in a fixed position within the rotating chamber 73-2 by way of a fastener 83 between the rotating chamber 73-2 and the nut 81.

[0296] The threaded plunger 79 is further configured to apply the outward force to the plunger 28 in response to the counter-rotational force of the object 12-3. The threaded plunger 79 is further configured to generate the outward force when the nut 81 counter-rotates on the threads of the threaded plunger 79 of as a result of the counter-rotational force of the object 12-3.

[0297] In an embodiment, the head unit 10-4 further includes a stop adjuster 87. The stop adjuster 87 is configured with the nut 81 to prevent the threaded plunger from traveling beyond a stop position of the stop adjuster 87. As a result, the door can only swing open to a maximum open position.

[0298] The head unit 10-4 further includes a cupped piston 55. The cupped piston is configured to facilitate flow of the STF 42 around the set of gates between the chamber 75 and the set of gates 57-1 and 57-2 from the front channel 26 through a slot set 63 of the cupped piston 55 to the back channel 24 to cause the STF 42 within the front channel 26 to provide the second range of shear rates when the set of gates is operating in the closed position. The cupped piston 55 is further configured to facilitate flow of the STF 42 from the back channel 24 through the slot set 63 of the cupped piston 55 to the front channel 26 to cause the set of gates to operate in the open position and to cause the STF 42 within the back channel 24 to provide the first range of shear rates when the set of gates is operating in the open position.

[0299] The cupped piston is further configured to facilitate flow of the STF 42 facilitate flow of the STF through the slot set 63 between the front channel 26 and the back channel 24 utilizing one or more varieties of slots. A first variety of the slot set is configured with at least one cylindrical tube with

substantially consistent diameter from one side of the cupped piston to an opposite side of the cupped piston. For example, a uniform hole.

[0300] A second variety of slot of the slot set is configured with at least one conical shaped tube with an increasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, a tapered hole.

[0301] A third variety of slot of the slot set is configured with the at least one conical shaped tube with a decreasing diameter from the one side of the cupped piston to the opposite side of the cupped piston. For example, another tapered hole in another direction.

[0302] A fourth variety of slot of the slot set is configured with at least one venturi shaped tube from the one side of the cupped piston to the opposite side of the cupped piston. For example, the venturi shaped tube to produce a venturi effect between opposite sides of the cupped piston.

[0303] The head unit 10-4 further includes a fixed bushing 77. The fixed bushing is configured in a fixed position relative to the chamber 75 to cause containment of the STF 42 in the back channel 24. In an embodiment, the fixed bushing 77 is further configured with one or more seals 47 provide further containment of the STF 42 within the back channel 24. In an embodiment, the fastener 83 stabilizes the fixed bushing 77 in the fixed position with regards to the chamber 75.

[0304] In an embodiment, the fixed bushing 77 further includes a dowel key 85 2 guide the plunger 28 through the fixed bushing 77 as the plunger 28 travels longitudinally through the chamber 75 without rotation.

[0305] The head unit 10-4 further includes a backstop 71. The backstop is configured with the chamber 75 to cause containment of the STF 42 in the front channel 26. In an embodiment, the backstop 71 includes one or more seals 47 when the backstop 71 is configured within the chamber 75 to prevent flow of the STF into the rotating chamber 73-1. The backstop 71 is secured to the rotating chamber 73-1 by way of another fastener 83.

[0306] The head unit 10-4 further includes a piston 36. The piston 36 is configured to travel longitudinally within the chamber 75 to cause containment of the STF 42 in the back channel 24. The piston 36 is further configured to compress a spring 45 within the chamber 75 between the piston 36 and the fixed bushing 77 relative to the chamber 75 when the STF 42 flows from the front channel 26 to the back channel 24 in response to the inward force to the plunger 28. The piston 36 is further configured to cause flow of the STF 42 in the back channel 24 in response to de-compression of the spring 45 further causing the set of gates 57-1 and 57-2 to operate in the open position and to further cause the STF 42 within the back channel 24 to provide the first range of shear rates when the set of gates is operating in the open position when the outward force is applied to the plunger 28.

[0307] In an embodiment, the head unit 10-4 further includes a chamber bypass 40 as discussed with reference to FIG. 1A between opposite ends of the chamber 75. The chamber bypass 40 facilitates flow of a portion of the STF 42 between the opposite ends of the chamber 75 when the set of gates 57-1 and 57-2 travels through the chamber 75 in an inward or an outward direction.

[0308] In an embodiment where the head unit system further includes the set of fluid flow sensors, the set of fluid

flow sensors are configured to provide a fluid response from the STF 42. The set of fluid flow sensors includes one or more of a valve opening detector associated with the slot set 63 of the cupped piston 55, a mechanical position sensor, an image sensor, a light sensor, and an audio sensor. The set of fluid flow sensors further includes a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0309] In an embodiment where the head unit system further includes the set of fluid manipulation emitters, the set of fluid manipulation emitters are positioned proximal to the chamber 75. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF 42, the set of gates 57-1 and 57-2, and the slot set 63 of the cupped piston 55 to provide the control of the motion of the object 12-3. The set of fluid manipulation emitters includes a variety of fluid manipulation emitters including one or more of a variable flow valve associated with the slot set 63, a mechanical vibration generator, an image generator, a light emitter, and an audio transducer. The set of fluid manipulation emitters further includes a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0310] FIG. 12C further illustrates an example of operation of the head unit system where, in a first step, the head unit system is in an open position. The object 12-3 is operably coupled to the rotating chamber 73-2 ready to exert the rotational force upon the rotating chamber 73-2. At idle, the threaded non-rotational plunger 79 and the plunger 28 plunger are at a full outward position such that the set of gates are in the closed position and are ready to exert force upon the STF 42 within the front channel 26. At idle, in the open position, the spring 45 is relaxed and the piston 36 is at a steady state with regards to spring pressure from the spring 45 and pressure of the STF within the back channel 24.

[0311] FIG. 12D further illustrates the example of operation of the head unit system where, in a second step, the head unit system moves from the open position to a rotational operation due to the rotational force from the object 12-3 applied to the rotating chamber 73-2. For example, the object 12-3 applies the rotational force to the rotating chamber 73-2 which turns the nut 81 causing the threaded non-rotation plunger 79 2 travel inward through the fixed bushing 77 exerting force on the operably coupled plunger 28 causing the plunger 28 to exert force on the cupped piston 55 and set of gates 57-1 and 57-2. When set of gates 57-1 and 57-2 travels through the chamber 75 in the inward direction away from the back channel 24 that produces a second shear threshold effect as the STF 42 within the front channel 26 is compressed by the closed set of gates and the cupped piston 55. The second shear threshold effect includes the second range of shear rates when the STF is configured to have the increasing viscosity.

[0312] The cupped piston 55 is further configured to facilitate flow of the STF 42 around the set of gates between the chamber and the set of gates from the front channel through the slot set 63 of the cupped piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position. For example, a smaller diameter slot set facilitates higher shear thresholds and a higher viscosity from the second range of shear rates such that a relatively small force from the object 12-3 causes an

abrupt slow down of the movement of the object 12-3 (e.g., stops the door from slamming the door jamb).

[0313] The cupped piston 55 is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the cupped piston to the back channel to cause the STF within the front channel to provide a sub-range of the second range of shear rates in accordance with a slot size configuration of the slot set when the set of gates is operating in the closed position. For example, a larger diameter slot set requires even more rotational force from the object 12-3 to reach the second range of shear rates to cause the slowing down of the movement of the object 12-3.

[0314] The second step of the example of operation of the head unit 10-4 ends with the nut 81 causing an end of travel of the threaded non-rotational plunger 79. Soon after reaching the end of travel, the pressure of the STF within the front channel 26 equalizes with the pressure of the STF within the back channel 24. As long as a maintaining force from the object 12-3 (e.g., the doors closed and latched by a door-knob) is applied to the plunger 28 the head unit 10-4 remains in this position with resistance to the force from the object 12-3 coming from this stored energy of the spring 45 and no longer from the STF 42.

[0315] FIG. 12E further illustrates the example of operation of the head unit system where, in a third step, the head unit system moves from the rotation operation (e.g., and idling with the cupped piston 55 fully extended within the front channel 26) to a counter-rotation operation as counter-rotation force applied from the object 12-3 (e.g., the doors opened). The set of gates 57-1 and 57-2 is further configured such that, when the set of gates is traveling through the chamber in the outward direction towards the back channel, a first shear threshold effect is produced. The first shear threshold effect includes the first range of shear rates when the STF is configured to have the decreasing viscosity.

[0316] As the counter-rotational force from the object 12-3 counter-rotates the nut 81 causing the threaded non-rotational plunger 79 to move outward within the rotating chamber 73-2 thus pulling the plunger 28 and the cupped piston 55 along with the gates 57-1 and 57-2 in the outward direction, the spring 45 is further configured within the chamber 75 to release the stored energy as a result of the STF 42 flowing from the back channel 24 through the cupped piston 55 and the open gates 57-1 and 57-2 into the front channel 26. The spring 45 is further configured to assist moving the STF 42 from the back channel 24 to the front channel 26 as the cupped piston 55 is drawn outward by the counter-rotation movement. As the set of gates 57-1 and 57-2 is pulled inward by the plunger 28 from the front channel 26 to the back channel 24, the force of the STF 42 serves to open the set of gates 57-1 and 57-2 causing even more STF 42 to flow through the cupped piston 55 with less resistance thus speeding up the counter-rotation movement of the object 12-3 (e.g., the door opens quickly).

[0317] The cupped piston 55 is further configured to facilitate the flow of the STF from the back channel 24 through the slot set 63 of the cupped piston 55 to the front channel 26 to cause the set of gates 57-1 and 57-2 to operate in the open position and to cause the STF 42 within the back channel to provide the first range of shear rates when the set of gates is operating in the open position. As such, an improvement to the operation of the head unit system is

provided where the rotating chamber 73-2 quickly rebounds enabling the object 12-3 to counter-rotate back to the open position.

[0318] FIG. 12F further illustrates the example of operation of the head unit system where, in a fourth step, the head unit system transitions from the counter-rotation operation to a stopping operation where the system finally comes to a steady-state rest as the object 12-3 returns to the open an idle position. For example, the spring 45 releases all of its energy against the piston 36 thus pushing the STF 42 through the set of gates 57-1 and 57-2 to a fully inward resting position where pressure of the STF equalizes between the back channel 24 and the front channel 26. The gates 57-1 and 57-2 begin to close when that pressure of the STF equalizes. As such, the head unit 10-4 is resetting to be ready for another cycle where the object 12-3 applies the rotational force to the rotating chamber 73-2.

[0319] FIGS. 13A-13D are schematic block diagrams of another embodiment of a mechanical head unit system to control an object. FIG. 13A illustrates the head unit system to include a shear thickening fluid (STF) module 206 to be coupled and mutually operated with a hinge module 204. To control motion of an object (e.g., a door coupled to the hinge module).

[0320] FIG. 13B illustrates an embodiment of an STF hinge unit 202 that includes the STF module 206 coupled to the hinge module 204 utilizing a retaining ring 210. The STF module 206 includes a shear thickening fluid (STF) 42. The STF 42 is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0321] The STF 42 includes a plurality of nanoparticles. The plurality of nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture thereof. The STF 42 further includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture thereof.

[0322] The STF module 206 further includes a chamber 16. The chamber 16 is configured to contain a portion of the STF 42. The chamber 16 includes a front channel 26 and a back channel 24.

[0323] In an embodiment, the STF module 206 further includes a set of gates 57-1 and 57-2. Alternatively, in another embodiment, the set of gates are replaced by functionality of slots of a piston as discussed below. The set of gates is configured to separate the front channel 26 and the back channel 24 to control velocity of flow of the STF 42 between the front channel 26 and the back channel 24.

[0324] The set of gates is further configured to enable the STF 42 to provide the first range of shear rates as the STF flows from the back channel 24 to the front channel 26 when the set of gates 57-1 and 57-2 is in an open position. The set of gates is further configured to enable the STF 42 to provide the second range of shear rates as the STF flows from the front channel 26 to the back channel 24 when the set of gates is in a closed position.

[0325] The STF module 206 further includes a piston (e.g., a cupped piston 55). The piston is housed at least partially radially within the chamber 16. The piston is configured to include a slot set. The piston is further

configured to facilitate flow of the STF 42 through the slot set between the front channel 26 and the back channel 24 when the piston travels longitudinally within the chamber 16.

[0326] The slot set includes one or more varieties of slots. A first slot of the slot set configured with at least one cylindrical tube with substantially consistent diameter from one side of the piston to an opposite side of the piston. A second slot of the slot set configured with at least one conical shaped tube with an increasing diameter from the one side of the piston to the opposite side of the piston. A third slot of the slot set configured with the at least one conical shaped tube with a decreasing diameter from the one side of the piston to the opposite side of the piston. A fourth slot of the slot set configured with at least one venturi shaped tube from the one side of the piston to the opposite side of the piston.

[0327] The STF module 206 further includes a plunger 28-1. The plunger 28-1 is housed at least partially radially within the back channel 24 of the chamber 16. The plunger 28-1 is configured to exert force on the piston in response to the motion of the object.

[0328] The STF module 206 further includes a bushing 53 and at least one seal 47 to provide a seal between the bushing 53 and the chamber 16. The bushing 53 configured in a fixed position relative to the chamber 16 to cause containment of the STF 42 in the back channel 24.

[0329] The STF module 206 further includes a backstop 51. The backstop 51 is configured with the chamber 16 to cause containment of the STF 42 in the front channel 26.

[0330] The STF module 206 further includes a cap 61. The cap 61 is configured to move in unison with the plunger 28-1 radially within the chamber 16.

[0331] The STF module 206 further includes a spring 45. The spring 45 is configured to store energy when the cap 61 moves in an inward direction in response to a force from the object. The spring is further configured to release the energy, absent the force from the object, such that the cap 61 and the plunger 28-1 move in an outward direction.

[0332] The hinge module 204 includes a threaded plunger (a threaded non-rotational plunger 79). The threaded plunger is configured to facilitate an inward force to the plunger 28-1 in response to a rotational force of the object (e.g., the door via half of a hinge object 12-3). The threaded plunger 79 is further configured to generate the inward force when a nut 81 rotates on threads of the threaded plunger as a result of the rotational force of the object (e.g., closing the door). The threaded plunger is further configured to facilitate an outward force to the plunger 28-1 in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object (e.g., opening the door).

[0333] The hinge module 204 further includes a fastener 83 to attach the nut 81 to the object 12-3 in a fixed relative position. As such, when the object 12-3 rotates versus object 12-2 (e.g., a portion of the hinge affixed to a doorjamb) the not anyone rotates around the threaded non-rotational plunger 79 thus moving the plunger laterally.

[0334] The hinge module 204 further includes a second plunger 28-2. The second plunger is configured to produce the inward force from the threaded plunger to the plunger of the STF module 206 when the nut rotates on threads of the threaded plunger as the result of the rotational force of the

object. The second plunger is further configured to produce the outward force from the threaded plunger to the plunger when the nut counter-rotates on the threads of the threaded plunger as the result of the counter-rotational force of the object.

[0335] In an embodiment, the inch module 204 further includes a stop adjuster 87. The stop adjuster 87 stops the threaded non-rotational plunger 79 from going too far at one end to facilitate a doorstep function.

[0336] The hinge module 204 further includes a fixed bushing 77 to guide the second plunger 28-2 from the threaded plunger to the plunger 28-1 and 61 of the STF module 206. In an embodiment, the fixed bushing 77 is attached to a rotating chamber 73-2 of the object 12-3 utilizing a fastener 83. In an embodiment, the object 12-2 further includes a fixed chamber 75 to encircle the fixed bushing 77. In another embodiment, the fixed chamber 75 is a separate component of the hinge module 204 and is not part of the object 12-2.

[0337] The plunger 79 is hard coupled to the plunger 28-2 and the plunger 28-2 utilizes a dowel key 85 within a slot of the fixed bushing 77 to prevent the combination of the threaded plunger and second plunger 28-2 from rotating and instead the rotation of the nut 81 around the threaded plunger causes the combination of the threaded plunger and the second plunger to move in a lateral fashion in the inward or outward directions through the fixed bushing 77 to make contact with the cap 61 and the plunger 28-1.

[0338] In an embodiment, the STF module 206 further includes a set of fluid flow sensors positioned proximal to the chamber as previously discussed. The set of fluid flow sensors is configured to provide a fluid response from the STF. The set of fluid flow sensors includes one or more of a valve opening detector associated with a slot set of the piston, a mechanical position sensor, an image sensor, a light sensor, an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor.

[0339] In an embodiment, the STF module 206 further includes a set of fluid manipulation emitters positioned proximal to the chamber. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF, the set of gates, and a slot set of the piston to provide the control of the motion of the object. The set of fluid manipulation emitters includes one or more of a variable flow valve associated with the slot set, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter.

[0340] The set of gates associated with the piston is further configured to, when the set of gates is traveling through the chamber in an outward direction towards the back channel 24, a first shear threshold effect includes the first range of shear rates when the STF is configured to have the decreasing viscosity. When the set of gates is traveling through the chamber in an inward direction away from the back channel 24, a second shear threshold effect includes the second range of shear rates when the STF is configured to have the increasing viscosity.

[0341] The gates 57-1 and 57-2 of the STF module 206 are configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The STF module 206 further

includes a set of hinges 59-1 and 59-2. The set of hinges is configured to enable the set of gates to swing to the open position from the closed position and to swing to the closed position from the open position.

[0342] FIG. 13C further illustrates the embodiment of the STF hinge unit 202 that includes the STF module 206 coupled to the hinge module 204 utilizing a retaining ring 210. In an example of operation, when the set of hinges 59-1 and 59-2 is configured to enable the set of gates to swing to the open position within the front channel, the STF module 206 enables the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the open position. The STF module 206 enables the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the closed position.

[0343] In another example of operation, when the set of hinges 59-1 and 59-2 is configured to enable the set of gates to swing to the open position within the front channel, the piston is further configured to facilitate the flow of the STF from the back channel through the slot set of the piston to the front channel to cause the set of gates to operate in the open position and to cause the STF within the back channel to provide the first range of shear rates when the set of gates is operating in the open position.

[0344] The piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position (e.g., as shown where rotation of object 2-3 turns the knot which turns the threaded plunger which pushes up on the second plunger which pushes up on the and plunger, which pushes up on the piston and gates into the front channel thus compressing the STF, and with a sufficient shear threshold, provides the second range of shear rates and a higher viscosity which slows down the moving parts including a door attached to object 12-3.

[0345] FIG. 13D illustrates another embodiment of a head unit system that includes a shear thickening fluid (STF) module 206-1, a second STF module 206-2, and a hinge module 204-1. The STF module 206-1 is coupled to the hinge module 204-1 utilizing a retaining ring 210-1. The hinge module 204-1 is coupled to the second STF module 206-2 utilizing a second retaining ring 210-2.

[0346] The STF module 206-1 includes the STF, where the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0347] The STF module 206 further includes a chamber, where the chamber configured to contain a portion of the STF. The chamber includes a front channel and a back channel. The STF 206 module further includes a set of gates, where the set of gates is configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position. The set of gates is further configured to enable the STF to provide the second range of shear rates

as the STF flows from the front channel to the back channel when the set of gates is in a closed position.

[0348] The STF module 206-1 further includes a piston (e.g., a cupped piston 55-1), where the piston housed at least partially radially within the chamber. The piston is configured to include a slot set. The piston is further configured to facilitate flow of the STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber.

[0349] The STF module 206-1 further includes a plunger 28-1, the plunger housed at least partially radially within the back channel of the chamber. The plunger is configured to exert force on the piston in response to the motion of the object.

[0350] The second STF module 206-2 further includes second shear thickening fluid (STF). The second STF is configured to have a decreasing viscosity in response to a third range of shear rates and an increasing viscosity in response to a fourth range of shear rates. The fourth range of shear rates are greater than the third range of shear rates.

[0351] The second STF module 206-2 further includes a second chamber. The second chamber is configured to contain a portion of the second STF. The second chamber includes a second front channel and a second back channel.

[0352] The second STF module 206-2 further includes a second set of gates. The second set of gates is configured to separate the second front channel and the second back channel to control velocity of flow of the second STF between the second front channel and the second back channel. The second set of gates is further configured to enable the second STF to provide the third range of shear rates as the second STF flows from the second back channel to the second front channel when the second set of gates is in an open position. The second set of gates is further configured to enable the second STF to provide the third range of shear rates as the second STF flows from the second front channel to the second back channel when the second set of gates is in a closed position.

[0353] The second STF module 206-2 further includes second piston (e.g., a cupped piston 55-3). The second piston housed at least partially radially within the second chamber. The second piston configured to include a second slot set. The piston is further configured to facilitate flow of the second STF through the second slot set between the second front channel and the second back channel when the piston travels longitudinally within the second chamber.

[0354] The second STF module 206-2 further includes a second plunger 28-4. The second plunger housed at least partially radially within the second back channel of the second chamber. The second plunger is configured to exert force on the second piston in response to the motion of the object.

[0355] The hinge module 204-1 includes a threaded plunger (e.g., a threaded non-rotational plunger 79-1). The threaded plunger is configured to facilitate an inward force to the plunger 28-1 via a plunger 28-2 of the hinge module 204-1 and a simultaneous outward force to the second plunger 28-4 via a plunger 28-3 in response to a rotational force of the object (e.g., the door). The threaded plunger is further configured to generate the inward force to the plunger and the simultaneous outward force to the second plunger when a nut 81 rotates on threads of the threaded plunger as a result of the rotational force of the object. The nut 81 is coupled to object 12-2 of the hinge module 204-1

utilizing a fastener 83 such that the nut 81 turns on the threaded plunger when a combination of the nut 81 and the object 12-2 rotates (e.g., closing the door).

[0356] The threaded plunger is further configured to facilitate an outward force to the plunger 28-1 via the plunger 28-2 and a simultaneous inward force to the second plunger 28-4 via the plunger 28-3 in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force to the plunger and the simultaneous inward force to the second plunger when the nut 81 counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object (e.g., object 12-2 and the door).

[0357] The hinge module 204-1 further includes object 12-3 (e.g., a portion of the hinge connected to a doorjamb). The object 12-3 is coupled to fixed bushings 77-1 and 77-2 at an upper and lower position of the object 12-3, where the fixed bushing 77-1 guides the plunger 28-2 and the fixed bushing 77-2 guides the plunger 28-3. The combination of the plunger 28-3, the threaded non-rotational plunger 79-1 and the plunger 28-2 is forced to travel in the inward and outward directions without rotation utilizing a dowel key 85-1 within a slot of the plunger 28-2 and another dowel key 85-2 within a slot of the plunger 28-3.

[0358] The STF of the STF modules 206-1 and 206-2 may have different properties to produce custom results for the overall operation of the head unit system. When the gates and cupped pistons are implemented facing both the outward ends of the head unit system, the STF of the STF module 206-1 may be utilized to provide the second range of shear rates to control the door closing and the STF of the STF module 206-2 may be utilized to provide the second range of shear rates to control the door opening.

[0359] In an alternative embodiment, the cupped piston 55-2 and the gates and hinges associated with the cupped piston 55-2 may be flipped such that both STF modules provide similar motion control when utilizing similar STF. Within the STF module 206-2, when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel of the STF module 206-2, the STF provides the second range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the closed position and the STF provides the first range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the open position.

[0360] Further within the STF module 206-2, when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel, the piston is further configured to facilitate the flow of the STF from the front channel through the slot set of the piston to the back channel to cause the set of gates to operate in the open position and to cause the STF within the front channel to provide the first range of shear rates when the set of gates is operating in the open position. The piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the back channel through the slot set of the piston to the front channel to cause the STF within the back channel to provide the second range of shear rates when the set of gates is operating in the closed position.

[0361] FIGS. 14A-14C are schematic block diagrams of another embodiment of a mechanical system (e.g., a head unit system) to control an object (e.g., a door). FIGS. 14A

and 14B illustrate embodiments where the head unit system includes the STF module 206-1 and the hinge module 204.

[0362] The STF module 206-1 includes a shear thickening fluid (STF). The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates.

[0363] The STF module 206-1 further includes a chamber 16, where the chamber is configured to contain a portion of the STF. The chamber includes a front channel and a back channel. A backstop 51-1 contains the STF in the front channel.

[0364] The STF module 206-1 further includes a set of gates, where the set of gates is configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position. The set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position.

[0365] The STF module 206-1 further includes a piston, where the piston is housed at least partially radially within the chamber 16. The piston is configured to include a slot set. The piston is further configured to facilitate flow of the STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber 16.

[0366] The STF module 206-1 further includes a plunger 28-1. The plunger is housed at least partially radially within the back channel of the chamber. The plunger is configured to exert force on the piston in response to the motion of the object. The STF module 206-1 further includes a cap 61 to operate in conjunction with the plunger 28-1 to exert pressure on the plunger. A spring 45 stores energy and releases the energy by pushing the cap 61 outward from the chamber 16 to pull the piston via the plunger from the front channel to the back channel.

[0367] The STF module 206-1 further includes a modular castle mount 220. The modular castle mount is configured to extend the chamber to facilitate the configuration of the plunger to exert the force on the piston in response to the motion of the object. The modular castle mount 220 includes a set of teeth and/or tabs constructed in a circle around the diameter of the chamber 16. Each of the set of teeth and/or tabs includes a retaining ring channel 212.

[0368] The hinge module 204 includes a threaded plunger, where the threaded plunger is configured to facilitate an inward force to the plunger in response to a rotational force of the object. The threaded plunger is further configured to generate the inward force when a nut rotates on threads of the threaded plunger as a result of the rotational force of the object. The threaded plunger is further configured to facilitate an outward force to the plunger in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object.

[0369] The hinge module 204 further includes a reciprocal modular castle mount 220, where the reciprocal modular castle mount is configured to couple the reciprocal modular

castle mount with the modular castle mount. The threaded plunger is configured to apply the inward force to the plunger in response to the rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount. The threaded plunger is further configured to apply the outward force to the plunger in response to the counter-rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount.

[0370] The reciprocal modular castle mount includes a reciprocal set of teeth and/or reciprocal tabs constructed in a circle around the diameter of a portion of the hinge module 204 (e.g., a rotating chamber 73-1 or a fixed chamber 75). Each of the set of teeth and/or tabs includes a retaining ring channel 212 such that a single common retaining ring channel is formed when the reciprocal modular castle mount is coupled with the modular castle mount. A retaining ring 210 configured within the single common retaining ring channel facilitates coupling of the STF module 206-1 and the hinge module 204 as illustrated in FIG. 14B.

[0371] The hinge module 204 further includes a lower castle mount 221 to facilitate coupling of the STF module 206-2 of FIG. 13D with the hinge module 204 utilizing another retaining ring 210-2 as illustrated in FIG. 14C. The STF module 206-2 further includes a backstop 51-2 to contain the STF of the STF module 206-2 within a front channel of a chamber 16-2.

[0372] The hinge module 204 further includes one or more fastener holes 222 to accept one or more fasteners 226 of FIG. 14B to secure the object 12-3 and to secure the object 12-2. The objects 12-3 and 12-2 further include one or more mounting holes 224 to secure the objects 12-3 and 12-2 to other objects.

[0373] FIGS. 15A-15C are schematic block diagrams of another embodiment of a mechanical system (e.g., a head unit system) to control an object. The head unit system includes the hinge module 204 and STF module 206-1 of FIG. 14A and one of accessory modules 208-1 through 208-7. The accessory modules 208-1 through 208-7 also couple to the hinge module 204 utilizing the castle mount 220 as discussed in reference to FIGS. 14A-C.

[0374] Each accessory module includes an accessory coupling mechanism that is configured to couple the accessory module to the hinge module. For example, the accessory coupling mechanism is configured with a second modular castle mount that couples the accessory module to the hinge module via a second reciprocal modular castle mount of the hinge.

[0375] Each accessory module further includes a computing device configured to operate in accordance with a first mode of the first range of shear rates. The computing device is further configured to operate in accordance with a second mode of the second range of shear rates when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism.

[0376] The computing device is configured with a processor for executing processor-executable instructions and a storage device for storing the processor-executable instructions. The processor-executable instructions include a first set of processor-executable instructions that are configured to facilitate the first mode of the first range of shear rates and

a second set of processor-executable instructions that are configured to facilitate the second mode of the second range of shear rates.

[0377] The first set of processor-executable instructions is further configured to facilitate the first mode of the first range of shear rates, when executed by the processor, to cause the processor to perform one or more of a variety of functions. A first function includes controlling a motor configured to accelerate the rotational force of the object. For example, the accessory module includes the motor and is activated when the object (e.g., a door) is moving slowly (e.g., the first range) to speed up closing the door.

[0378] A second function includes determining a position of the object. For example, the processor calculates the position of the door from sensor data when the accessory module includes a position sensor. A third function includes determining a movement vector of the object. For example, the processor calculates movement vector of the door based on time and the position of the door from the sensor data when the accessory module includes the position sensor.

[0379] A fourth function includes identifying a secondary object associated with the object. For example, the processor identifies a person or a pet (e.g., near the door) from an image sensor output and subsequently activates the motor to contain the pet to a particular area. A fifth function includes interpreting a request to change the movement vector of the object. For example, the processor decodes an IoT command via a wireless link to stop the door or even close the door. A sixth function includes indicating a status of the object. The status of the object includes one or more of open (e.g., door open), closed, locked, unlocked, safe (e.g., when no one is blocking the door, or no fire on the other side of the door, etc.), not safe (e.g., person in way, pet in way, intruder in next room, active shooter in next room or nearby, fire or other hazard present), and emergency fast closing when the object is a door.

[0380] In another embodiment, a seventh function includes controlling a set of fluid manipulation emitters positioned proximal to the chamber to facilitate the first range of shear rates. The set of fluid manipulation emitters provide a fluid activation to at least one of the STF, the set of gates, and a slot set of the piston to provide the control of the motion of the object (e.g., to quickly close the door).

[0381] The second set of processor-executable instructions is further configured to facilitate the second mode of the second range of shear rates (e.g., higher viscosity of the STF), when executed by the processor, to cause the processor to perform one or more of a variety of other functions. A first function includes controlling the motor configured to decelerate the rotational force of the object (e.g., stop the door for safety reasons or slow close to allow someone to enter within a short timeframe). A second function includes detecting an end-position of the object (e.g., the door has closed). A third function includes determining an end-movement vector of the object (e.g., based on speed, position, and time of just finishing closing the door).

[0382] A fourth function includes determining a movement vector of the secondary object associated with the object (e.g., estimating speed and position of the pet or person based on an image sensor output). A fifth function includes interpreting a request to change the end-movement vector of the object (e.g., the processor decodes another IoT command via the wireless link to slow down the closing of the door even more or keep it just open).

[0383] A sixth function includes indicating a further status of the object. The further status of the object includes one or more of emergency slow closing and locking, when the object is the door. For instance, in response to a wireless message from a device initiated by a teacher, the processor activates the motor to quickly close a classroom door and indicate the emergency closing status. In an embodiment, a seventh function includes controlling the set of fluid manipulation emitters to facilitate the second range of shear rates (e.g., to slow close the door).

[0384] A technological improvement is provided to the STF module **206-1** with regards to controlling motion of the object by utilizing one of the accessory modules **208-1** through **208-7**. The STF module **206-1** and accessory module share the hinge module **204** and operate within proximity. In an embodiment, each of the STF module **206-1** and the accessory modules **208-1** through **208-7** include the computing device **100-1** of FIG. 3 such that the STF module **206-1** communicates with the accessory module via wireless communication to facilitate optimization of the controlling of the motion of the object.

[0385] The accessory modules **208-1** through **208-7** represent a few examples of numerous embodiments of the accessory module to provide one or more specialty sensors and/or actuators with regards to the operation of the head unit system. For example, the accessory module **208-1** includes a media sensor such as one or more of an image sensor and a microphone.

[0386] As another example, the accessory module **208-2** includes a set of wireless communication modems to provide communications between modules sharing the hinge module **204** depicted in FIG. 14C and as discussed with reference to FIG. 15B and with other modules associated with other hinge modules as discussed further with reference to FIG. 15C. For instance, an infrared wireless communication path is established through the hinge module **204** between the STF module **206-1** mounted on the castle mount **220** and the accessory module **208-2** mounted on another castle mount **220** of the hinge module **204**. In another instance, the infrared wireless communication path is further established to module associated with another hinge module **204**. In yet another instance, a Wi-Fi wireless communication path is established from the accessory module **208-2** to another module in another room.

[0387] As another example, the accessory module **208-3** includes visual user output for status and other purposes via a set of status light emitting diodes (LED). For example, illumination of the LED with a red color indicates that an associated door is open while elimination of the LED with a green color indicates that the associated door is closed.

[0388] As another example, the accessory module **208-4** includes an energy harvesting apparatus to produce minute amounts of energy that is temporarily stored (e.g., in a battery, in a capacitor, etc.) as a result of motion of the object. The energy may be utilized to power the computing device **100-1** of the same module. In another embodiment, the energy may be utilized power another module coupled to the same hinge module **204**, where an isolated center conductor that runs through the hinge module **204** transfers the energy from the energy affiliated accessory module **208-4** to the other module.

[0389] As yet another example, the accessory module **208-5** includes an Internet of Things (IoT) functionality to both sense operations of the STF module **206-1** and control

the STF module **206-1**. In an instance, the IoT functionality includes providing status of a door associated with the head unit system to a digital twin platform. In another instance, the IoT functionality includes providing a security service for the door associated with the head unit system (e.g., triggering an alarm for an unexpected opening of the door, indicating when the second range of shear rates are utilized within the STF module **206-1**, etc.).

[0390] As a further example, the accessory module **208-6** includes an electric motor such that the electric motor can affect the position and motion of the object. For instance, the electric motor is activated to further slow down the closing of the door in addition to a safety response from the STF module **206-1**. In another instance, the electric motor is activated to open a closed door to assist an individual that normally needs assistance to open doors. In yet another instance, the motor stops the travel of all the mechanicals when the motor is implemented as a solenoid (e.g., a coil around a solenoid plunger) where the solenoid plunger travels into any rotating portion of the head unit system to immediately stop rotation stopping movement of the object (e.g., holding a closed door closed/locked, preventing a door from opening more than a desired maximum amount, etc.).

[0391] As a still further example, the accessory module **208-7** includes a position sensor to provide information with regards to the position and/or motion of the object. For instance, the position sensor indicates how far open the door for comparison to a security threshold level of opening of the door. In another instance, the position sensor is utilized to track exactly how the head unit system operates when the second range of shear rates are active (e.g., how effective is the STF module **206-1** at providing a desired safe and quiet closing of the door).

[0392] FIG. 15B illustrates scenario examples of utilization of the STF module **206-1**, the hinge module **204**, and one of a variety of accessory modules that shares the hinge module **204** in common with the STF module. The head unit system for controlling motion of an object includes a shear thickening fluid (STF) module, a hinge module, and an accessory module.

[0393] The STF module includes a shear thickening fluid (STF), where the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates. The second range of shear rates are greater than the first range of shear rates. The STF module further includes a chamber, where the chamber is configured to contain a portion of the STF. The chamber includes a front channel and a back channel as previously discussed. The STF module further includes a set of gates, where the set of gates is configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel. The set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position, wherein the set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position.

[0394] The STF module further includes a piston, where the piston is housed at least partially radially within the chamber. The piston is further configured to include a slot set. The piston is further configured to facilitate flow of the

STF through the slot set between the front channel and the back channel when the piston travels longitudinally within the chamber.

[0395] The STF module further includes a plunger, where the plunger is housed at least partially radially within the back channel of the chamber. The plunger is configured to exert force on the piston in response to the motion of the object.

[0396] The hinge module **204** includes a coupling mechanism, where the coupling mechanism is configured to couple the STF module to the hinge module. The hinge module further includes a threaded plunger, where the threaded plunger is configured to facilitate an inward force to the plunger in response to a rotational force of the object. The threaded plunger is further configured to generate the inward force when a nut rotates on threads of the threaded plunger as a result of the rotational force of the object. The threaded plunger is further configured to facilitate an outward force to the plunger in response to a counter-rotational force of the object. The threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object.

[0397] The accessory module includes an accessory coupling mechanism, where the accessory coupling mechanism is configured to couple the accessory module to the hinge module. The accessory module further includes a computing device, where the computing device is configured to operate in accordance with a first mode of the first range of shear rates when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism. For example, the computing device monitors the operation of the head unit system when the first mode is to monitor operations while the STF is operating in the first range of shear rates.

[0398] In an embodiment, the accessory module further includes a motor, where the motor is coupled to the threaded plunger and configured to exert a force on the threaded plunger (e.g., to open or close the hinge module and hence open or close a door when a door is the object). The computing device is configured to operate in accordance with another first mode of the first range of shear rates to control the motor to accelerate the rotational force of the object (e.g., to quickly close the door).

[0399] The computing device is further configured to operate in accordance with a second mode of the second range of shear rates when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism. For example, the computing device actively controls parameters of the STF module when the STF is operating in the second range of shear rates.

[0400] In an embodiment, the computing device is further configured to operate in accordance with another second mode of the second range of shear rates to control the motor to decelerate the rotational force of the object (e.g., slow close the door) when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism.

[0401] A first example scenario of utilization of the head unit system, that includes the status LED accessory module

and the STF module, includes the status LED indicating one or more of open/closed door, locked/unlocked lock, safe/not safe condition, and facilitating monitoring and control of every door closing utilizing each available STF threshold zone. To accommodate multiple status scenarios parameters of the LEDs are modified in correlation with a particular status. For example, a green LED illumination indicates that the door is closed while a red LED illumination indicates that the door is open. A flashing yellow LED indicates that an unsafe condition exists while a blue LED indicates that the doorway is safe. Flash patterns may also be utilized in addition to an illumination level of the LED. For example, a fast flash pattern indicates that the second STF threshold zone has been utilized while a slow flash pattern indicates that only the first STF threshold zone has been utilized.

[0402] A second example scenario of utilization of the head unit system includes the IoT module and the STF module. The IoT logic can be a wireless communication modem powered by an internal door movement energy harvesting device provides IoT logic and communications of door status with other similar devices and determines door control functions for storage and execution. For example, the IoT module is preprogrammed with parameters of the STF of the STF module such that when monitoring position and velocity of door movement the IoT module can predict which zone the STF is operating in to report STF utilization information to another computing device (e.g., of a facility where the head unit system is utilized along with numerous other head unit systems for controlling doors).

[0403] A third example scenario of utilization of the head unit system includes the motor module and the STF module. An electric motor functions to rotate one of the objects associated with the hinge objects to affect repositioning of an associated door. Movement via the electric motor is coordinated with the parameters of the STF module to provide a composite desire door functionality with regards to speed of closing the door and overall safety functions. In an instance, the computing device associated with the motor module interprets the movement of the door over numerous door opening and closing cycles to estimate the parameters of the STF (e.g., shear threshold ranges and zones versus viscosity). Once the parameters are determined, the computing device of the motor module determines when and how to activate the electric motor to assist the STF module to provide an overall desired opening and closing pattern for safety and convenience purposes.

[0404] A fourth example scenario of utilization of the head unit system includes the position sensor and the STF module. For example, a precision linear potentiometer is utilized within the position sensor module such that a computing device associated with the precision linear potentiometer interprets an output of the precision linear potentiometer to produce position, velocity, and acceleration information with regards to all physical aspects of the hinge module **204**. In an instance, computing device of the position sensor monitors movement of the hinge module over numerous opening and closing cycles to compare the response of the STF module for each of the opening and closings. The comparisons are interpreted to determine whether the STF module is still operating within desired operational parameters or if the STF module needs an adjustment or replacement. When a threshold level of undesired performance is reached, the position sensor module provides an indication

to the head unit system and/or to at least one other head unit system for potential corrective action.

[0405] In another instance, the computing device of the position sensor module further monitors the movement of the hinge module and compares a position and velocity signature of such openings and closings to other correlated information (e.g., day of week, time of day, traffic patterns, individual people coming through the doorway, etc.) to provide an output that is an estimate of an outcome associated with the other correlated information. For example, the position sensor module indicates that Joe has just walked through the door based on patterns of utilization of the door by Joe over the last **200** utilizations by Joe.

[0406] FIG. 15C illustrates scenario examples of utilization of at least two of the STF module **206-1**, the hinge module **204**, and one of a variety of accessory modules that shares the hinge module **204**. For example, the head unit system for controlling motion of an object includes a first STF module, a first hinge module coupled to the object and the first STF module, a second hinge module also coupled to the object, and an accessory module coupled to the second hinge module, where the accessory module is configured to interact with the first STF module (e.g., communicate, share power sources, control the object in a coordinated way, etc.)

[0407] A first scenario example of the utilization of two or more hinge modules along with multiple other modules includes the status LED module, the IoT module, the media sensor module, and the STF module. For example, the IoT module interprets image sensor imagery to provide improved status for display by the status LED and control of the door via adapting the STF. For instance, the image sensor mounted on the lower hinge constantly views foot traffic associated with people utilizing the door to produce statistics (e.g., traffic pattern versus time of day, velocity of entering the doorway, velocity of exiting the doorway, force utilized by each person to operate the door in either direction, etc.) associated with the aggregate of people utilizing the door. The IoT module determines implications of the traffic patterns on safety factors associated with utilizing the door and issues status information to the status LED module for display. For instance, the status LED module flashes the yellow LED with a fast pattern to indicate that parameters of the STF module may not be able to safely meet the needs of an eminent door opening sequence.

[0408] A second scenario example of the utilization of the two or more hinge modules along with multiple other modules includes the motor module, the position sensor module, the media sensor module, and the STF module. For example, electric motor module interprets output from the position sensor module and output from the image sensor of the media sensor module to provide improved adaptation of the STF of the STF module. For instance, the computing device of the motor module interprets the position sensor output over time and compares a desired response of the system versus actual to identify gaps in operation where the electric motor can assist by applying an opening or a closing force to the door.

[0409] A third scenario example of the utilization of the two or more hinge modules along with multiple other modules includes the media sensor module, the IoT module, the motor module, and the STF module, where the media sensor module is mounted at an upper hinge location. For example, an image sensor associated with the upper mounted media sensor module is utilized to perform routine

facial recognition of people utilizing the doorway for correlation with typical parameters associated with operation of the door (e.g., force of opening, force of closing, speed of door operation, etc.). The IoT module acquires further door utilization information from another head unit system associated with another door for a common person that utilizes both doorways. The computing device of the electric motor module summarizes the door utilization information for the common person to produce adaptation commands for the electric motor to assist the STF module in the operation of the door to further benefit the common person as they utilize this particular door. For instance, recognizing that Joe is approaching the door utilizing the media sensor, the motor module applies energy to the electric motor to assist in pre-positioning the door in conjunction with the STF module to provide a better experience for Joe and utilizing the door as Joe desires.

[0410] In another instance of the third scenario example, the IoT module interprets an emergency lock signal from another wireless device (e.g., a wireless pendent utilized by a school teacher). When the emergency lock signal is detected, the motor module operates a lockdown of the door movement (e.g., closed a hold, hold).

[0411] In another scenario, the media sensor output is interpreted by the computing device of the media sensor to identify an unsafe local condition (e.g., fire, unauthorized access, water on the floor, someone standing in the way of the way the door swings on the other side of the door, etc.). The IoT module communicates the unsafe local condition to at least one other head unit system associated with another door for subsequent optimization. The electric motor of the motor module prevents the door from opening due to the unsafe local condition.

[0412] In yet another scenario, the computing device associated with any of the modules (e.g., the STF module) interprets fluid response from a set of fluid flow sensors associated with the chamber to produce a piston velocity and a piston position of the piston associated with the STF module. The computing device determines a shear force based on the piston velocity and the piston position. The computing device determines a desired response of the STF based on one or more of the shear force, the piston velocity, and the piston position. The computing device activates an accessory function in accordance with the desired response for the STF to cause selection of one of the first range of shear rates and the second range of shear rates. Examples of the accessory function include a status LED, and IoT function, a motor control function, a position determination function, an image processing function, a sound processing function, etc.

[0413] FIG. 15D illustrates a further scenario example of utilization of at least two of the STF module 206-1, the hinge module 204, and one of a variety of accessory modules that shares the hinge module 204 on three doors 302-1 through 302-3 within a hallway 300. For example, the three head unit systems each include an STF module, a status LED module, a motor module, and a media sensor module. The scenario includes an active shooter situation where the shooter is closer to doors 302-1 and 302-2 and further away from door 302-3. The scenario further includes a person near door 302-3.

[0414] The media sensor module includes a detection function to detect the active shooter scenario. The detection function includes at least one of a gunshot detector (e.g.,

matching live audio samples to known gunshot audio patterns while excluding routine sounds) and an image matcher to identify a weapon and/or the shooter holding the weapon (e.g., matching live images/video samples to stored images/video of weapons and shooters with weapons).

[0415] The processor associated with the media sensor of door 302-1 detects that a safety threat exists within a first proximity threshold of the object. The safety threat includes anything that threatens the safety of people or property. Examples include an active shooter, a fire, a building collapse, and a person intent on causing bodily harm. The detecting includes local detection by sensing and receiving a message. For example, a gunshot is detected by the processor associated with the media sensor of door 302-1 and receives a wireless message from the processor associated with the media sensor of door 302-3 that the gunshot was detected slightly later.

[0416] The processor associated with the motor of the door 302-1 controls the motor configured to accelerate the rotational force of the object when the object 302-1 is the door while the door closes with a first velocity (e.g., very fast relative to normal operation). The processor associated with the status LED module of door 302-1 indicates a status of the object to include closing of the door with the first velocity (e.g., flashes red to telegraph the urgency of closing the door very fast relative to normal operation). The modules associated with the door 302-2 operate in a similar way as described with reference to the modules associated with the door 302-1.

[0417] The processor associated with the media sensor of door 302-3 detects that the safety threat exists within a second proximity threshold of the object (e.g., third door). The second proximity threshold is a greater distance than the first proximity threshold. For example, the gunshot is detected by the processor associated with the media sensor of door 302-3 and receives the wireless message from the processor associated with the media sensor of door 302-1 that the gunshot was detected slightly sooner at door 1 as compared to at door 3.

[0418] The processor associated with the motor of the door 302-3 controls the motor configured to deaccelerate the rotational force of the object when the object 302-1 is the door while the door closes with a second velocity (e.g., similar to normal operation). The processor associated with the status LED module of door 302-3 indicates a status of the object to include closing of the door with the second velocity (e.g., flashes yellow to telegraph the closing the door due to the safety threat).

[0419] In another example of operation, the media sensor of the door 302-3 detects that a person, not the shooter, is nearby the third door and that a safety improvement can be provided by enabling that person to pass through the third door to better relative safety. Having detected the person, the processor of the motor of the door 302-3 controls the motor to accommodate getting that person through the door and then quickly closing the door to protect those behind door 3 from the active shooter. Once the door is closed, the processor of the status LED causes the LED to indicate the door locked (e.g., steady red).

[0420] It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a

number of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as ‘data’).

[0421] As may be used herein, the terms “substantially” and “approximately” provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than $\pm 1\%$). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

[0422] As may also be used herein, the term(s) “configured to”, “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as “coupled to”.

[0423] As may even further be used herein, the term “configured to”, “operable to”, “coupled to”, or “operably coupled to” indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term “associated with”, includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

[0424] As may be used herein, the term “compares favorably”, indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired relationship is that signal 1 has a greater magnitude than signal 2, a favorable comparison may be achieved when the magnitude of signal 1 is greater than that of signal 2 or when the magnitude of signal 2 is less than that of signal 1. As may be used herein, the term “compares unfavorably”, indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

[0425] As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase “at least one of a, b, and c” or of this generic form “at least one of a, b, or c”, with more or less elements than “a”, “b”, and “c”. In either phrasing, the phrases are to be interpreted identically. In particular, “at least one of a, b, and c” is equivalent to “at least one of a, b, or c” and shall mean a, b,

and/or c. As an example, it means: “a” only, “b” only, “c” only, “a” and “b”, “a” and “c”, “b” and “c”, and/or “a”, “b”, and “c”.

[0426] As may also be used herein, the terms “processing module”, “processing circuit”, “processor”, “processing circuitry”, and/or “processing unit” may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

[0427] One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.

[0428] To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and

still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules, and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.

[0429] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0430] The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

[0431] Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

[0432] The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

[0433] As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive, server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent storage (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

[0434] While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

What is claimed is:

1. A head unit system for controlling motion of an object, the head unit system comprising:

a shear thickening fluid (STF) module, wherein the STF module includes:

a shear thickening fluid (STF), wherein the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates;

a chamber, the chamber configured to contain a portion of the STF, wherein the chamber includes a front channel and a back channel;

a piston, the piston housed at least partially radially within the chamber, the piston configured to include a slot set, wherein the piston is further configured to facilitate flow of the STF through the slot set

- between the front channel and the back channel when the piston travels longitudinally within the chamber; and
- a plunger, the plunger housed at least partially radially within the back channel of the chamber, wherein the plunger is configured to exert force on the piston in response to the motion of the object;
 - a hinge module, wherein the hinge module includes:
 - a coupling mechanism, the coupling mechanism configured to couple the STF module to the hinge module; and
 - a threaded plunger, the threaded plunger configured to facilitate an inward force to the plunger in response to a rotational force of the object, wherein the threaded plunger is further configured to generate the inward force when a nut rotates on threads of the threaded plunger as a result of the rotational force of the object, wherein the threaded plunger is further configured to facilitate an outward force to the plunger in response to a counter-rotational force of the object, wherein the threaded plunger is further configured to generate the outward force when the nut counter-rotates on the threads of the threaded plunger as a result of the counter-rotational force of the object; and
 - an accessory module, wherein the accessory module includes:
 - an accessory coupling mechanism, the accessory coupling mechanism configured to couple the accessory module to the hinge module;
 - a motor, the motor coupled to the threaded plunger and configured to exert a force on the threaded plunger; and
 - a computing device, the computing device includes a wireless communication modem and is configured to operate in accordance with a first mode of the first range of shear rates to control the motor to accelerate the rotational force of the object in response to a wireless message received via the wireless communication modem from another computing device, wherein the computing device is further configured to operate in accordance with a second mode of the second range of shear rates to control the motor to decelerate the rotational force of the object in response to the wireless message when the STF module is coupled to the hinge module via the coupling mechanism and the hinge module is simultaneously coupled to the accessory module via the accessory coupling mechanism.
2. The head unit system of claim 1, wherein the STF module further comprises:
- a modular castle mount, the modular castle mount configured to extend the chamber to facilitate the configuration of the plunger to exert the force on the piston in response to the motion of the object.
3. The head unit system of claim 2, wherein the hinge module further comprises:
- a reciprocal modular castle mount, the reciprocal modular castle mount configured to couple the reciprocal modular castle mount with the modular castle mount, wherein the threaded plunger is configured to apply the inward force to the plunger in response to the rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount, wherein the threaded plunger is further configured to apply the outward force to the plunger in response to the counter-rotational force of the object when the reciprocal modular castle mount is coupled with the modular castle mount.
4. The head unit system of claim 3, wherein the accessory module further comprises:
- the accessory coupling mechanism configured with a second modular castle mount, wherein the second modular castle mount couples the accessory module to the hinge module via a second reciprocal modular castle mount of the hinge module.
5. The head unit system of claim 1, wherein the slot set further comprises one or more of:
- a first slot of the slot set configured with at least one cylindrical tube with substantially consistent diameter from one side of the piston to an opposite side of the piston;
 - a second slot of the slot set configured with at least one conical shaped tube with an increasing diameter from the one side of the piston to the opposite side of the piston;
 - a third slot of the slot set configured with the at least one conical shaped tube with a decreasing diameter from the one side of the piston to the opposite side of the piston; and
 - a fourth slot of the slot set configured with at least one venturi shaped tube from the one side of the piston to the opposite side of the piston.
6. The head unit system of claim 1, wherein the STF module further comprises:
- a set of gates, the set of gates configured to separate the front channel and the back channel to control velocity of flow of the STF between the front channel and the back channel, wherein the set of gates is further configured to enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in an open position, wherein the set of gates is further configured to enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in a closed position;
- the set of gates is further configured to:
- when the set of gates is traveling through the chamber in an outward direction towards the back channel, a first shear threshold effect includes:
 - the first range of shear rates when the STF is configured to have the decreasing viscosity; and
 - when the set of gates is traveling through the chamber in an inward direction away from the back channel, a second shear threshold effect includes:
 - the second range of shear rates when the STF is configured to have the increasing viscosity; and
- a set of hinges, the set of hinges configured to enable the set of gates to swing to the open position from the closed position and to swing to the closed position from the open position; and
- wherein the set of gates is further configured to:
- when the set of hinges is configured to enable the set of gates to swing to the open position within the front channel;

- enable the STF to provide the first range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the open position; and
- enable the STF to provide the second range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the closed position; and
- when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel:
 - enable the STF to provide the second range of shear rates as the STF flows from the back channel to the front channel when the set of gates is in the closed position; and
 - enable the STF to provide the first range of shear rates as the STF flows from the front channel to the back channel when the set of gates is in the open position.
- 7. The head unit system of claim 6, wherein the STF module further comprises:
 - when the set of hinges is configured to enable the set of gates to swing to the open position within the front channel:
 - the piston is further configured to facilitate the flow of the STF from the back channel through the slot set of the piston to the front channel to cause the set of gates to operate in the open position and to cause the STF within the back channel to provide the first range of shear rates when the set of gates is operating in the open position, wherein the piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the front channel through the slot set of the piston to the back channel to cause the STF within the front channel to provide the second range of shear rates when the set of gates is operating in the closed position; and
 - when the set of hinges is configured to enable the set of gates to swing to the open position within the back channel:
 - the piston is further configured to facilitate the flow of the STF from the front channel through the slot set of the piston to the back channel to cause the set of gates to operate in the open position and to cause the STF within the front channel to provide the first range of shear rates when the set of gates is operating in the open position, wherein the piston is further configured to facilitate flow of the STF around the set of gates between the chamber and the set of gates from the back channel through the slot set of the piston to the front channel to cause the STF within the back channel to provide the second range of shear rates when the set of gates is operating in the closed position.
- 8. The head unit system of claim 1, wherein the STF module further comprises:
 - a bushing, the bushing configured in a fixed position relative to the chamber to cause containment of the STF in the back channel; and
 - a backstop, the backstop configured with the chamber to cause containment of the STF in the front channel.
- 9. The head unit system of claim 1, wherein the STF module further comprises:
 - a cap, the cap configured to move in unison with the plunger radially within the chamber; and
 - a spring, the spring configured to store energy when the cap moves in an inward direction in response to a force from the object, wherein the spring is further configured to release the energy, absent the force from the object, such that the cap and the plunger move in an outward direction.
- 10. The head unit system of claim 1, wherein the hinge module further comprises:
 - a second plunger, the second plunger configured to produce the inward force from the threaded plunger to the plunger when the nut rotates on threads of the threaded plunger as the result of the rotational force of the object, wherein the second plunger is further configured to produce the outward force from the threaded plunger to the plunger when the nut counter-rotates on the threads of the threaded plunger as the result of the counter-rotational force of the object.
- 11. The head unit system of claim 1, wherein the STF comprises:
 - a plurality of nanoparticles, wherein the plurality of nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture thereof.
- 12. The head unit system of claim 1, wherein the STF further comprises:
 - one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture thereof.
- 13. The head unit system of claim 1, wherein the STF module further comprises:
 - the computing device configured with:
 - a processor for executing processor-executable instructions,
 - the wireless communication modem configured to communicate a set of wireless messages with a set of other computing devices,
 - an image sensor configured to provide imagery, and
 - a storage device for storing the processor-executable instructions, wherein the processor-executable instructions include:
 - a first set of processor-executable instructions, the first set of processor-executable instructions configured to communicate a first wireless message with regards to first imagery, wherein the set of wireless messages includes the first wireless message,
 - a second set of processor-executable instructions, the second set of processor-executable instructions configured to communicate a second wireless message to control the motor, wherein the set of wireless messages includes the second wireless message, and
 - a third set of processor-executable instructions, the third set of processor-executable instructions configured to communicate a third wireless message to control another motor associated with another accessory module, wherein the set of wireless messages includes the third wireless message.
- 14. The head unit system of claim 1, wherein the STF module further comprises:

a set of fluid flow sensors positioned proximal to the chamber, wherein the set of fluid flow sensors are configured to provide a fluid response from the STF, wherein the set of fluid flow sensors includes one or more of:

- a valve opening detector associated with a slot set of the piston,
- a mechanical position sensor,
- an image sensor,
- a light sensor,
- an audio sensor,
- a microphone,
- an ultrasonic sound sensor,
- an electric field sensor,
- a magnetic field sensor, and
- a radio frequency wireless field sensor.

15. The head unit system of claim **1**, wherein the STF module further comprises:

- a set of fluid manipulation emitters positioned proximal to the chamber, wherein the set of fluid manipulation emitters provide a fluid activation to at least one of the STF, a set of gates, and a slot set of the piston to provide the control of the motion of the object, wherein the set of fluid manipulation emitters includes one or more of:
 - a variable flow valve associated with the slot set,
 - a mechanical vibration generator,
 - an image generator,
 - a light emitter,
 - an audio transducer,
 - a speaker,
 - an ultrasonic sound transducer,
 - an electric field generator,
 - a magnetic field generator, and
 - a radio frequency wireless field transmitter.

16. The head unit system of claim **1**, wherein the accessory module further comprises:

- the computing device configured with:
 - a processor for executing processor-executable instructions,
- the wireless communication modem configured to communicate the wireless message with the other computing device, and
- a storage device for storing the processor-executable instructions, wherein the processor-executable instructions include:
 - a first set of processor-executable instructions, the first set of processor-executable instructions configured to facilitate the first mode of the first range of shear rates, and
 - a second set of processor-executable instructions, the second set of processor-executable instructions configured to facilitate the second mode of the second range of shear rates.

17. The head unit system of claim **16**, wherein the accessory module further comprises:

- the first set of processor-executable instructions is further configured to facilitate the first mode of the first range of shear rates, when executed by the processor, to cause the processor to perform one or more of:
 - controlling the motor configured to accelerate the rotational force of the object,

- determining a position of the object,
- determining a movement vector of the object,
- identifying a secondary object associated with the object,

- interpreting a request to change the movement vector of the object,

- indicating a status of the object, wherein the status of the object includes one or more of open, closed, locked, unlocked, safe, not safe, and emergency fast closing when the object is a door, and

- controlling a set of fluid manipulation emitters positioned proximal to the chamber to facilitate the first range of shear rates, wherein the set of fluid manipulation emitters provide a fluid activation to at least one of the STF, the set of gates, and a slot set of the piston to provide the control of the motion of the object; and

the second set of processor-executable instructions is further configured to facilitate the second mode of the second range of shear rates, when executed by the processor, to cause the processor to perform one or more of:

- controlling the motor configured to deaccelerate the rotational force of the object,

- detecting an end-position of the object,

- determining an end-movement vector of the object,

- determining a movement vector of the secondary object associated with the object,

- interpreting a request to change the end-movement vector of the object,

- indicating a further status of the object, wherein the further status of the object includes one or more of open, closed, locked, unlocked, safe, not safe, and emergency slow closing when the object is a door when the object is the door, and

- controlling the set of fluid manipulation emitters to facilitate the second range of shear rates.

18. The head unit system of claim **16**, wherein the accessory module further comprises:

- the first set of processor-executable instructions is further configured to facilitate the first mode of the first range of shear rates, when executed by the processor, to cause the processor to perform one or more of:

- detecting that a safety threat exists within a first proximity threshold of the object, controlling the motor configured to accelerate the rotational force of the object when the object is a door while the door closes with a first velocity, and

- indicating a status of the object to include closing of the door with the first velocity; and

- the second set of processor-executable instructions is further configured to facilitate the second mode of the second range of shear rates, when executed by the processor, to cause the processor to perform one or more of:

- detecting that the safety threat exists within a second proximity threshold of the object, wherein the second proximity threshold is a greater distance than the first proximity threshold,

controlling the motor configured to deaccelerate the rotational force of the object when the object is the door while the door closes with a second velocity, wherein the second velocity is less than the first velocity, and
indicating the status of the object to include closing of the door with the second velocity.

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